UNIVERSIDADE DA CORUÑA

Rnasa Imedir

Unit 1.3

Support Vector Machines (SVM)

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Machine Learning I

fic facultade de informática da coruña

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Support Vector Machines

- SVM
  - Classification technique
  - They are derived from the statistical learning theory postulated by Vapnik and Chervonenkis
  - They were introduced in 1992, and have received much interest in recent years for:
    - Very high classification accuracy
    - Very firm mathematical basis
    - In many fields they have shown to outperform the most successful systems (ANNs)
  - Classifier in two classes
    - Generalizable to more classes
    - Extendable for regression problems

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Support Vector Machines

- From a series of observations, each consisting of a pair of data:
  - A vector

$$x_i \in R^n, i = 1,\ldots,l$$

  - A label

$$y_i \in \{+1,-1\}$$

- Suppose we have a hyperplane separating the positive (+1) from the negative (-1) samples.
  - Linearly separable problems
  - Example, in R^2:

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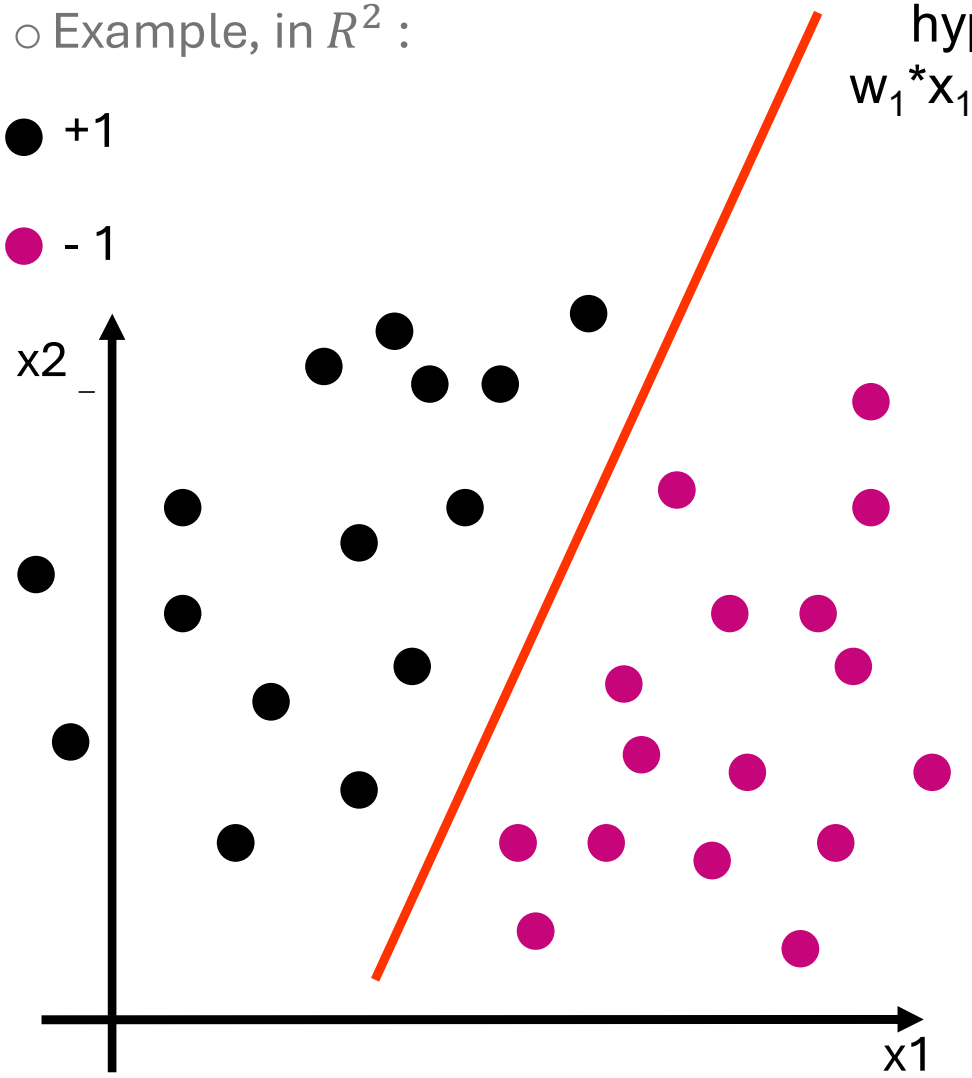
Support Vector Machines

- Linearly separable problems
  - Example, in R^2 :



hyperplane :
$w_1 * x_1 + w_2 * x_2 + b$

$$w = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$$

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\mathbf{w}^T \mathbf{x} + \mathbf{b} = 0$$

All samples in the hyperplane satisfy:
w*x+b = 0

All samples meet: ●
w*x+b > 0

All samples meet: ●
w*x+b < 0

(in this image the subscripts of x_1 and x_2 denote the dimension, not that they are samples 1 and 2)

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Support Vector Machines

- From a series of observations, each consisting of a pair of data:
  - A vector $x_i \in R^n, i = 1,\ldots,l$
  - A label $y_i \in \{+1,-1\}$
- Suppose we have a hyperplane separating the positive (+1) from the negative (-1) samples.
  - Linearly separable problems
  - Hyperplane, general case:
    - The points xi that are on the hyperplane satisfy $w^T x + b = 0$
    - The points xi that are "on one side" of the hyperplane satisfy $w^T x + b > 0$
    - The points xi that are "on the other side" of the hyperplane satisfy $w^T x + b < 0$

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Support Vector Machines

- In the real world, the vast majority of problems are not linearly separable.
- Study of the SVMs in three different cases:
  - Linearly separable problems
    - Linear SVMs
    - Separation region: hyperplane
  - Nonlinearly separable problems
    - Linear SVMs
    - Separation region: hyperplane
    - Some error in classification is allowed
  - Nonlinear SVM
    - Even admitting some error, it is not possible to separate the data
    - More complex separation regions

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Support Vector Machines

- Linearly separable problems
  - We want to find the hyperplane $w \cdot x + b = 0$
  - defined by the pair (w, b), such that the point xi can be separated according to the function
  - i.e. $$f(x_i) = \begin{cases} 1 & w \cdot x_i + b > 0 \quad \text{and } y_i = 1 \\ -1 & w \cdot x_i + b < 0 \quad \text{and } y_i = -1 \end{cases}$$
  - where $x \in R^n$ and $b \in R$
    - That is, the function f classifies each point as +1 or -1

$$f(x_i) = sign(w \cdot x_i + b) = \begin{cases} 1 \text{ and } y_i = 1 \\ -1 \text{ and } y_i = -1 \end{cases}$$

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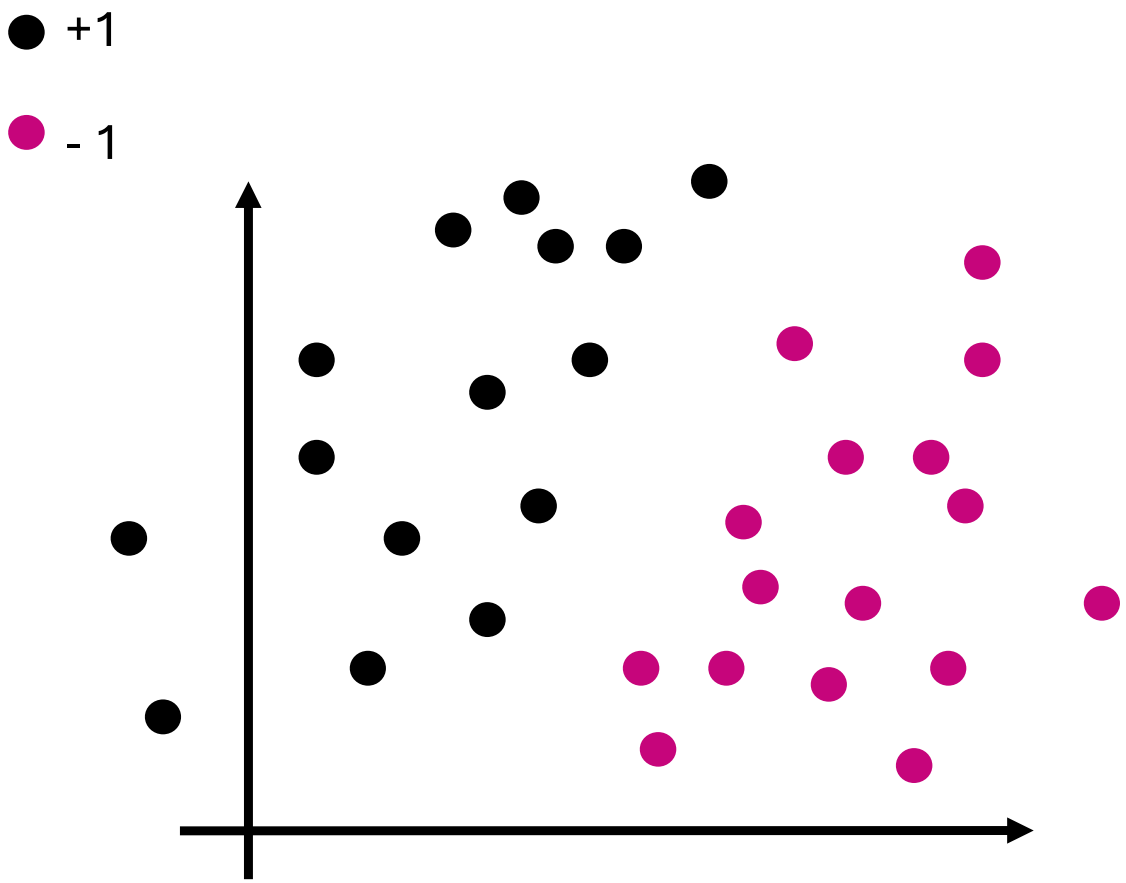
Support Vector Machines

- Linearly separable problems



How to classify this data?

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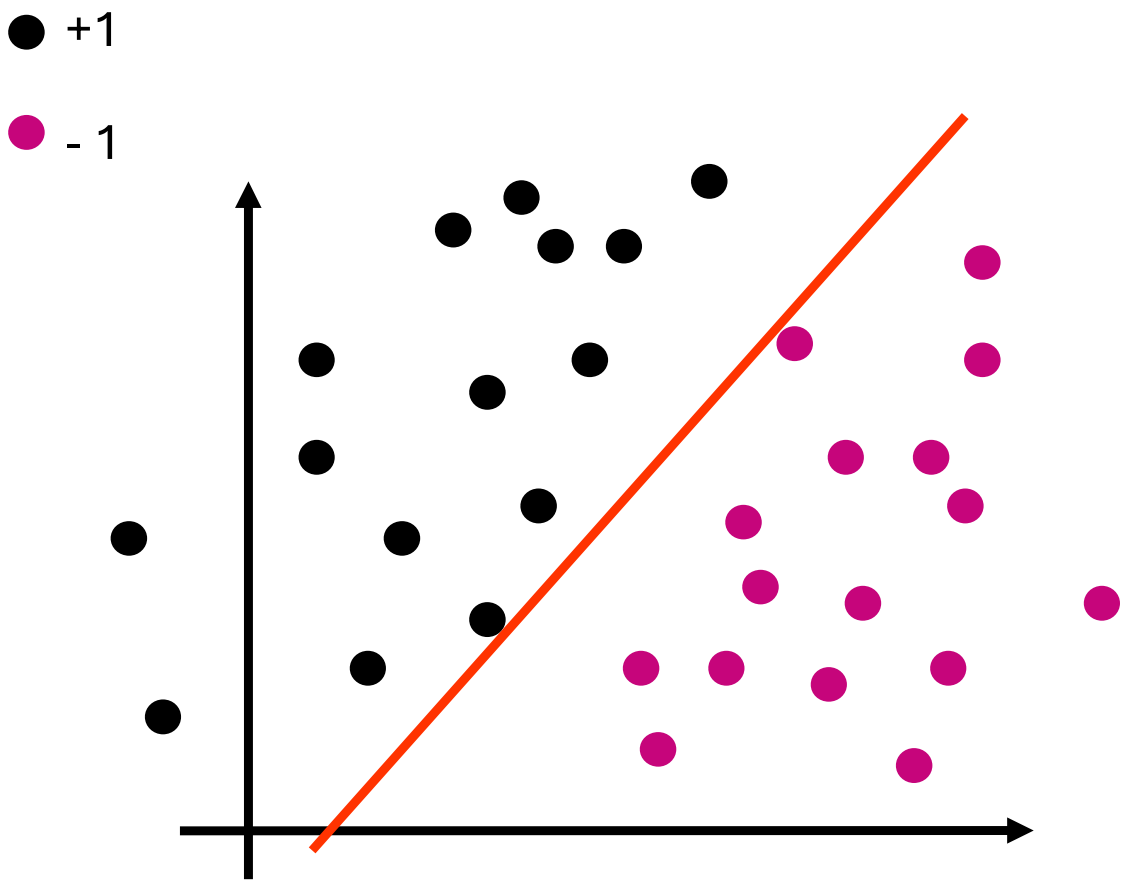
Support Vector Machines

- Linearly separable problems



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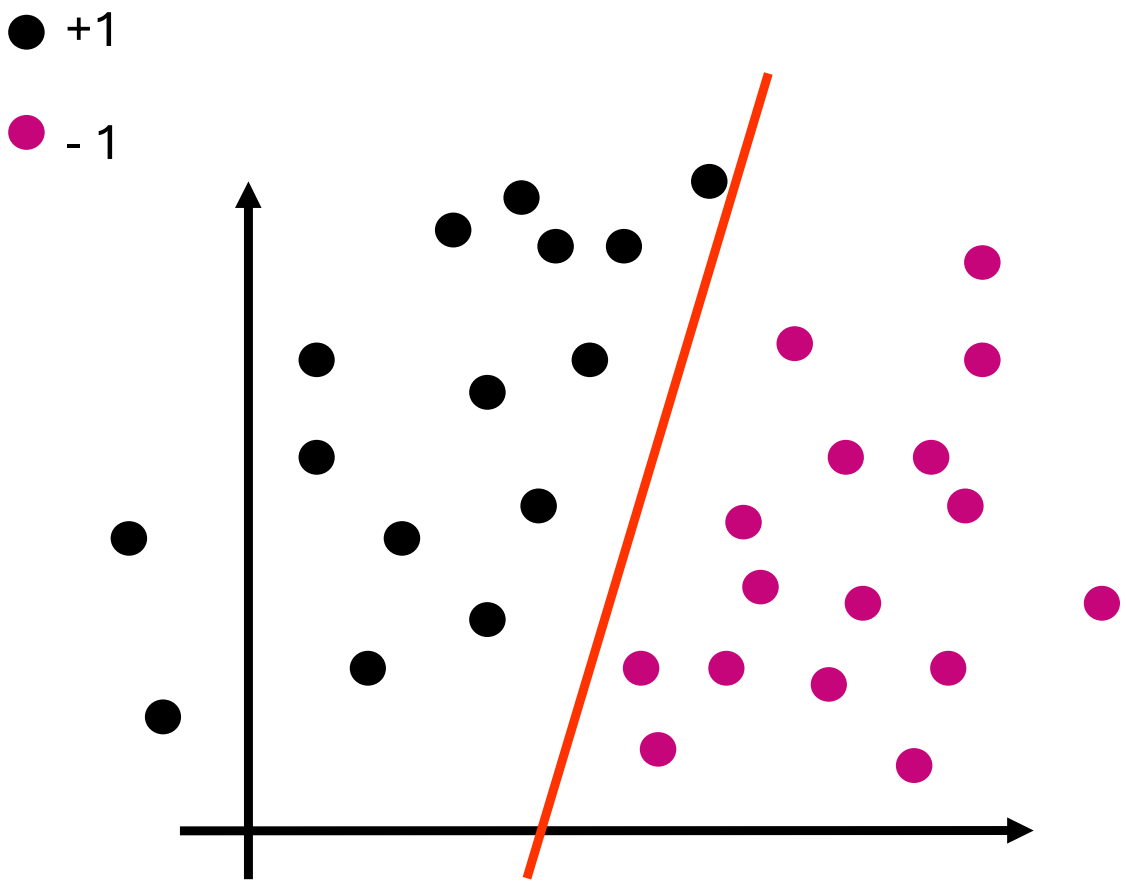
Support Vector Machines

- Linearly separable problems



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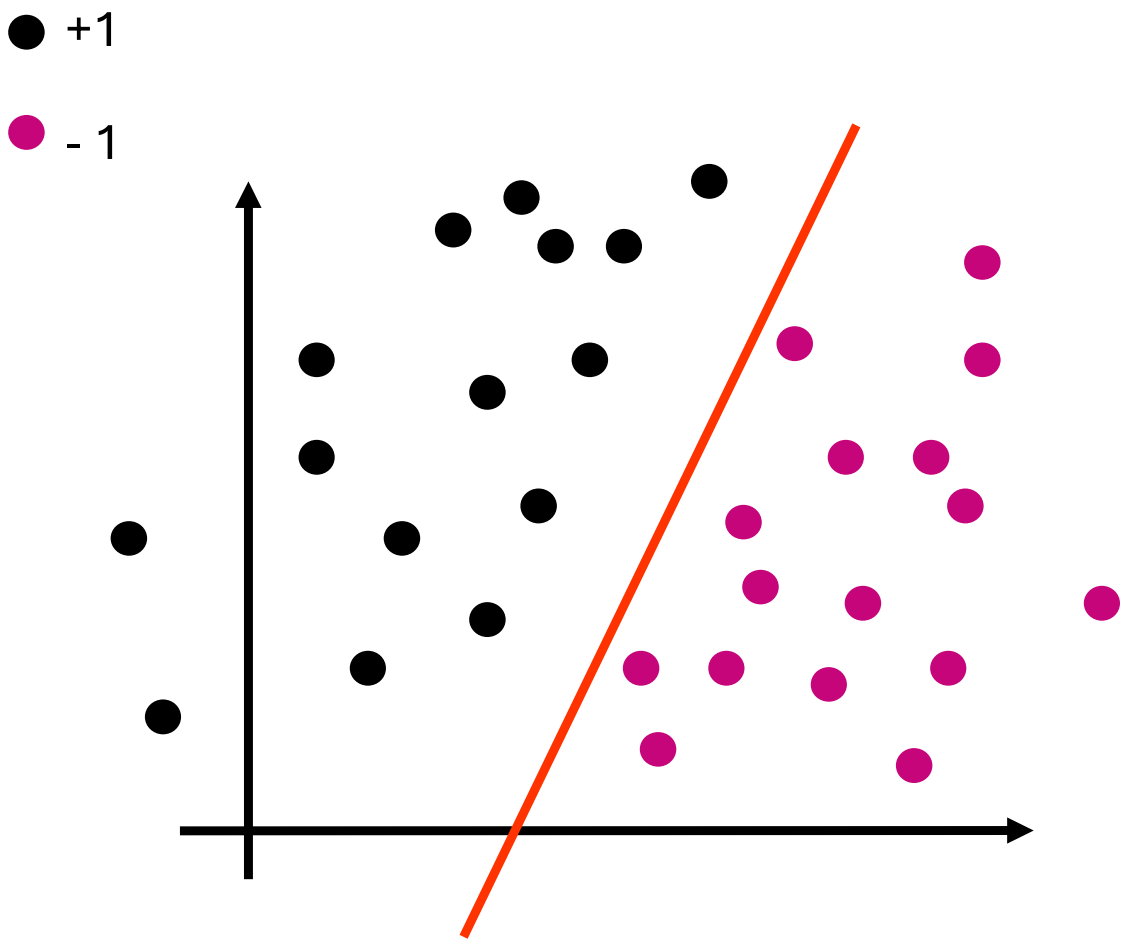
Support Vector Machines

- Linearly separable problems



Any can be good, but which is the best?

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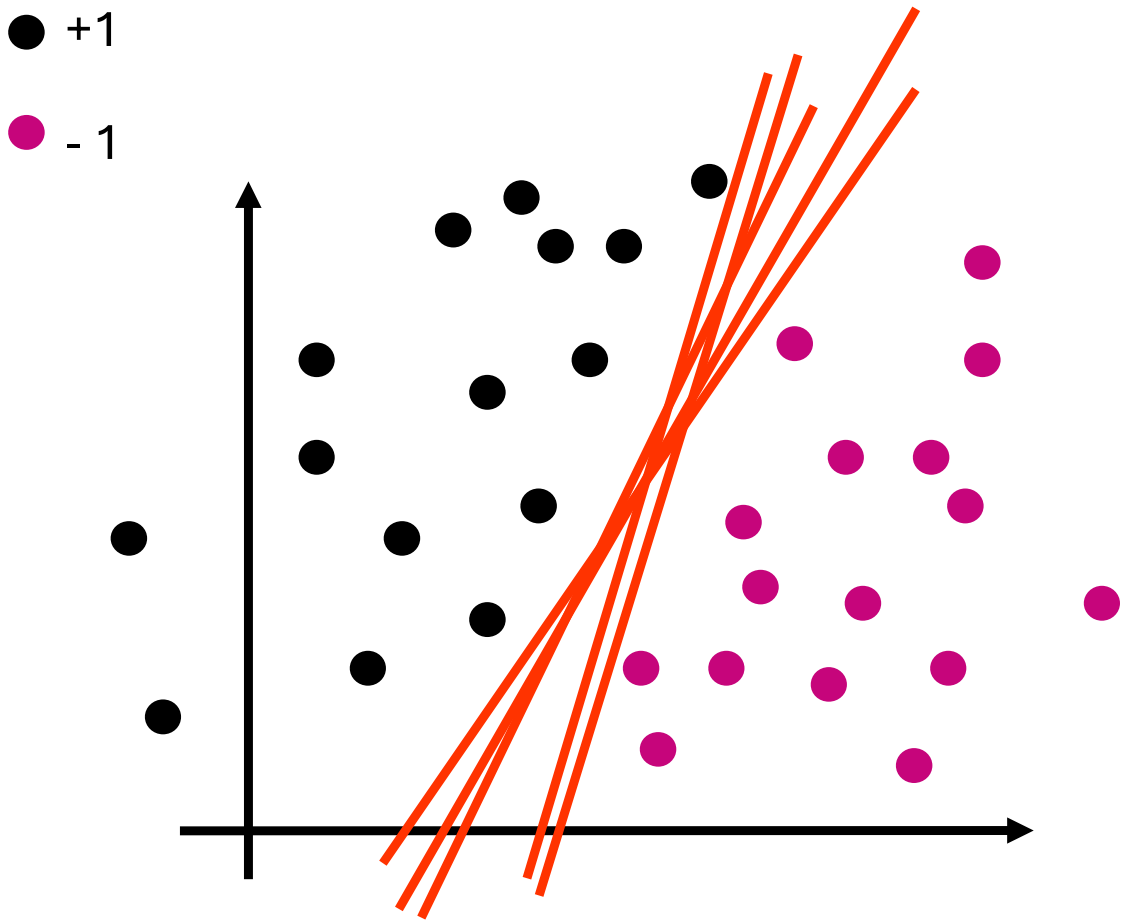
Support Vector Machines

- Linearly separable problems



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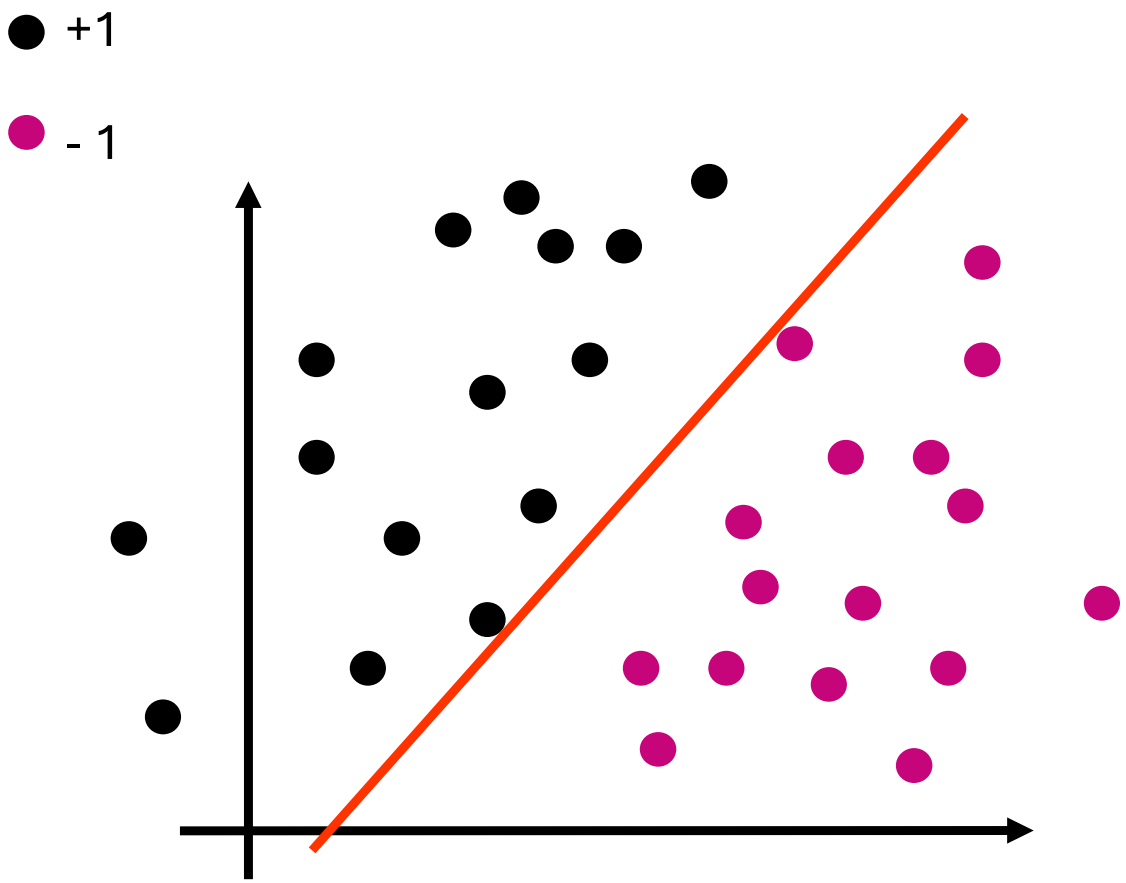
Support Vector Machines

- Linearly separable problems



If this hyperplane is chosen

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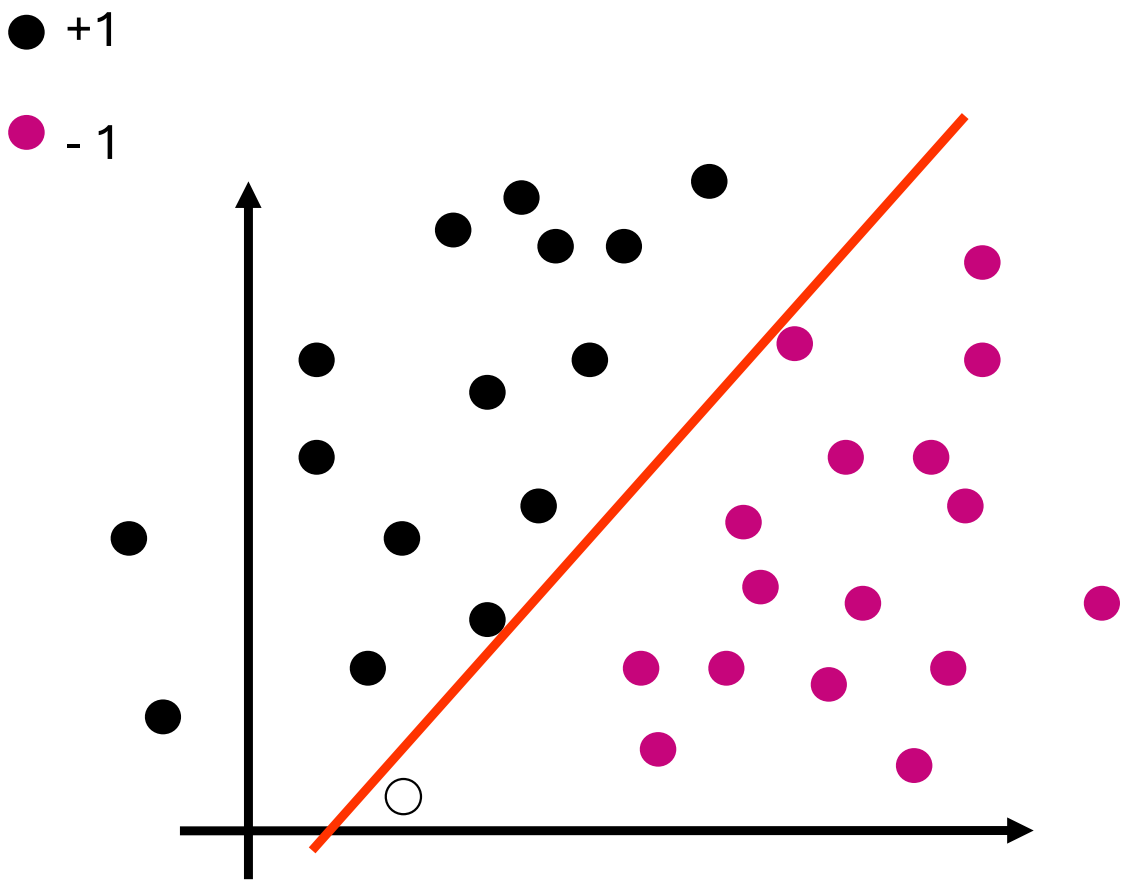
Support Vector Machines

- Linearly separable problems



If this hyperplane is chosen, given this new instance…

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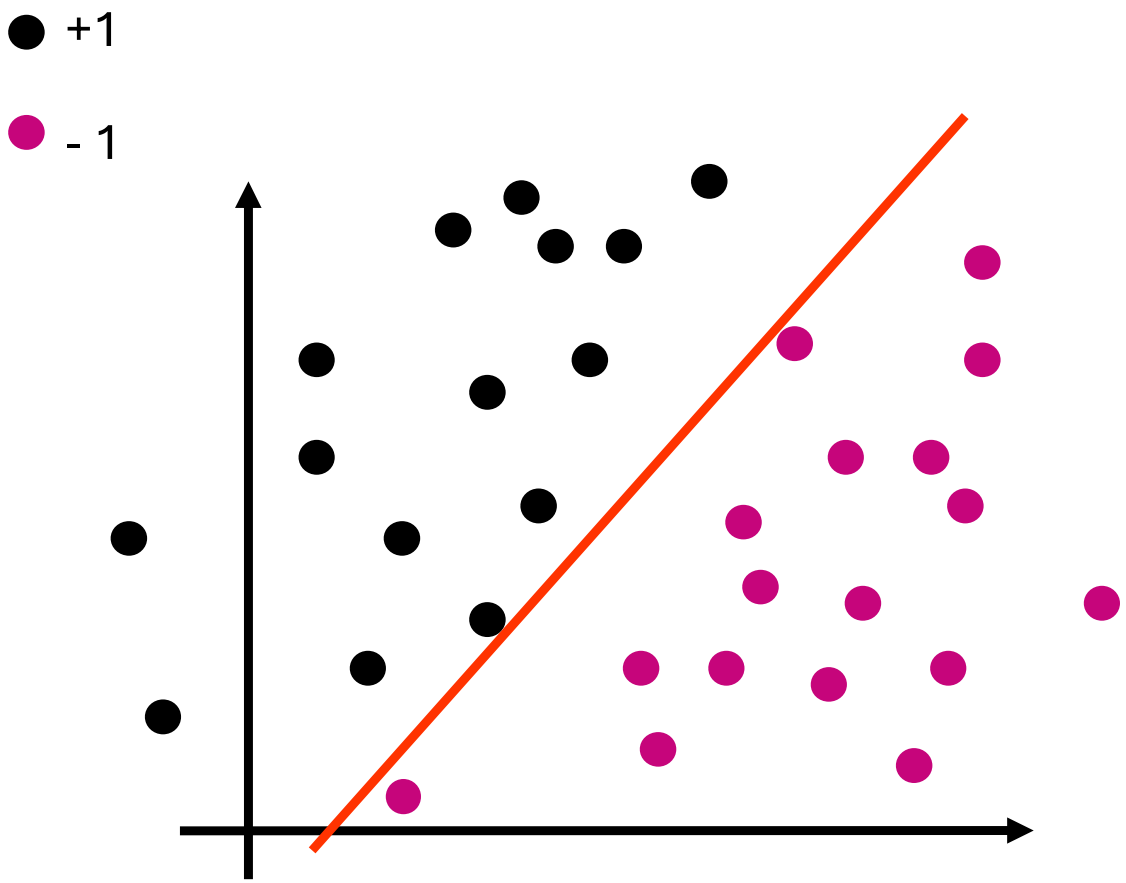
Support Vector Machines

- Linearly separable problems



If this hyperplane is chosen, given this new instance…
it is classified as -1

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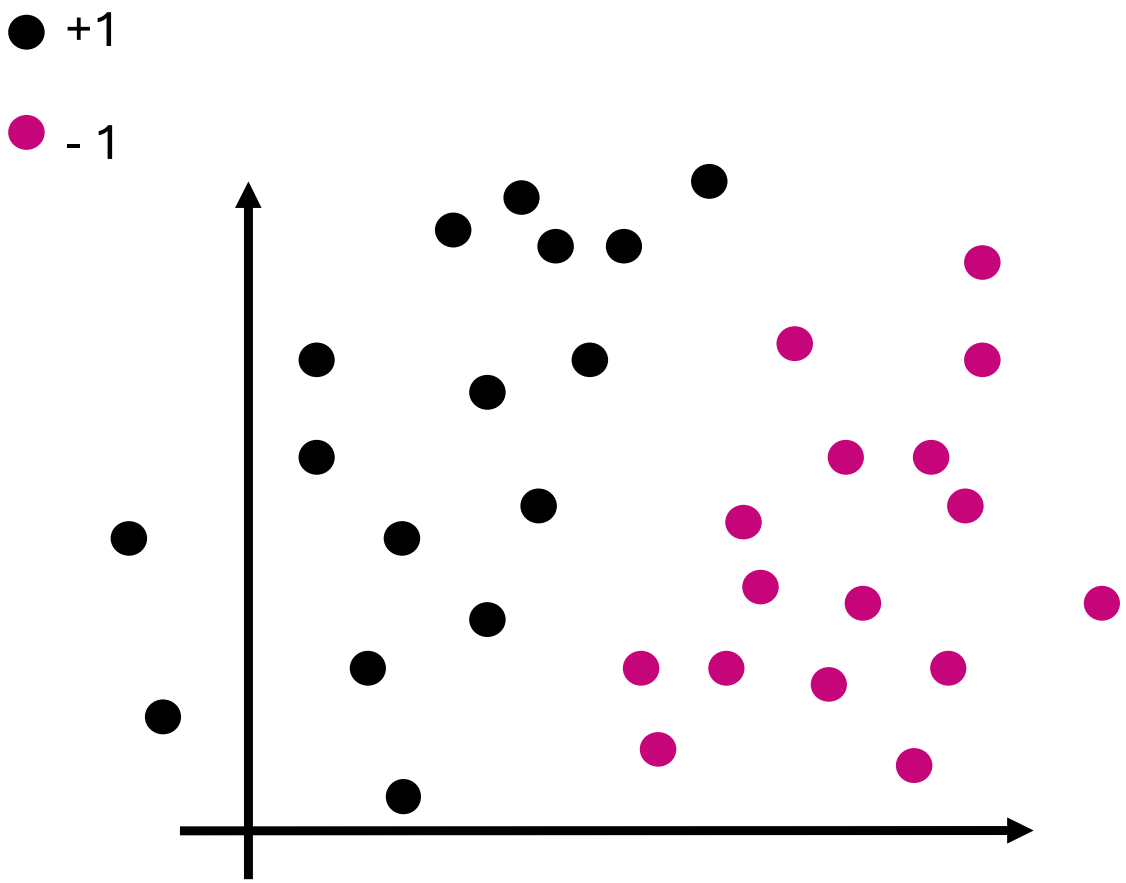
Support Vector Machines

- Linearly separable problems



If this hyperplane is chosen, given this new instance…
it is classified as -1
when it should be +1

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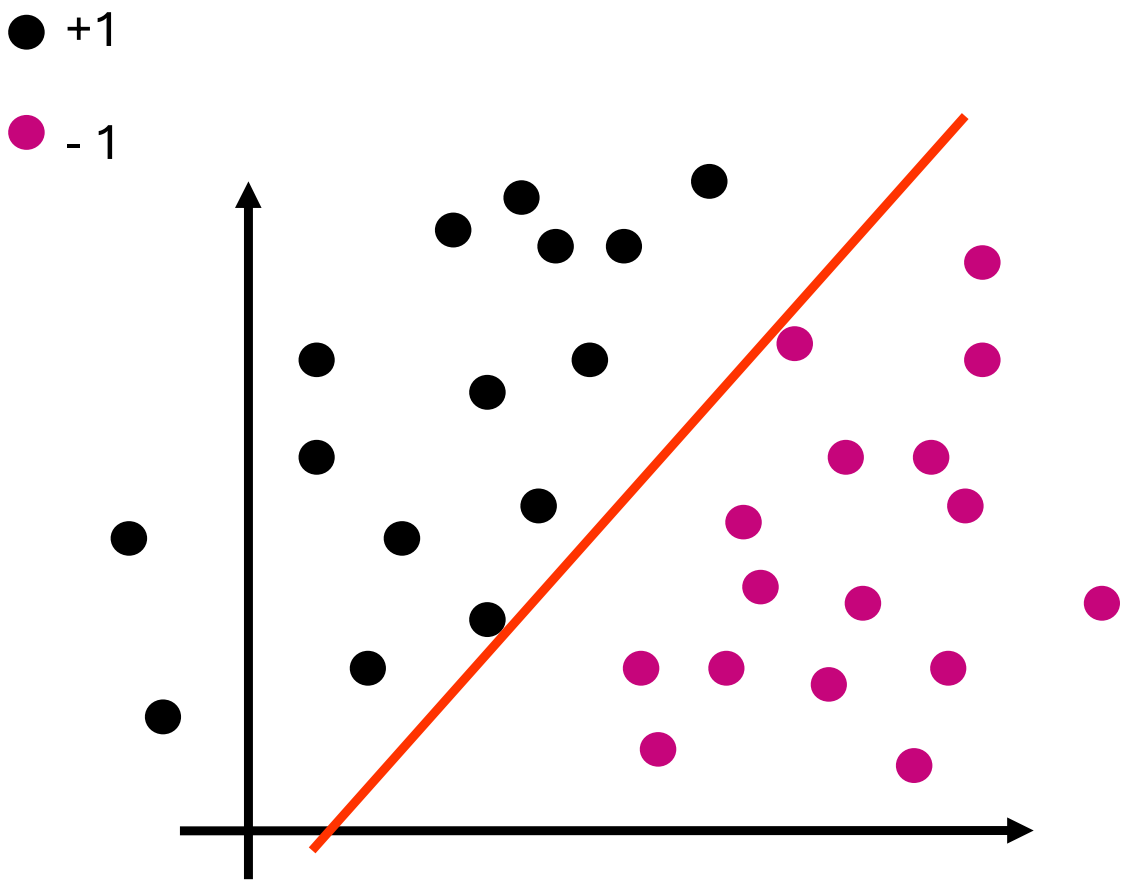
Support Vector Machines

- Linearly separable problems



Problem of the chosen hyperplane: it is too close to the samples

A sample close to the hyperplane could "fall" to either side

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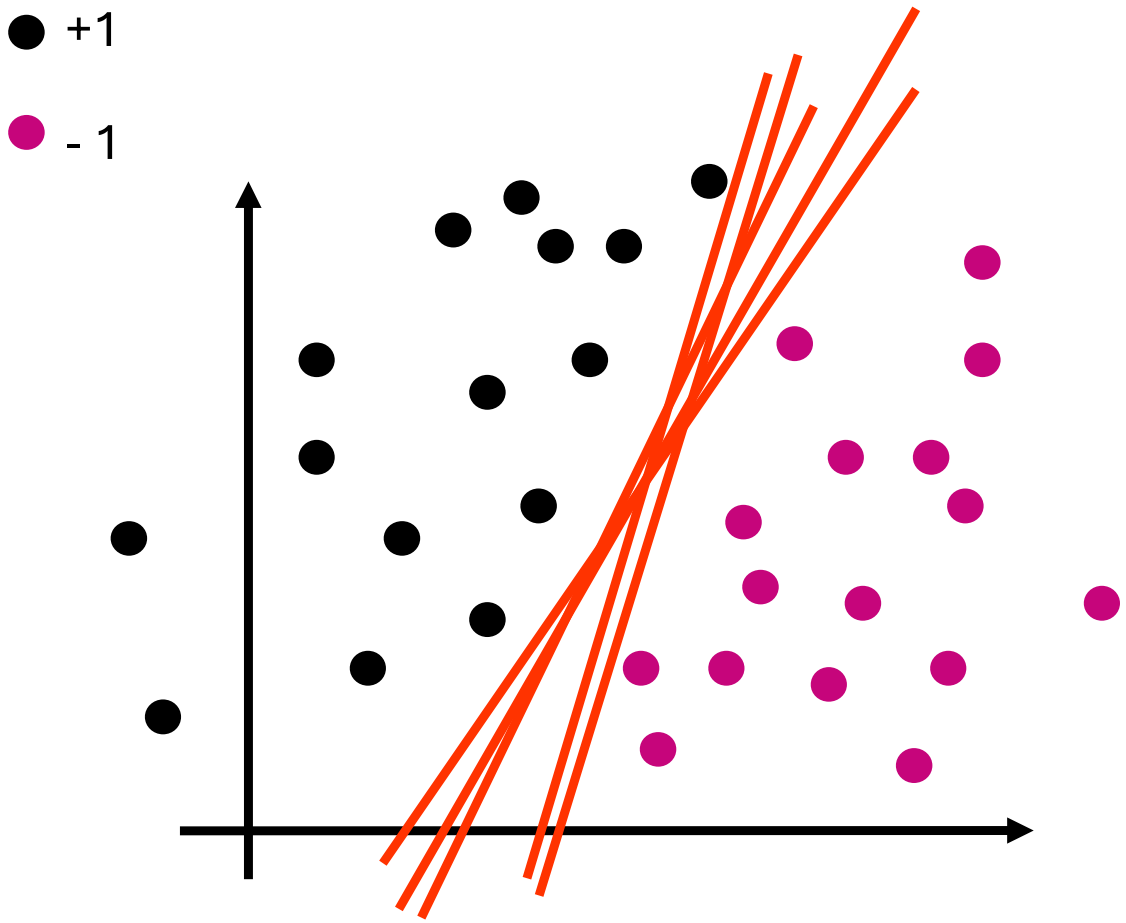
Support Vector Machines

- Linearly separable problems



So, which hyperplane best classifies new cases?

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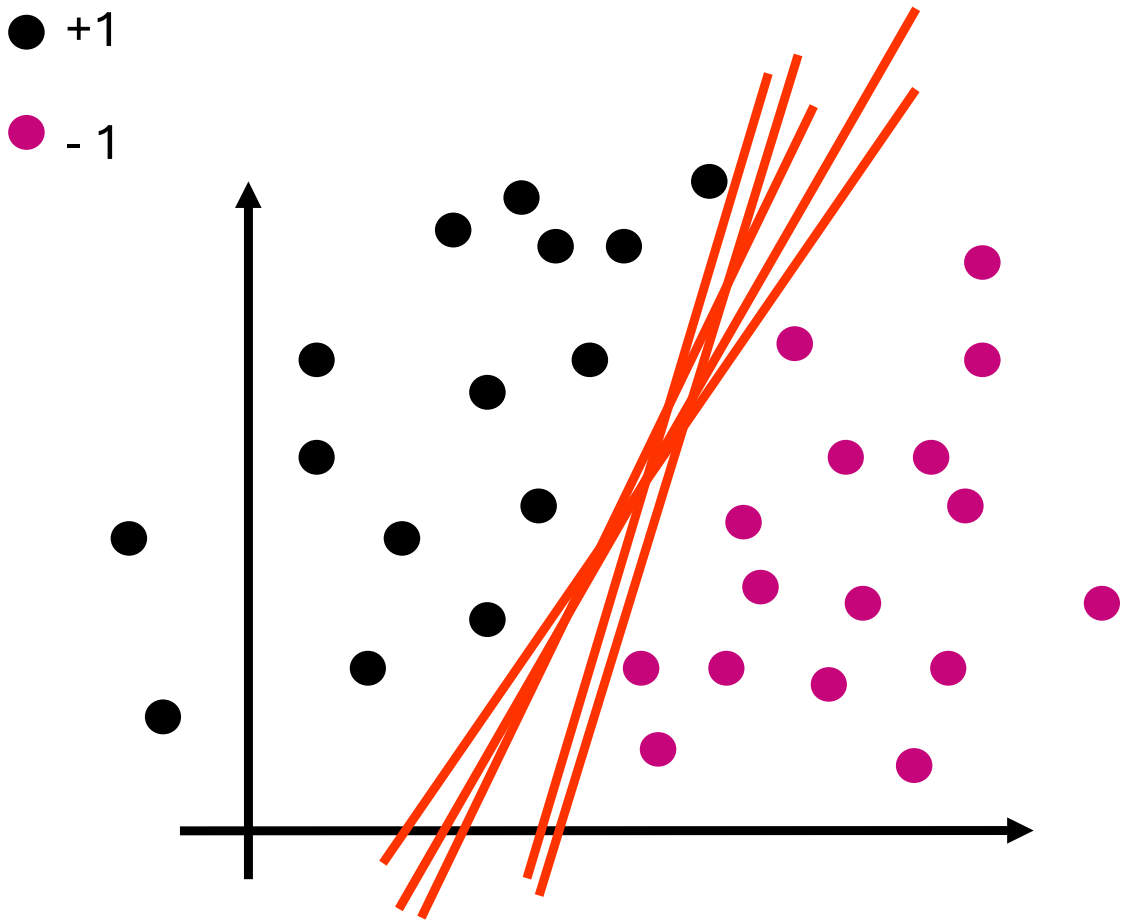
Support Vector Machines

- Linearly separable problems



So, which hyperplane best classifies new cases?

The one that is more separated from the samples of both classes

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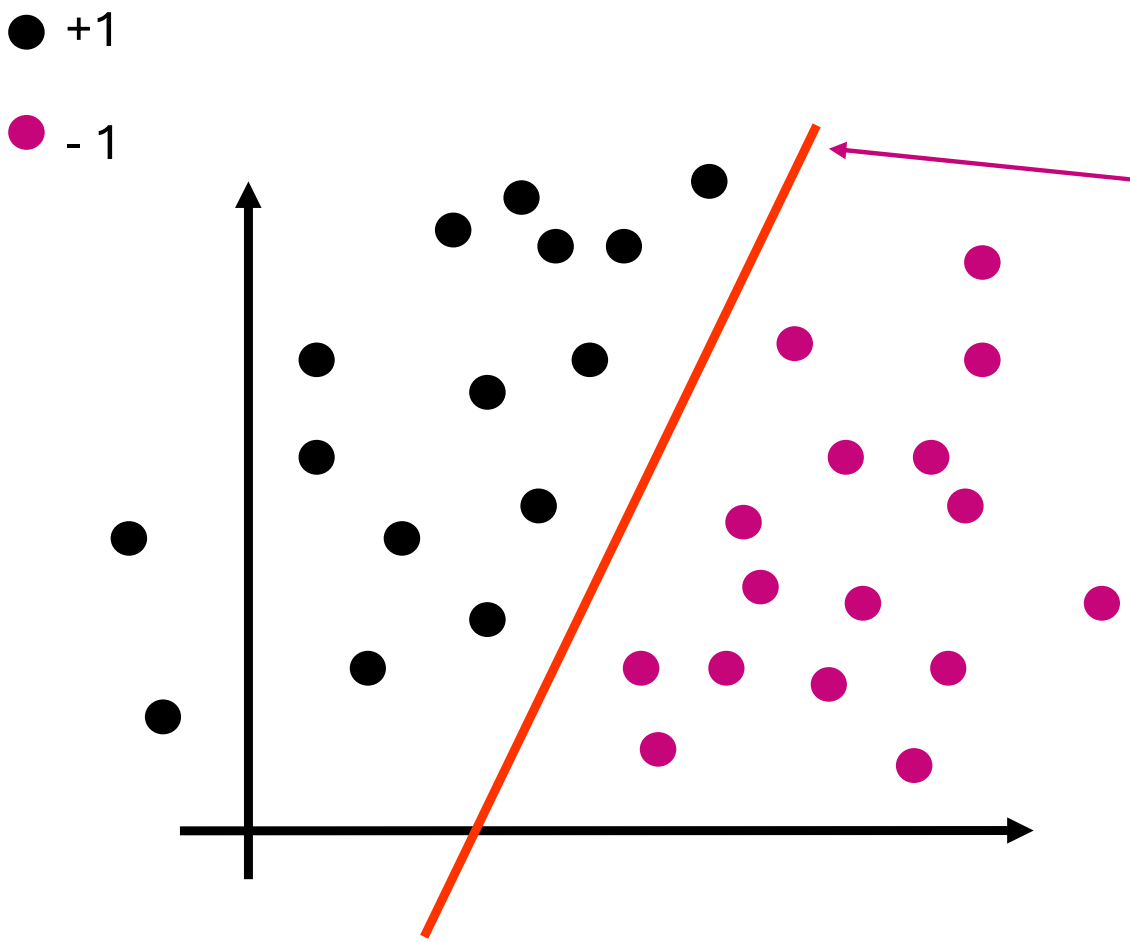
Support Vector Machines

- Linearly separable problems



We define the hyperplane

$$\boldsymbol{w}\, x + b = \mathbf{0}$$

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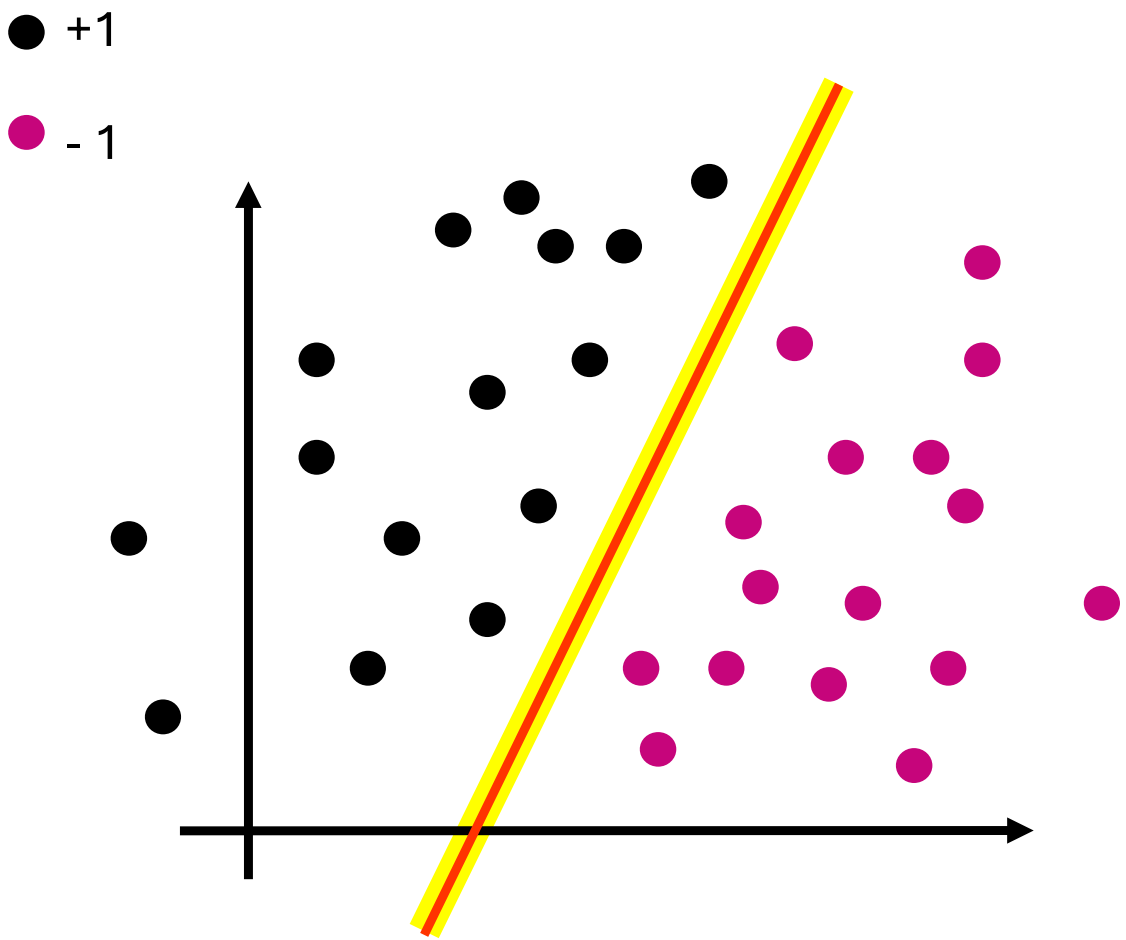
Support Vector Machines

- Linearly separable problems



We define the margin

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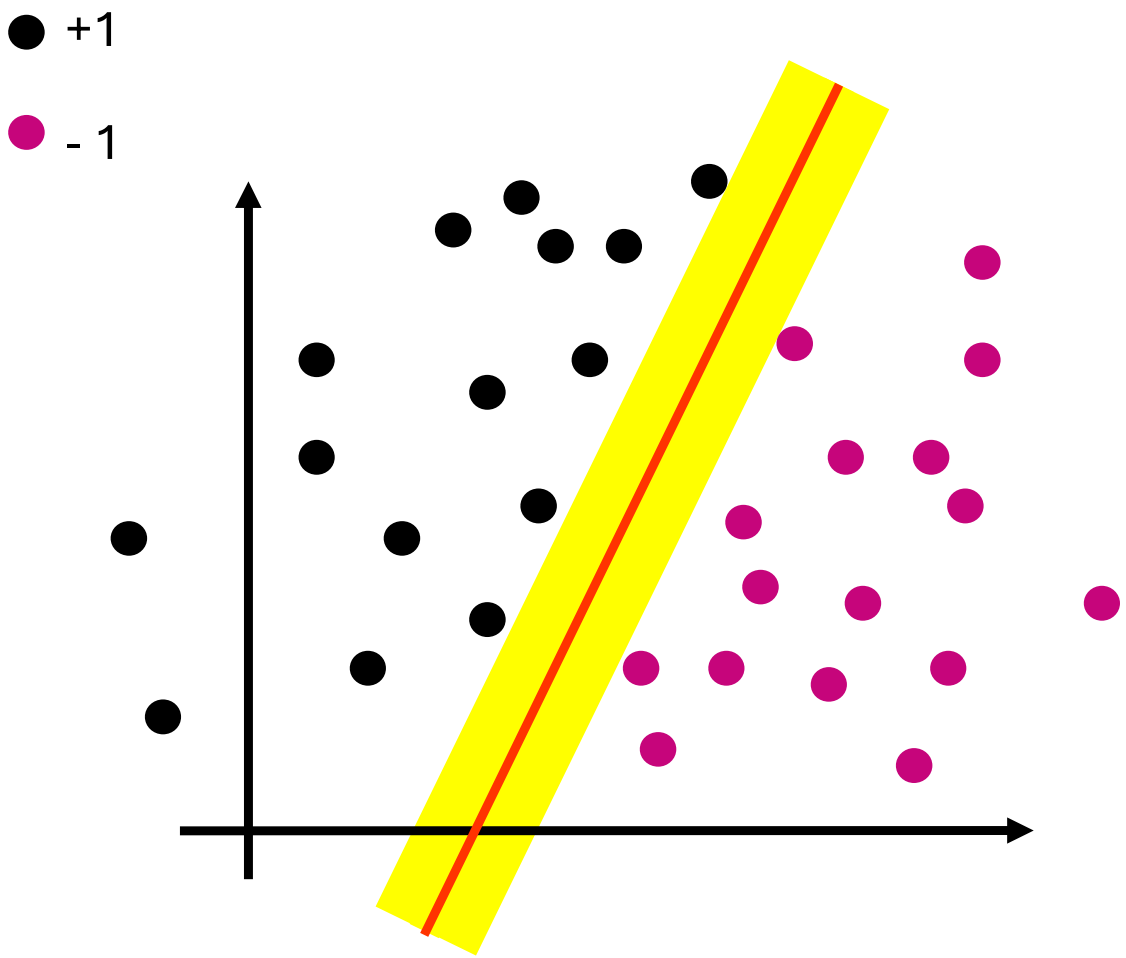
Support Vector Machines

- Linearly separable problems



The key idea is to maximize the margin

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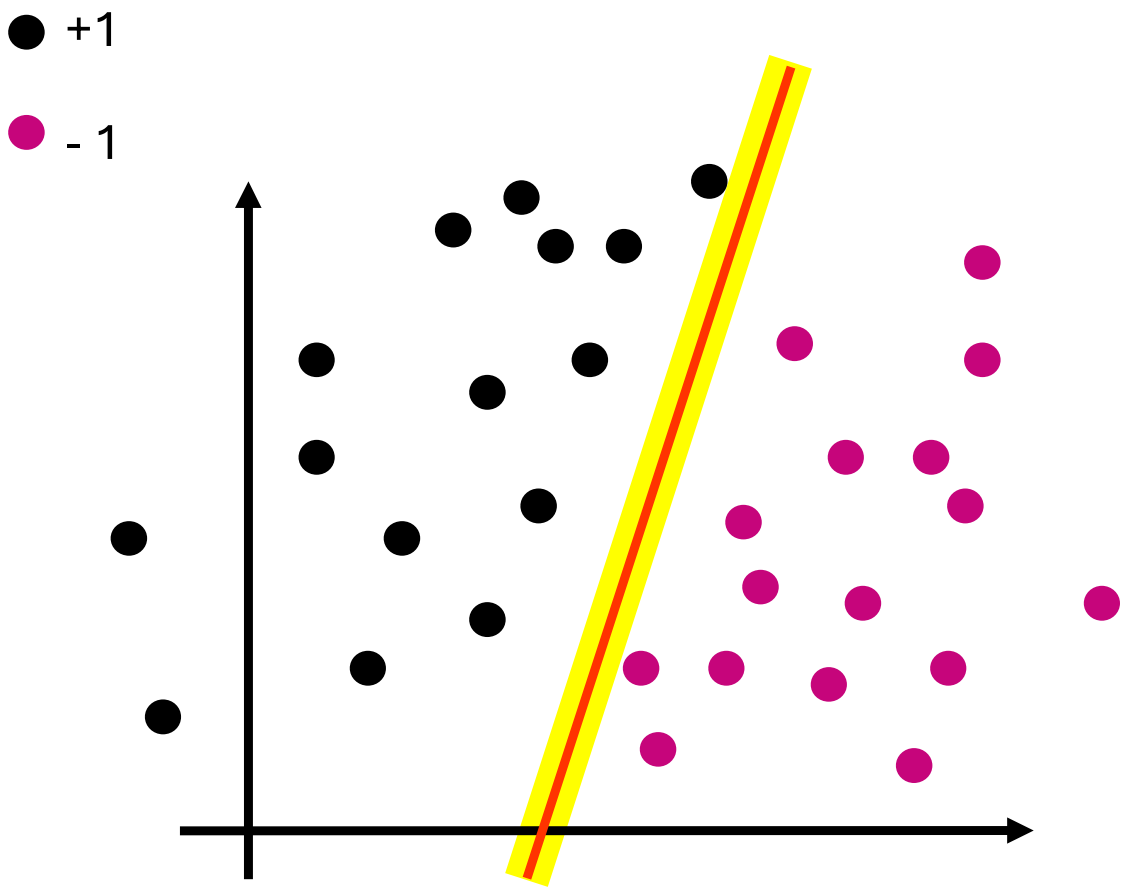
Support Vector Machines

- Linearly separable problems



Other hyperplanes with other maximized margins would be possible

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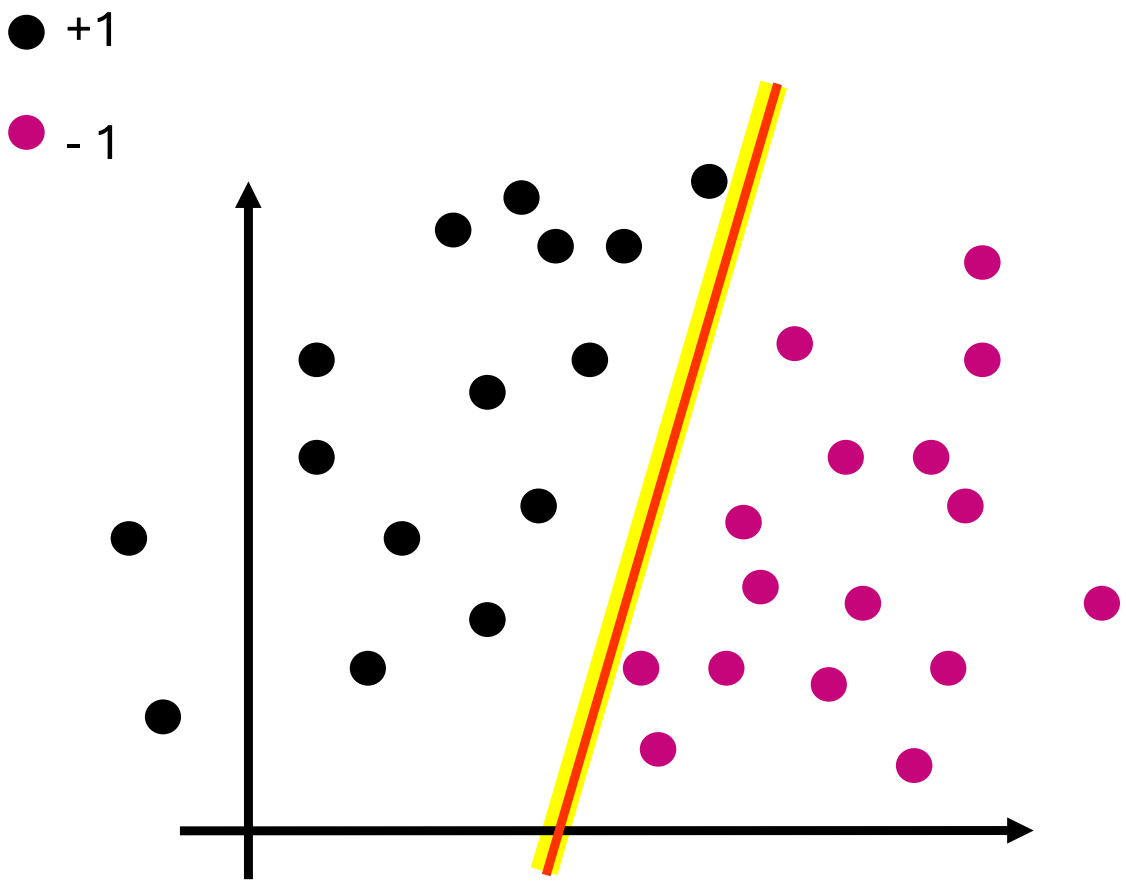
Support Vector Machines

- Linearly separable problems



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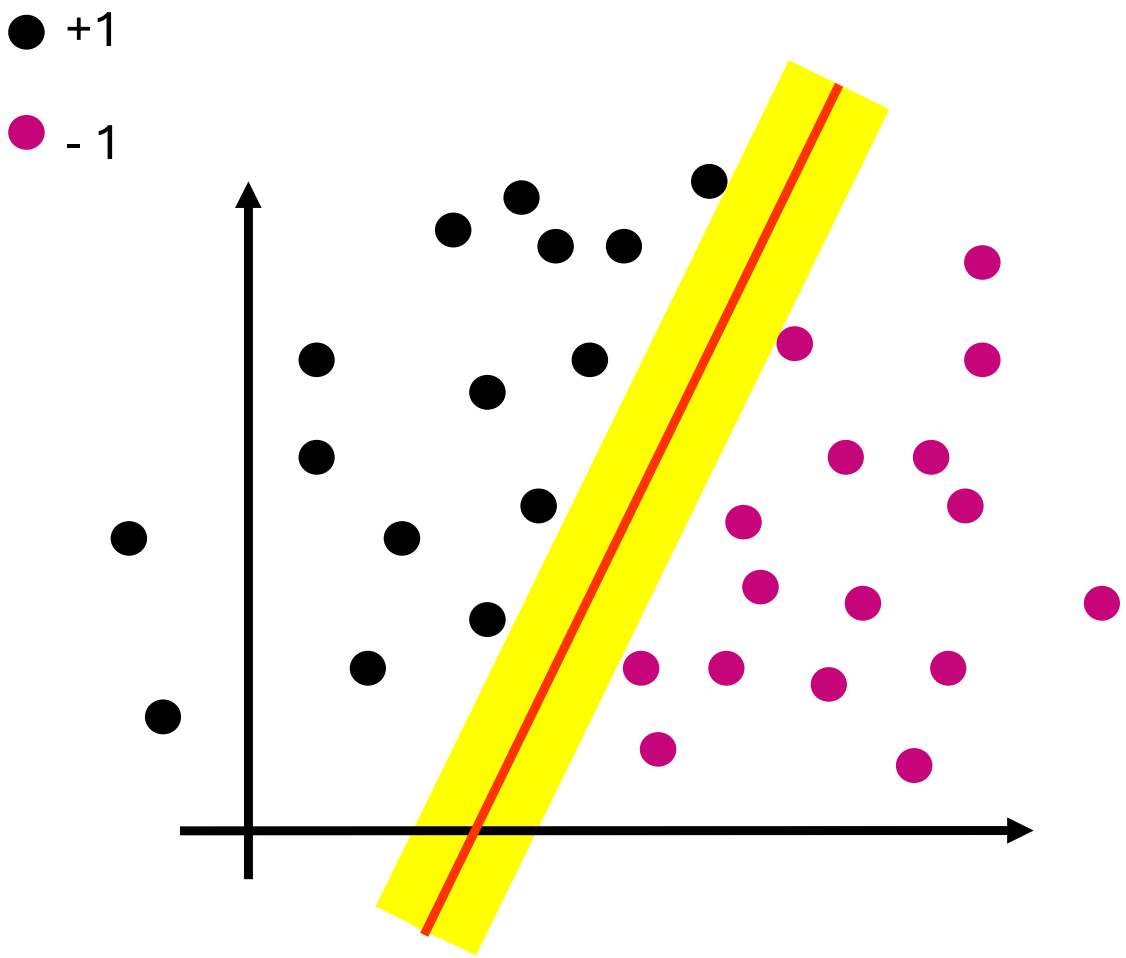

The hyperplane with the largest margin is the best classifier of the data.

This is the simplest class of SVM, the LSVM.

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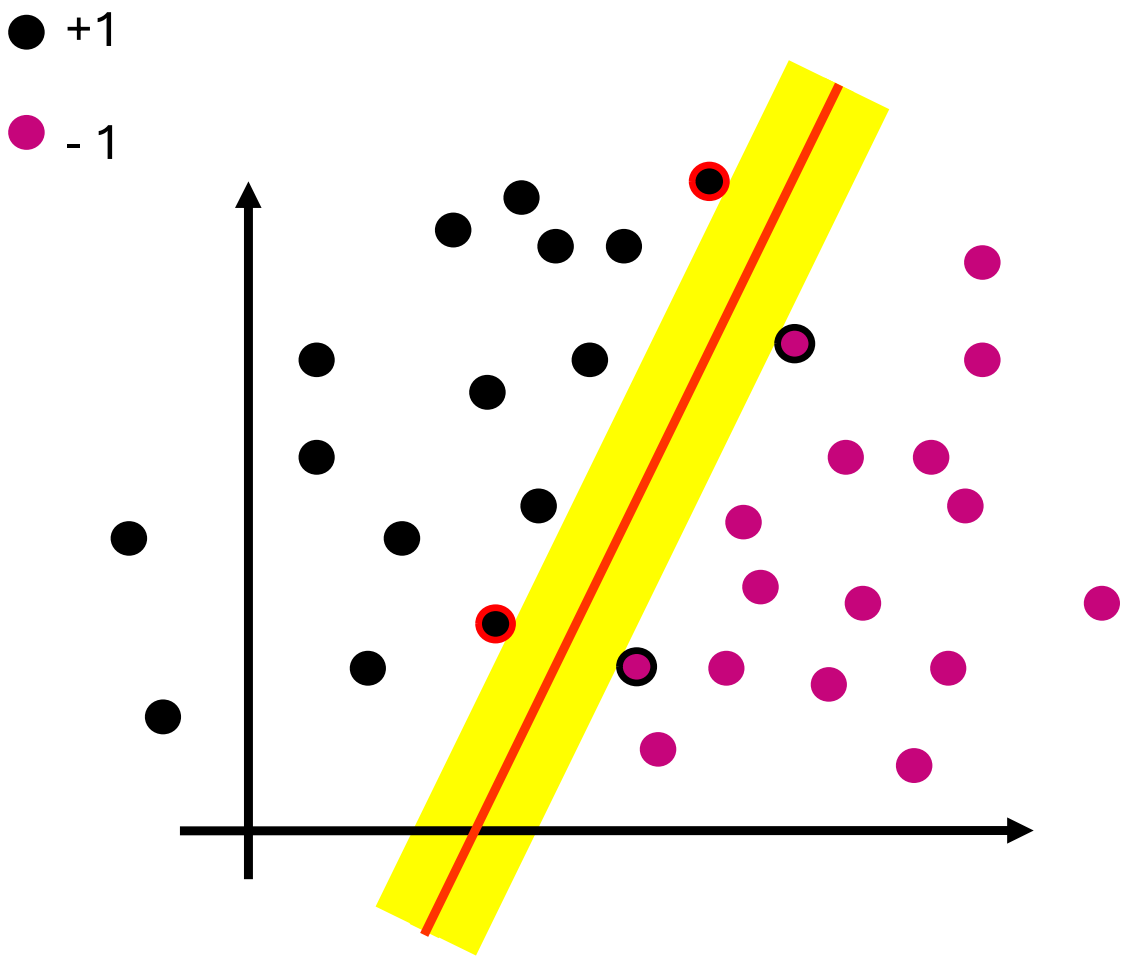
Support Vector Machines

- Linearly separable problems



Support vectors are the samples that touch the margin boundary.

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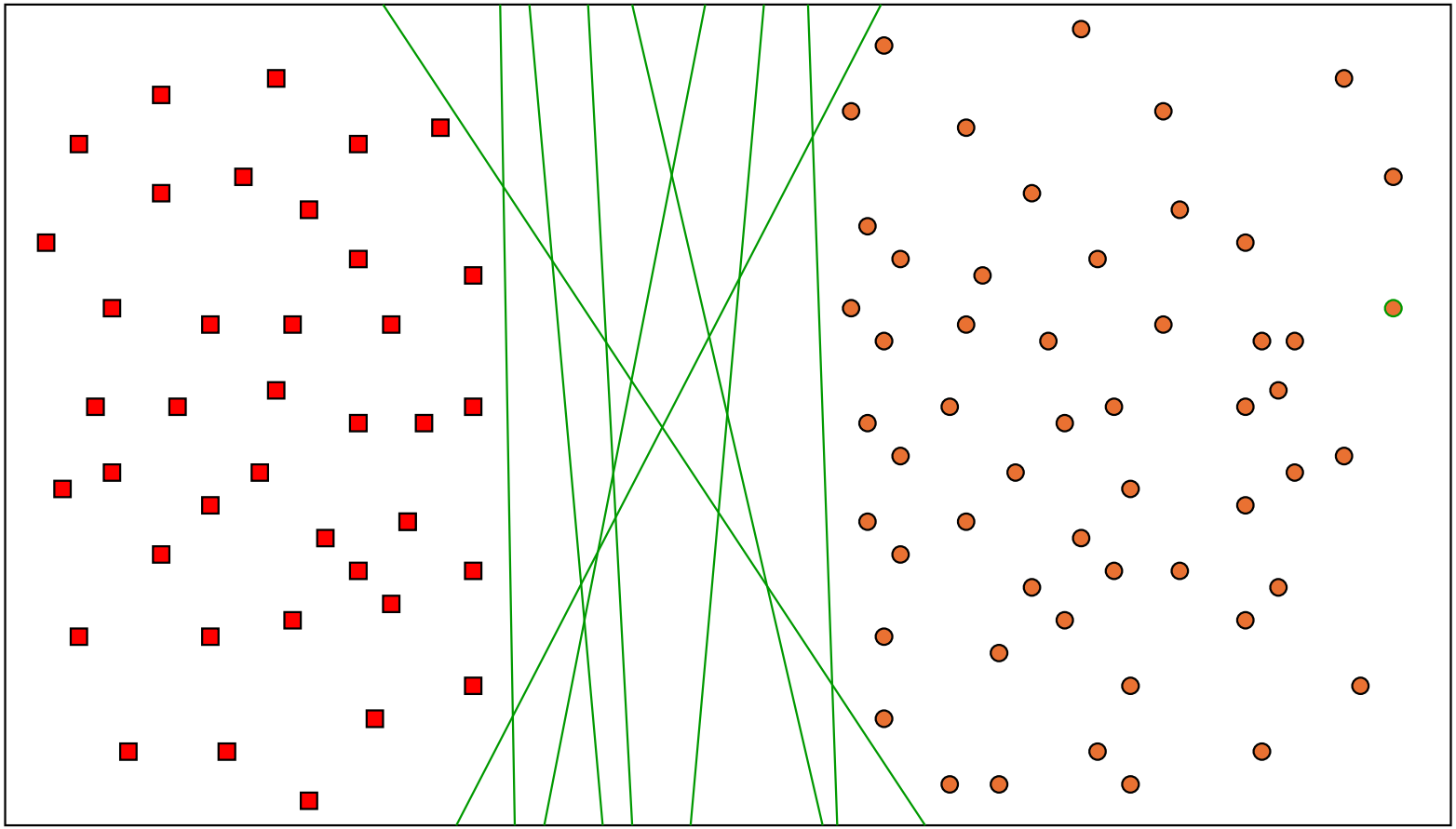
Support Vector Machines

- Linearly separable problems



Possible decision frontiers for linearly separable data

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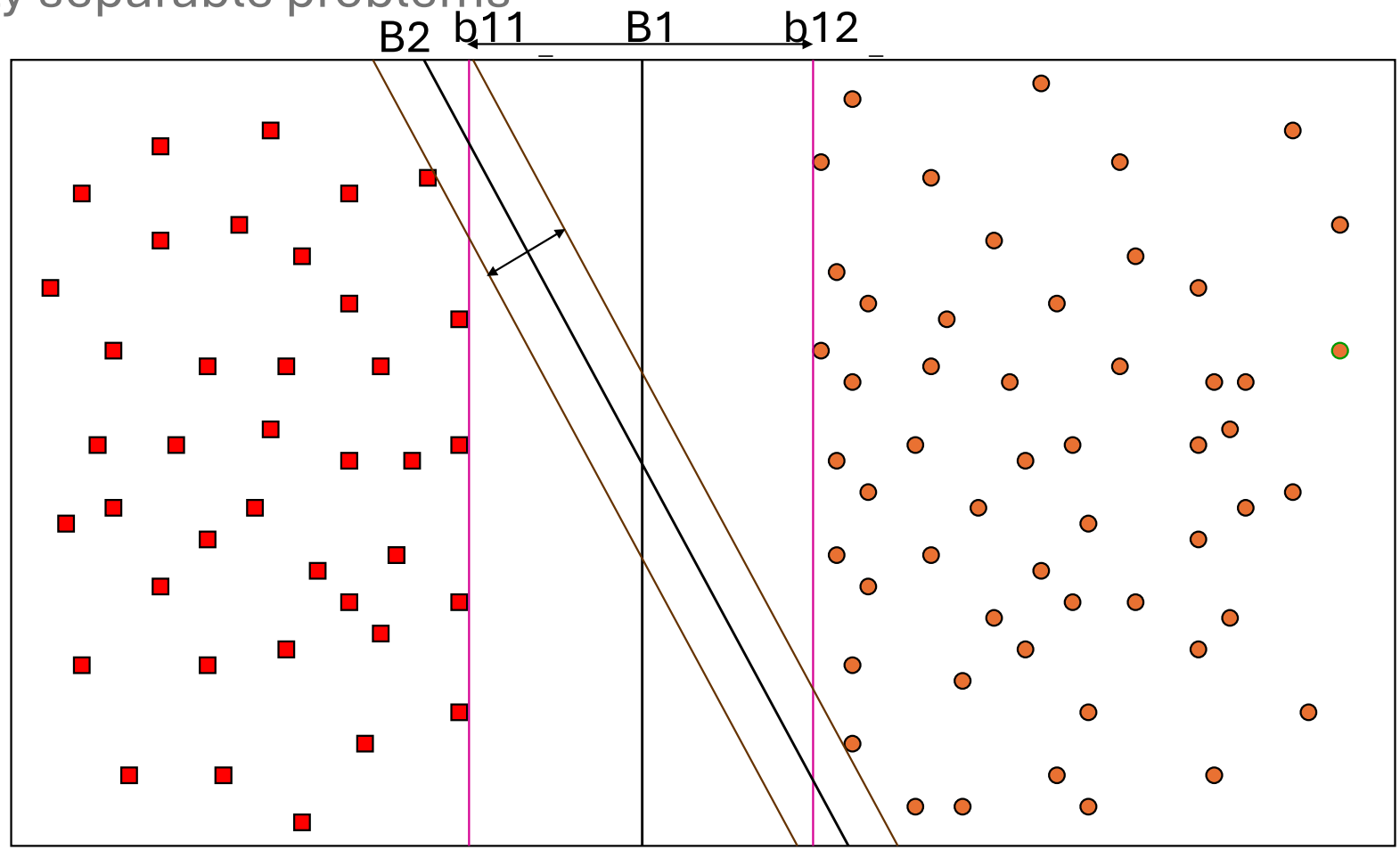
Support Vector Machines

- Linearly separable problems



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Support Vector Machines

- Linearly separable problems
  - Given a set of training examples,
  - Build a hyperplane "$w * x + b$" as a decision surface such that the **separation of the two classes is maximum** (generalization principle)
    - Decision margin as wide as possible
    - Training seeks to find w and b

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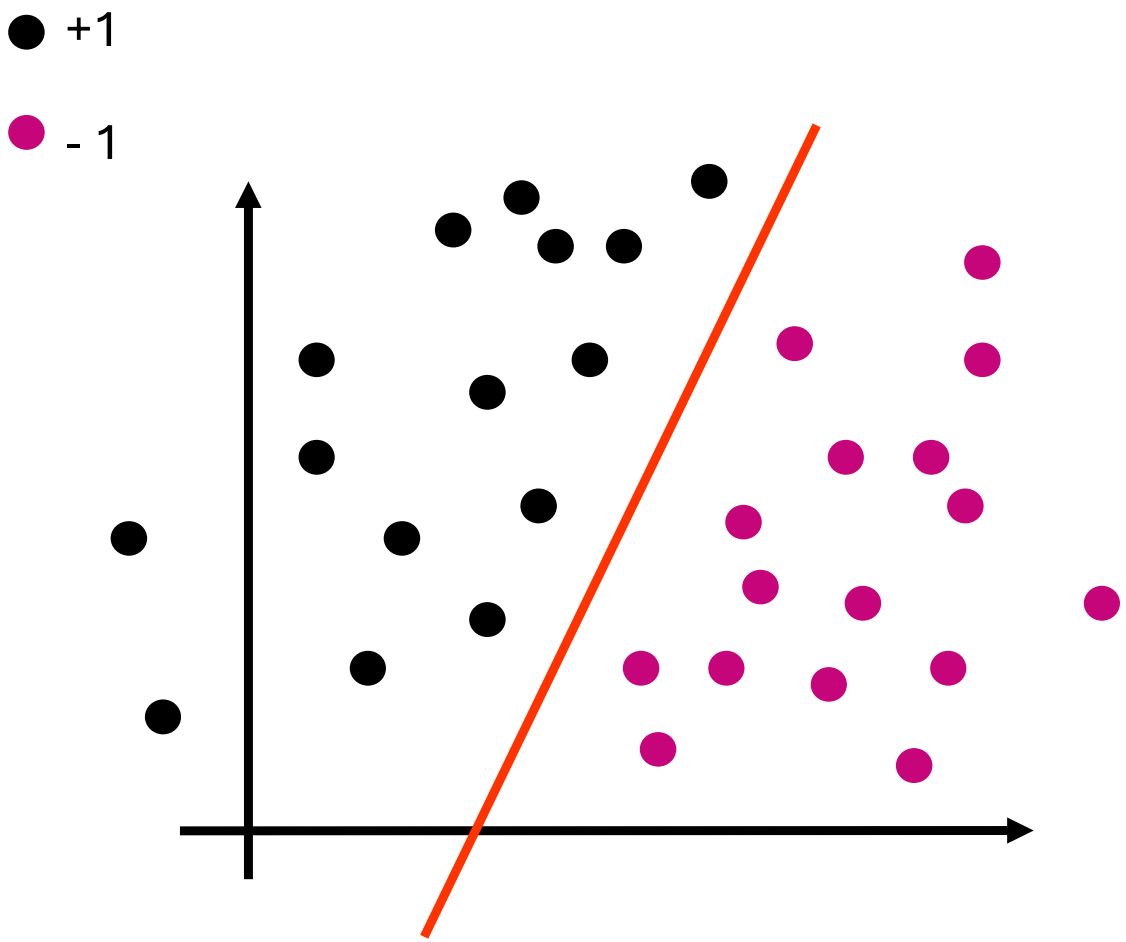
Support Vector Machines

- Linearly separable problems
  - hyperplanes of the margins are defined :



$$\boldsymbol{w} * \boldsymbol{x} + \boldsymbol{b} = \mathbf{0}$$

$$w * xi_{+} b \geq 0 \text{ for } y_i = +1$$

$$w * xi_{+} b \leq 0 \text{ for } y_i = -1$$

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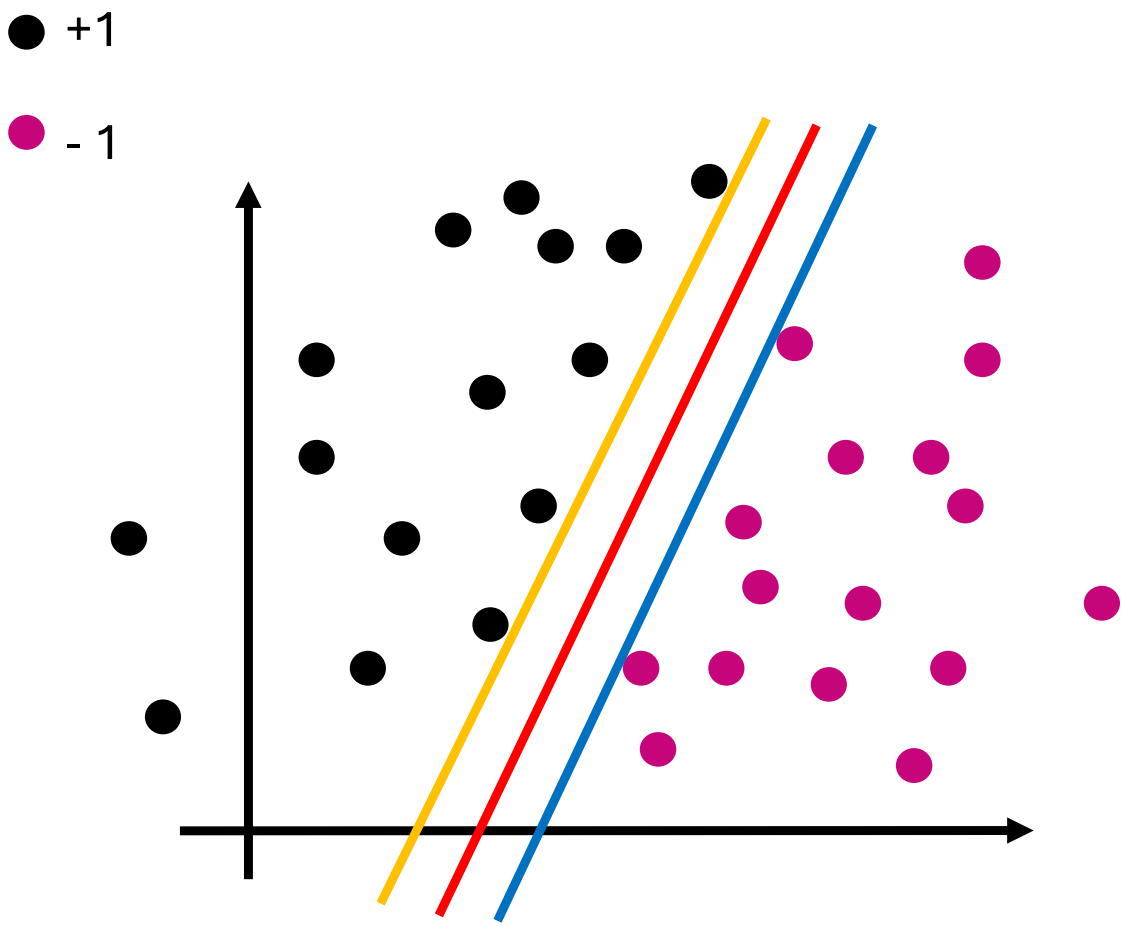
Support Vector Machines

- Linearly separable problems



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$$w * xi_{+} b \geq 0 \text{ for } y_{i} = +1$$

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"positive" hyperplane:

$$\boldsymbol{w} * x + b = +1$$

“negative” hyperplane:

$$\boldsymbol{w} * x + b = -1$$

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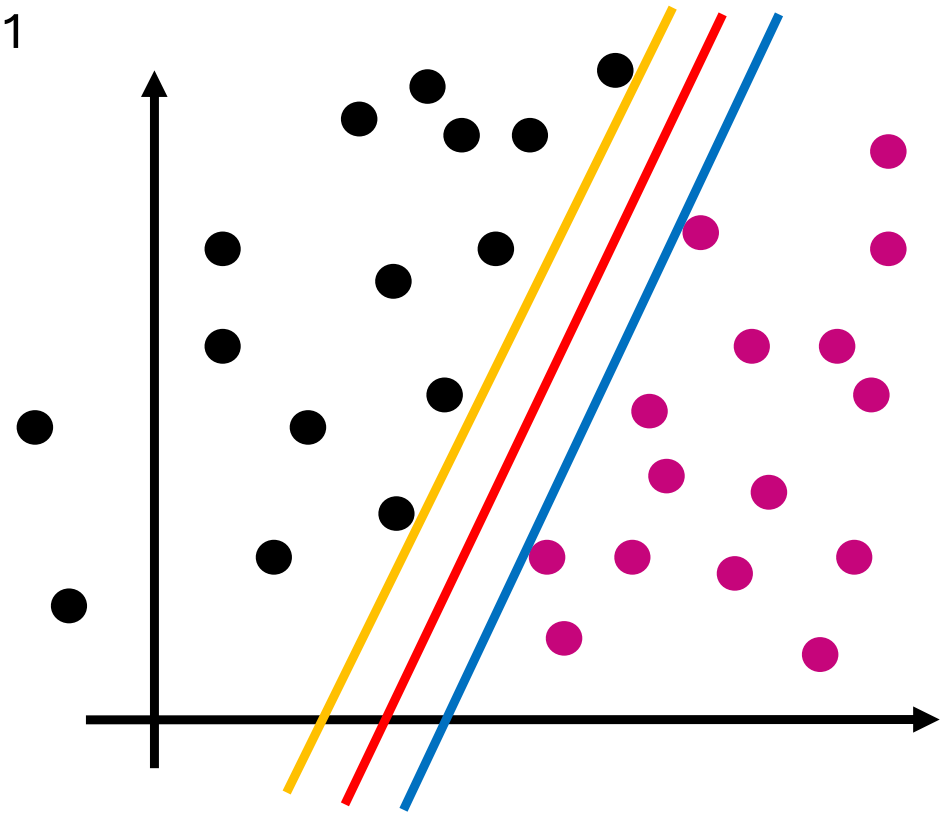
Support Vector Machines

- Linearly separable problems

● +1

● - 1



$$\boldsymbol{w} * \boldsymbol{x} + \boldsymbol{b} = \mathbf{0}$$

$$w * xi + b \geq 0 \text{ for } y_i = +1$$

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"positive" hyperplane:

$$\boldsymbol{w} * x + b = +1$$

“negative” hyperplane:

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It is desired that within the margin there are no instances

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Support Vector Machines

- Linearly separable problems

● +1

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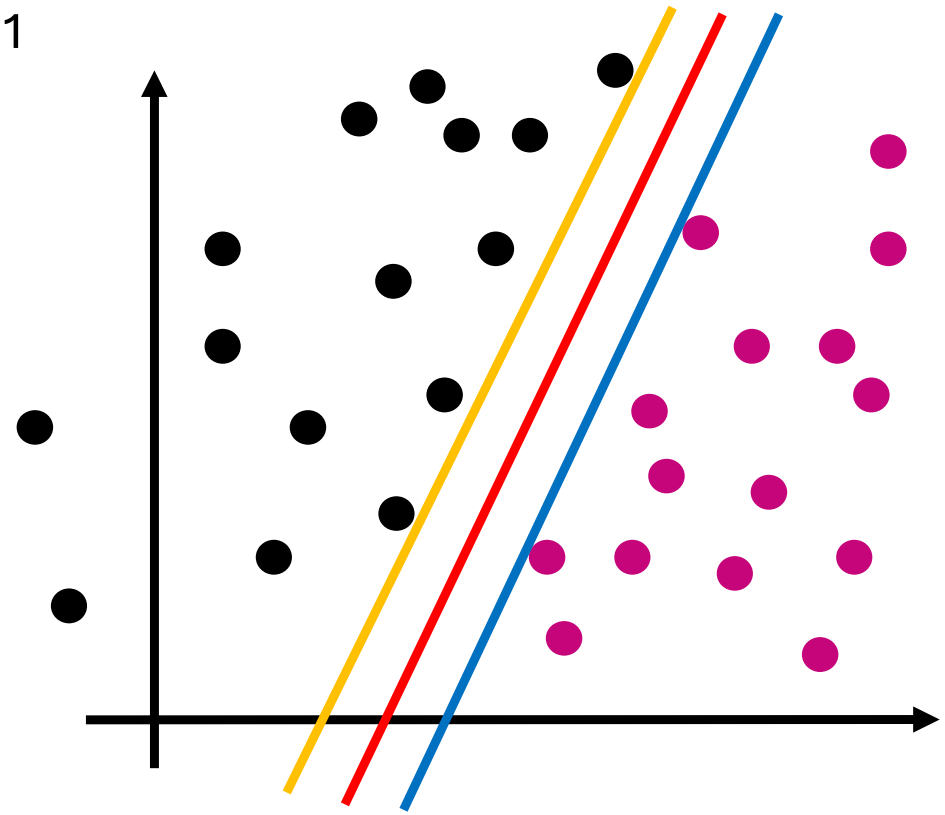

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"positive" hyperplane:

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It is desired that within the margin there are no instances

All samples ● meet:

$$\boldsymbol{w}\, x + \boldsymbol{b} \geq +1$$

All samples ● meet:

$$\boldsymbol{w}\, x + \boldsymbol{b} \leq -1$$

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Support Vector Machines

- Linearly separable problems
  - If we work with a set of vectors (samples) S, we say that it is linearly separable if there exists (w,b) such that the inequalities

$$\begin{cases} (w \cdot x_i + b) \geq 1 & y_i = 1 \text{ (positive cases)} \\ (w \cdot x_i + b) \leq -1 & y_i = -1 \text{ (negative cases)} \end{cases} \quad i=1,\ldots,L$$

  - are valid for all elements of the set S
    - For the linearly separable case of S, a **unique optimal hyperplane can be found**, for which the margin between the training points of two different classes is maximized

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Support Vector Machines

- Linearly separable problems
  - These equations

$$\begin{cases} (w \cdot x_i + b) \geq 1 & y_i = 1 \\ (w \cdot x_i + b) \leq -1 & y_i = -1 \end{cases} \qquad i=1,\ldots,L$$

  - can be reformulated to

$$y_i (w \cdot x_i + b) \geq 1 \qquad i=1,\ldots,L$$

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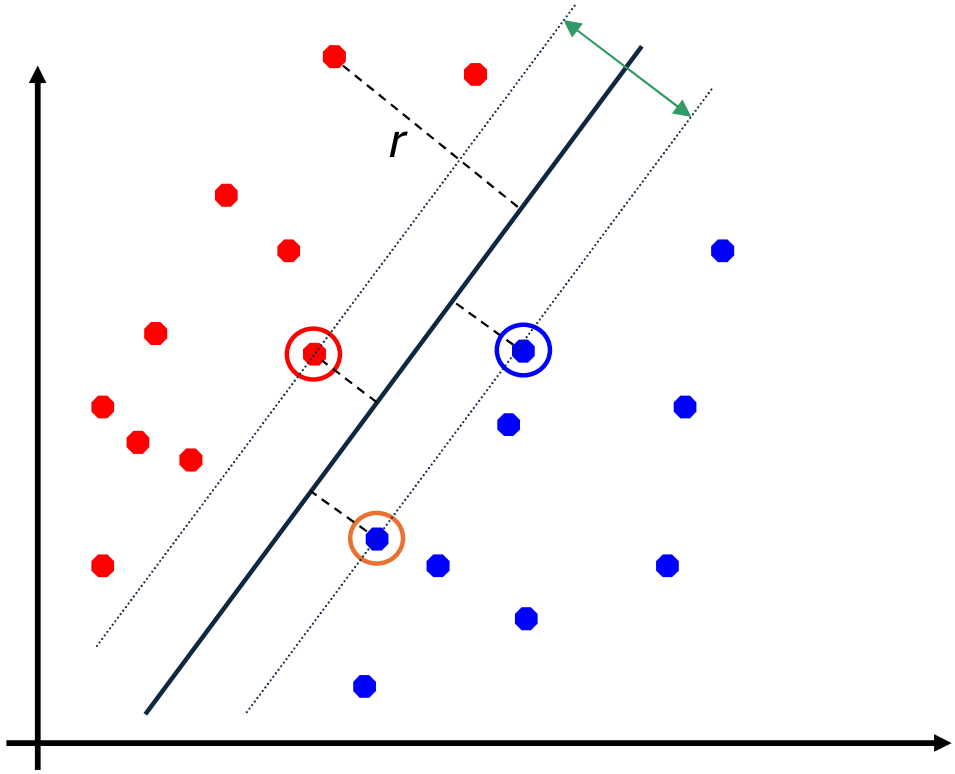
Support Vector Machines

- Linearly separable problems
  - The closest examples to the hyperplane are the so-called support vectors
    - Only these matter, the rest of the training examples can be ignored



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Support Vector Machines

- Linearly separable problems

● +1

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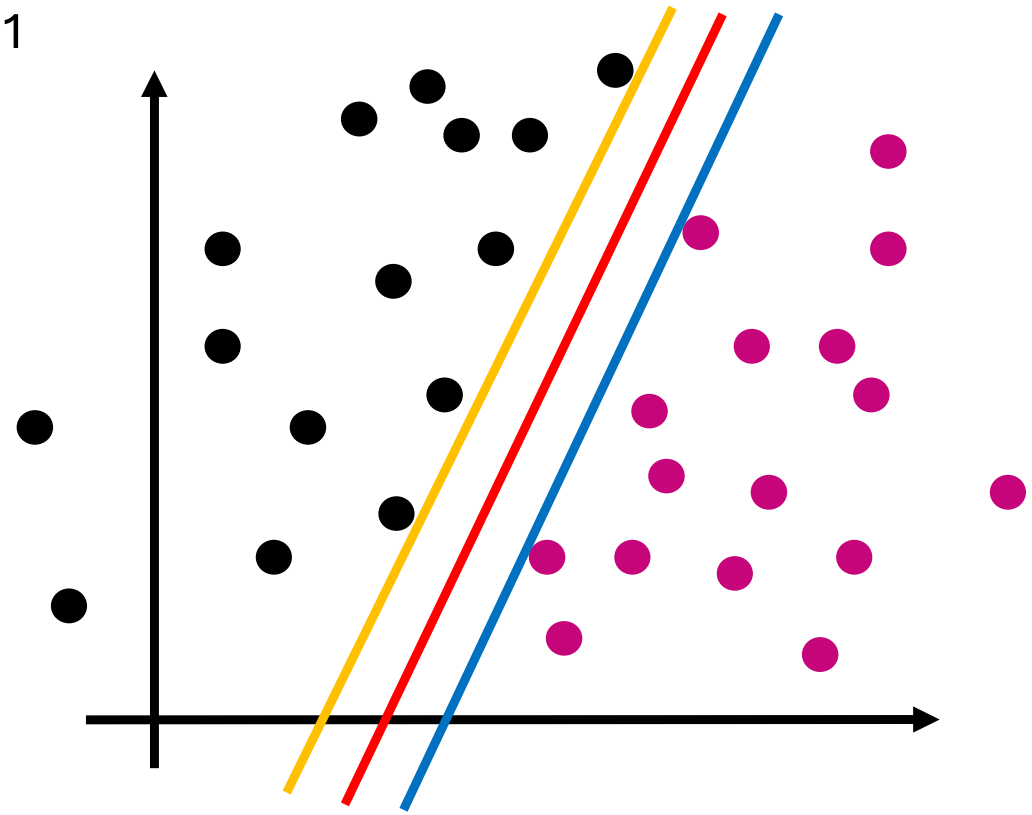

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"positive" hyperplane:

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“negative” hyperplane:

$$\boldsymbol{w} * x + b = -1$$

All samples ● meet:

$$\boldsymbol{w}\, x + \boldsymbol{b} \geq +1$$

All samples ● meet:

$$\boldsymbol{w}\, x + \boldsymbol{b} \leq -1$$

The support vectors satisfy:

$$\boldsymbol{w}\, x + b = +1$$
$$\boldsymbol{w}\, x + b = -1$$

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Support Vector Machines

- Linearly separable problems

"positive" hyperplane:

$$\boldsymbol{w} * x + b = +1$$

“negative” hyperplane:

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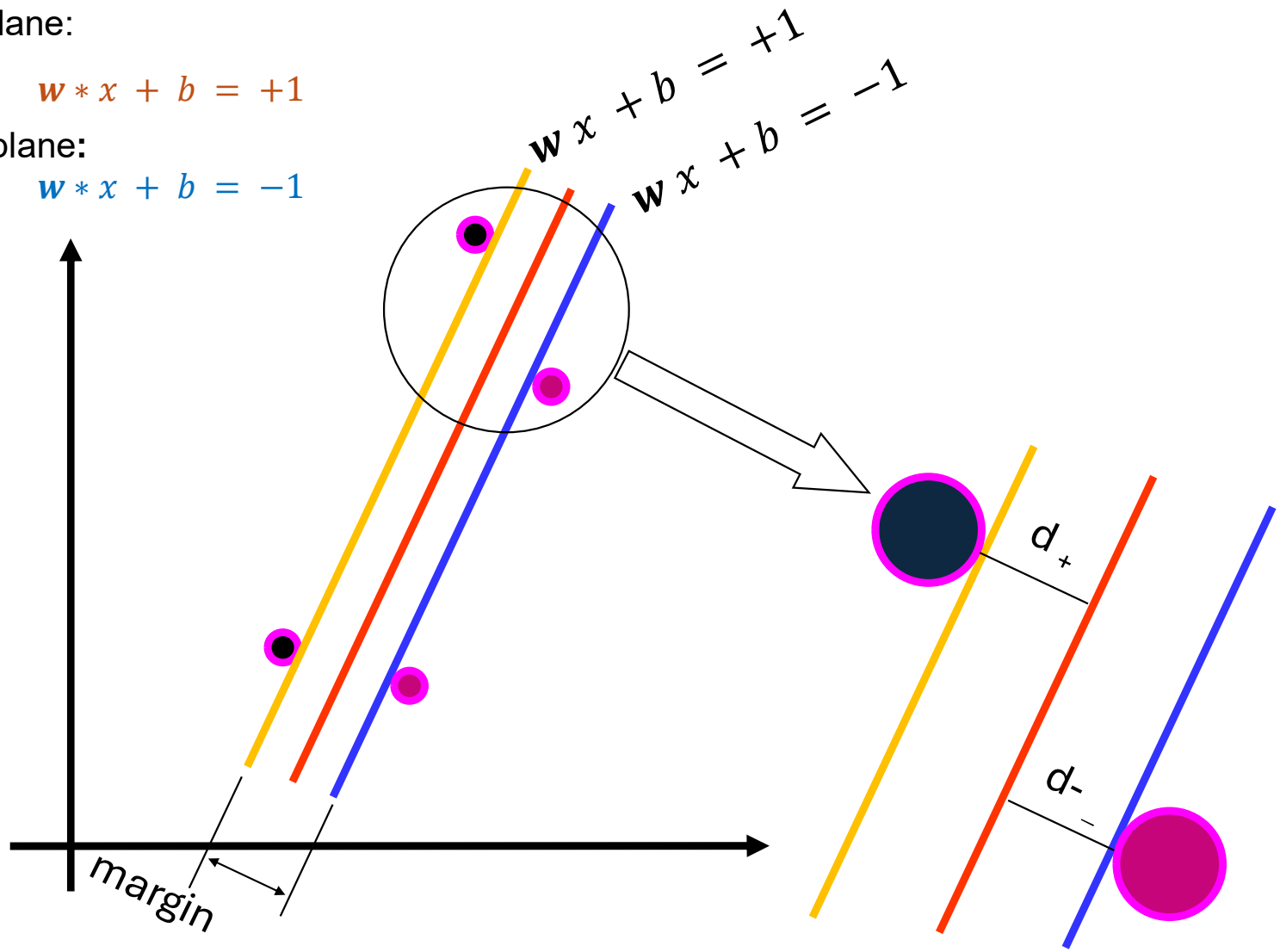

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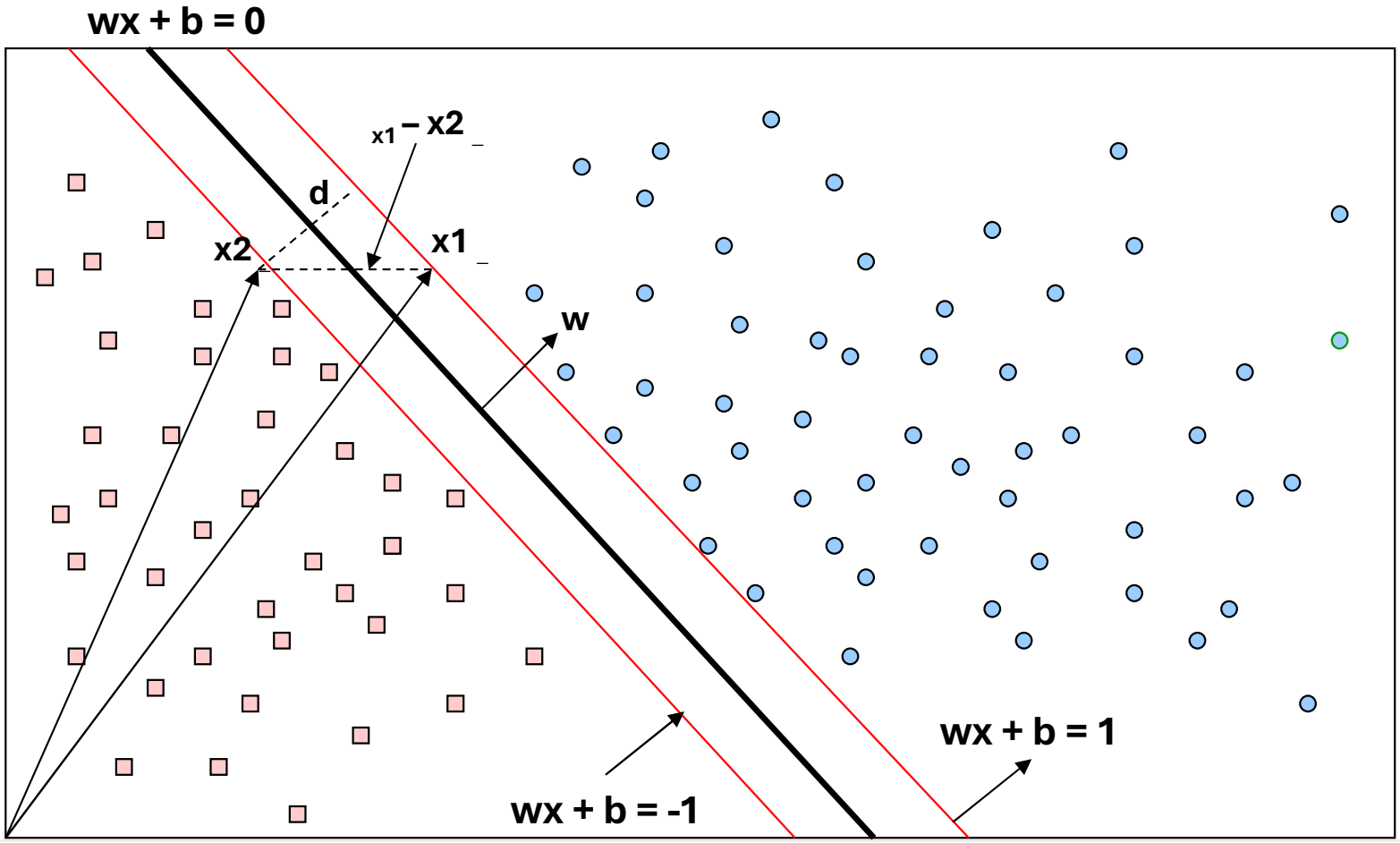
Support Vector Machines

- Linearly separable problems
  - Maximize margin width, d

Decision frontier and margin of SVM



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Support Vector Machines

- Linearly separable problems
  - We want to find w and b such that the margin is maximized
  - Let the "margin" d be the distance between the "positive" and "negative" hyperplanes
    - It is desired to maximize that value d
    - For each training example:

$$w^T x_i + b \leq -d/2 \qquad \text{if } y_i = -1$$

$$w^T x_i + b \geq d/2 \qquad \text{if } y_i = 1$$

    - Namely:

$$y_i\left(w^T x_i + b\right) \geq d/2$$

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Support Vector Machines

- Linearly separable problems
  - For each support vector, the above inequality is an equality

$y_i\left(w^T x_i + b\right) = d/2$ can be transformed to the previous formulation $y_i\left(w^T x_i + b\right) = 1$

by rescaling w and b by d/2

- The distance from an example xi to the separating plane is $r = \dfrac{\mathbf{w}^T \mathbf{x}_i + b}{\|\mathbf{w}\|}$
- A support vector is at the margin boundary
  - Putting both equations together:

$$r = \frac{y_i(\mathbf{w}^T \mathbf{x}_i + b)}{\|\mathbf{w}\|} = \frac{1}{\|\mathbf{w}\|}$$

- Therefore, the separation margin to be maximized is:

$$d = 2r = \frac{2}{\|\mathbf{w}\|}$$

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Support Vector Machines

- Linearly separable problems
  - The margin is equal to: $^{2}/_{\|w\|}$
  - The idea is to find a hyperplane with the maximum "margin"
    - This is an optimization problem:

$$\text{maximize: } {}^{2}/_{\|w\|} \text{ subjecto to: } y_i(w^T x_i + b) \geq 1$$

    - which can also be expressed as

$$\text{minimize: } \frac{1}{2}\|w\|^2 \text{ subjecto to: } y_i(w^T x_i + b) \geq 1$$

      - where $\|w\|^2$ is the Euclidean norm of w
      - That is, we need to optimize a quadratic function subject to linear constraints
        - One constraint for each sample

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Support Vector Machines

- Linearly separable problems
  - Quadratic optimization problems are a well-known class of mathematical problems for which several algorithms exist.
  - But the problem can be transformed to make it easier to handle:
    - Lagrange multipliers (α_i) are associated with each of the inequalities of the original problem
    - The dual problem is constructed:

$$L_P \equiv \tfrac{1}{2}\|\mathbf{w}\|^2 - \sum_{i=1}^{l} \alpha_i \left[y_i \left(\mathbf{w} \cdot \mathbf{x}_i + b \right) - 1 \right]$$

$$\alpha_i \geq 0 \quad \text{i=1,...,L}$$

  - A value α_i is associated to each sample x_i and to each inequality

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Support Vector Machines

- Linearly separable problems
  - To construct the dual problem, a value α_i is associated to each inequality (and, therefore, to each sample), to include the constraints within the expression
    - Original problem:

$$\text{minimize: } \frac{1}{2}\|w\|^2 \quad \text{subjecto to: } y_i(w^T x_i + b) \geq 1$$

    - Dual problem:

$$L_P \equiv \frac{1}{2}\|\mathbf{w}\|^2 - \sum_{i=1}^{l} \alpha_i \left[y_i \left(\mathbf{w} \cdot \mathbf{x}_i + b \right) - 1 \right]$$

$$\alpha_i \geq 0 \quad \text{i=1,...,L}$$

    - This problem must be minimized with respect to w and b, and maximized with respect to α_i >= 0

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Support Vector Machines

- Linearly separable problems
  - Dual problem minimized with respect to w and b:
    - Calculate the derivative L_p with respect to w and b and set it equal to 0

$$\frac{\partial L_P}{\partial b} = 0 \quad \Rightarrow \quad \sum_{i=1}^{l} \alpha_i y_i = 0$$

$$\frac{\partial L_P}{\partial w} = 0 \quad \Rightarrow \quad w = \sum_{i=1}^{l} \alpha_i y_i x_i$$

    - Substitute this expression into the above equation:

$$L_P \equiv \tfrac{1}{2}\|\mathbf{w}\|^2 - \sum_{i=1}^{l} \alpha_i \left[y_i \left(\mathbf{w} \cdot \mathbf{x}_i + b \right) - 1 \right] \qquad w = \sum_{i=1}^{l} \alpha_i y_i x_i \quad \Bigg] \quad L_D = \sum_{i=1}^{l} \alpha_i - \frac{1}{2} \sum_{i=1}^{l} \sum_{j=1}^{l} \alpha_i \alpha_j y_i y_j \mathbf{x}_i^T \mathbf{x}_j$$

    - which must be maximized with respect to α_i, with the constraint $\sum_{i=1}^{l} \alpha_i y_i = 0$

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Support Vector Machines

- Linearly separable problems
  - The equation to optimize is therefore

maximize: $$L_D = \sum_{i=1}^{l} \alpha_i - \frac{1}{2} \sum_{i=1}^{l} \sum_{j=1}^{l} \alpha_i \alpha_j y_i y_j \mathbf{x}_i^T \mathbf{x}_j$$

subject to: $$\sum_{i=1}^{l} \alpha_i y_i = 0 \qquad \alpha_i \geq 0 \quad \text{i=1,...,L}$$

  - In this expression, w and b no longer appear, only the α_i and y_i
    - one for each sample

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Support Vector Machines

- Linearly separable problems
  - The equation to optimize is therefore

maximize: $$L_D = \sum_{i=1}^{l} \alpha_i - \frac{1}{2} \sum_{i=1}^{l} \sum_{j=1}^{l} \alpha_i \alpha_j y_i y_j \mathbf{x}_i^T \mathbf{x}_j$$

subject to: $$\sum_{i=1}^{l} \alpha_i y_i = 0 \qquad \alpha_i \geq 0 \quad \text{i=1,...,L}$$

  - This optimization can be solved by quadratic programming (QP) techniques (Bertsekas 1995)

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Support Vector Machines

- Linearly separable problems
  - Given a solution to that problem, the solution to the "primary" problem would be, for w:

$$w = \sum_{i=1}^{l} \alpha_i y_i \mathbf{x}_i$$

    - Previous expression
    - Sum for each training sample
    - Each α_i not equal to 0 indicates that the corresponding x_i is a **support vector**
      - If α_i is 0, that vector x_i has not influence on the calculation of w (product by 0)

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Support Vector Machines

- Linearly separable problems
  - Given a solution to that problem, the solution to the "primary" problem would be, for b:
    - A support vector satisfies

$$y_k(w^T x_k + b) = 1$$

    - Then the parameter b is calculated:

$$y_k(w^T x_k + b) = 1 \Rightarrow w^T x_k + b = 1/y_k = y_k \Rightarrow b = y_k - w^T x_k$$

    - for any support vector $\alpha_k > 0$
      - For anyone it will give the same value
        - In practice, the value of b is averaged to avoid rounding problems:

$$b = \frac{1}{N_{VS}} \sum_{i=1}^{N_{VS}} (y_i - w^T x_i)$$

for each support vector
N_{VS} : number of support vectors

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Support Vector Machines

- Linearly separable problems
  - Therefore, the solution is:

$$w = \sum_{i=1}^{l} \alpha_i y_i \mathbf{x}_i \qquad b = y_k - w^T x_k$$

$$b = y_k - \sum_{i=1}^{l} \alpha_i y_i x_i^T x_k$$

for any $\alpha_k > 0$

  - In this way, the classification function is:

$$f(x) = \sum_{i=1}^{l} \alpha_i y_i \boxed{x_i^T x} + b$$

- It is not necessary to calculate the w
- In this function, a scalar product is performed between the point to classify and each of the support vectors
  - Instances that are not support vectors have $\alpha_k = 0$

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Support Vector Machines

- Linearly separable problems
  - Furthermore, if the $\alpha_i \geq 0$, Karush Kuhn Tucker conditions can be applied to convert the inequalities:

$$y_i(w^T x_i + b) \geq 1$$

  - into several equality equations for each example x_i:

$$\alpha_i(y_i(w^T x_i + b) - 1) = 0$$

  - All examples where $\alpha_i > 0$ are the support vectors
    - If $\alpha_i \neq 0$, then

$$y_i(w^T x_i + b) - 1 = 0 \Rightarrow y_i(w^T x_i + b) = 1$$

    - and therefore x_i is at the limit of the decision margin

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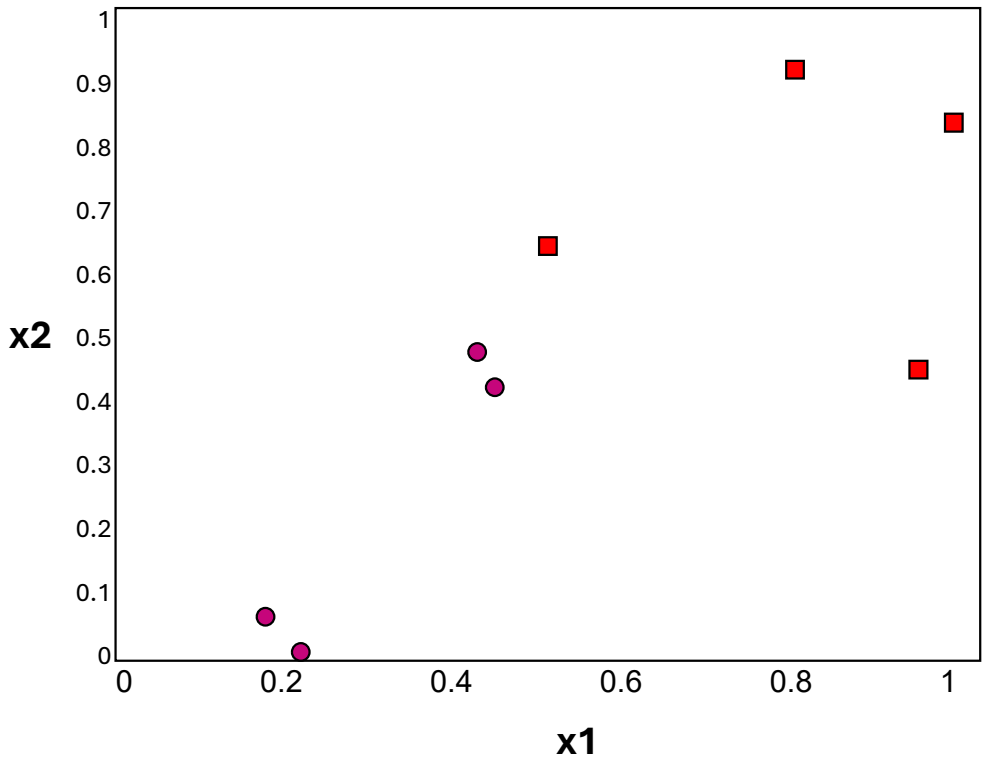
Support Vector Machines

- Linearly Separable Problems: Example:



x1	x2	Y
0.3858	0.4687	1
0.4871	0.611	-1
0.9218	0.4103	-1
0.7382	0.8936	-1
0.1763	0.0579	1
0.4057	0.3529	1
0.9355	0.8132	-1
0.2146	0.0099	1

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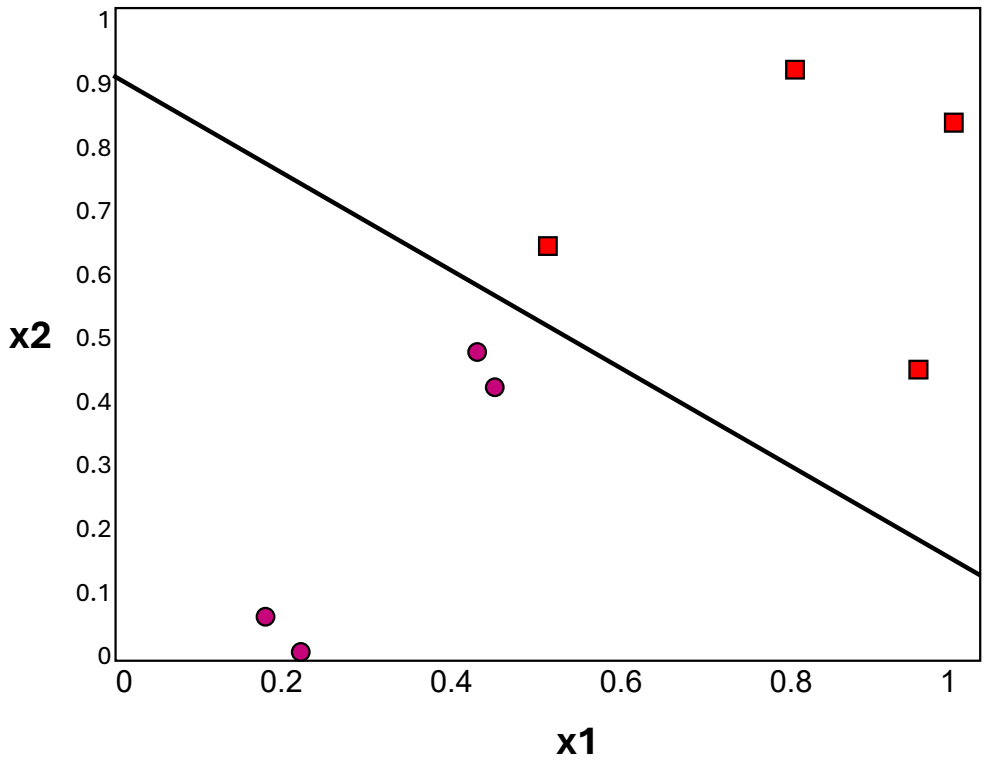
Support Vector Machines

- Linearly Separable Problems: Example:



x1	x2	Y	α_i
0.3858	0.4687	1	65.5261
0.4871	0.611	-1	65.5261
0.9218	0.4103	-1	0
0.7382	0.8936	-1	0
0.1763	0.0579	1	0
0.4057	0.3529	1	0
0.9355	0.8132	-1	0
0.2146	0.0099	1	0

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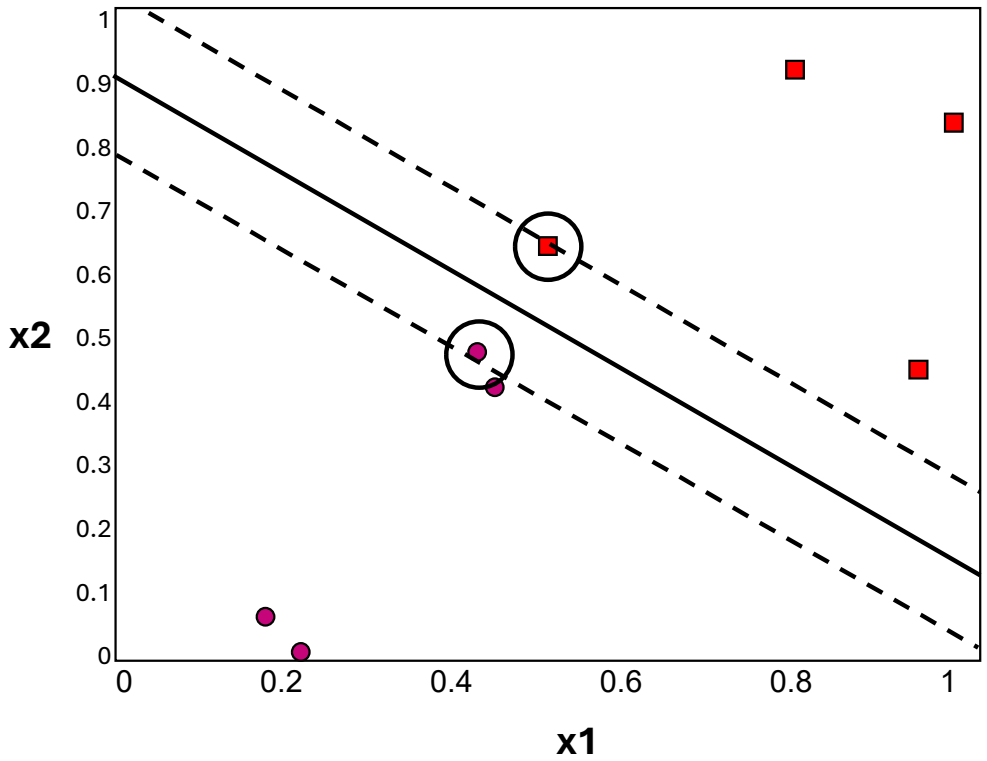
Support Vector Machines

- Linearly Separable Problems: Example:



x1	x2	Y	α_i
0.3858	**0.4687**	**1**	65.5261
0.4871	**0.611**	**-1**	65.5261
0.9218	0.4103	-1	0
0.7382	0.8936	-1	0
0.1763	0.0579	1	0
0.4057	0.3529	1	0
0.9355	0.8132	-1	0
0.2146	0.0099	1	0

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Support Vector Machines

- Linearly Separable Problems: Example:
  - Solution:
    - Calculation of w:

$$w = \sum_{i=1}^{l} \alpha_i y_i \mathbf{x}_i = 65.5621*1*\begin{bmatrix} 0.3858 \\ 0.4687 \end{bmatrix} + 65.5621*(-1)*\begin{bmatrix} 0.4871 \\ 0.611 \end{bmatrix} = \begin{bmatrix} -6.64 \\ -9.32 \end{bmatrix}$$

    - Calculation of b:
      - With the first instance:

$$b = y_k - w^T x_k = 1 - \begin{bmatrix} -6.64 & -9.32 \end{bmatrix} * \begin{bmatrix} 0.3858 \\ 0.4687 \end{bmatrix} = 1 - (-6.929996) = 7.929996$$

      - With the second instance:

$$b = y_k - w^T x_k = -1 - \begin{bmatrix} -6.64 & -9.32 \end{bmatrix} * \begin{bmatrix} 0.4871 \\ 0.611 \end{bmatrix} = -1 - (-8.928862) = 7.928864$$

      - Both values are averaged: b = 7.929

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Support Vector Machines

- Linearly Separable Problems: Example:
  - To classify a new sample x:

$$f(x) = w \cdot x + b$$

or

$$f(x) = \sum_{i=1}^{l} \alpha_i y_i x_i^T x + b$$

- Classification:
  - If $f(x) > 0$ ⇒ 1
  - If $f(x) < 0$ ⇒ -1
  - $f(x)$ gives a level of **confidence** about the membership of that sample to one of the classes
    - The larger its absolute value, the more confident
      - Further away from the region of separation

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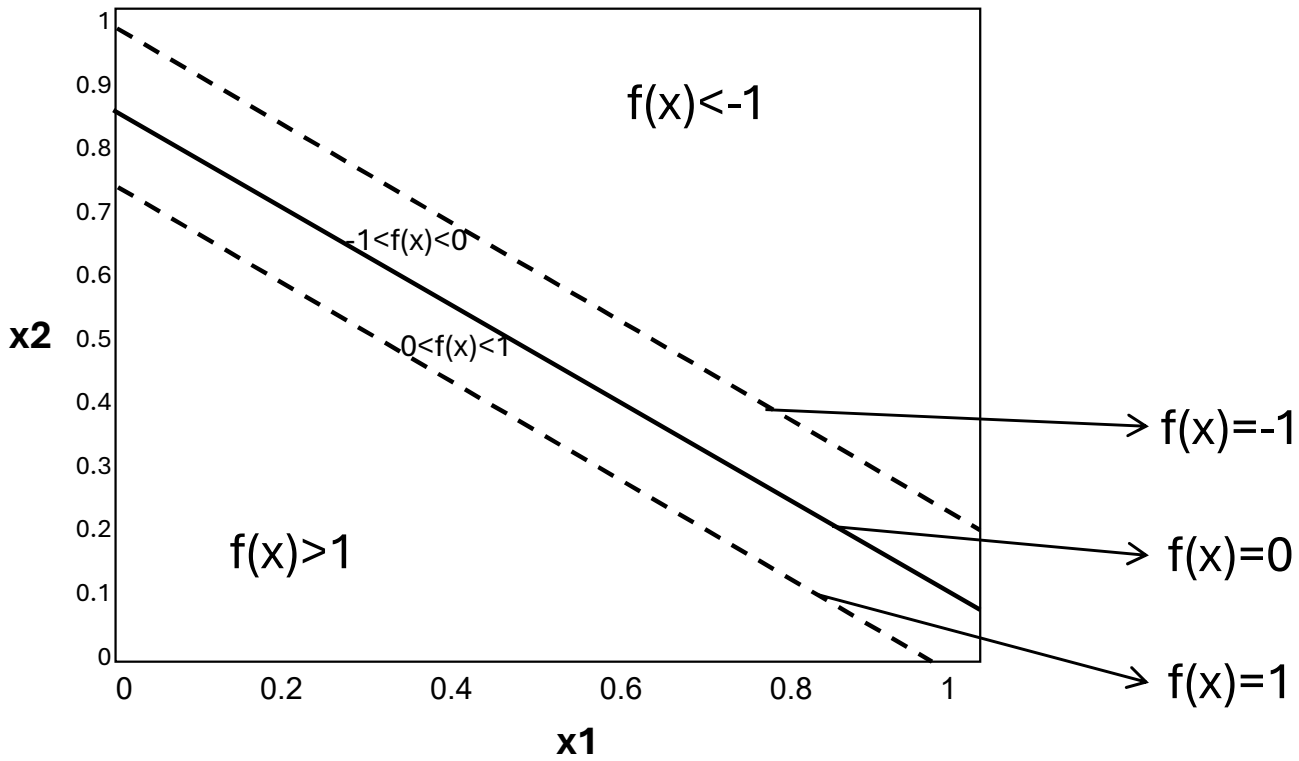
Support Vector Machines

- Linearly Separable Problems: Example:
  - To classify a new sample x:



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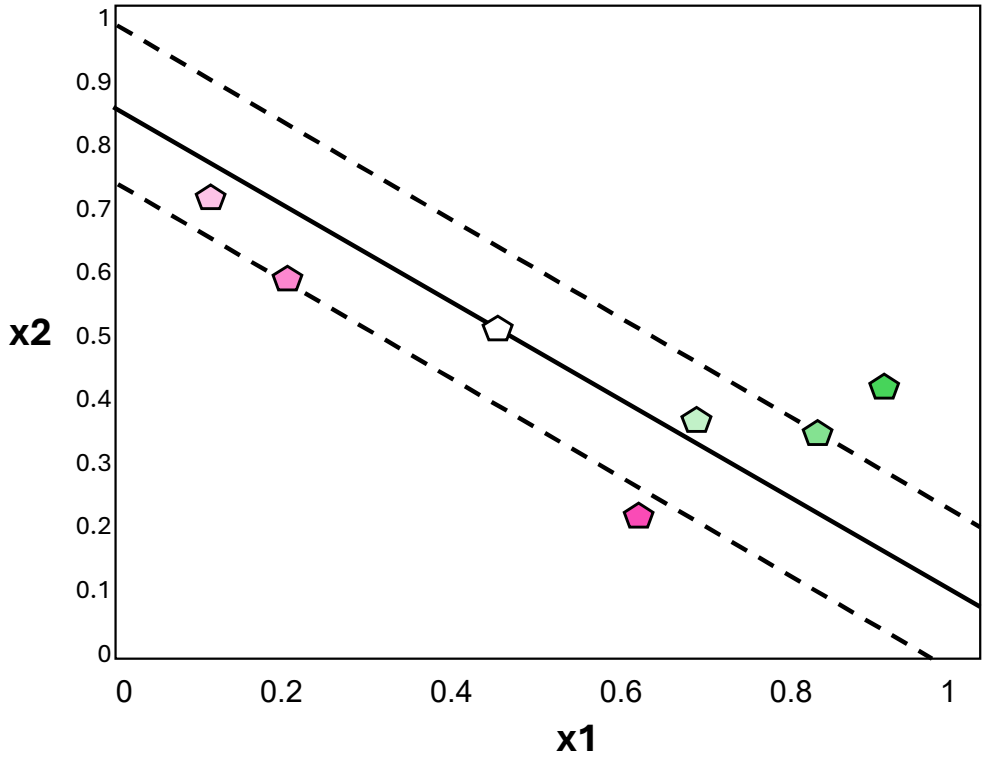
Support Vector Machines

- Linearly Separable Problems: Example:
  - To classify a new sample x:
    - Examples:



	x1	x2	f(x)
	0.4	0.5658	0
	0.15	0.7042	0.37
	0.2	0.601	1
	0.6	0.2623	1.5
	0.65	0.4091	-0.2
	0.8	0.3881	-1
	0.9	0.4563	-1.8

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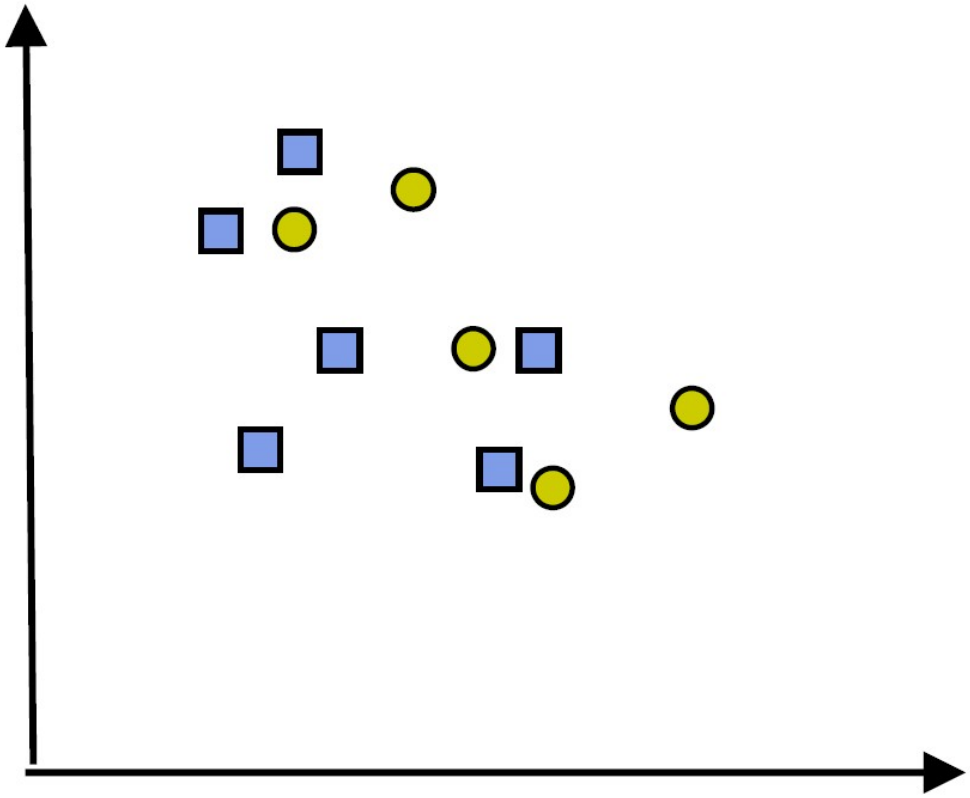
Support Vector Machines

- Nonlinearly separable problems
  - If the set S is not linearly separable, some classification errors must be allowed in the formulation



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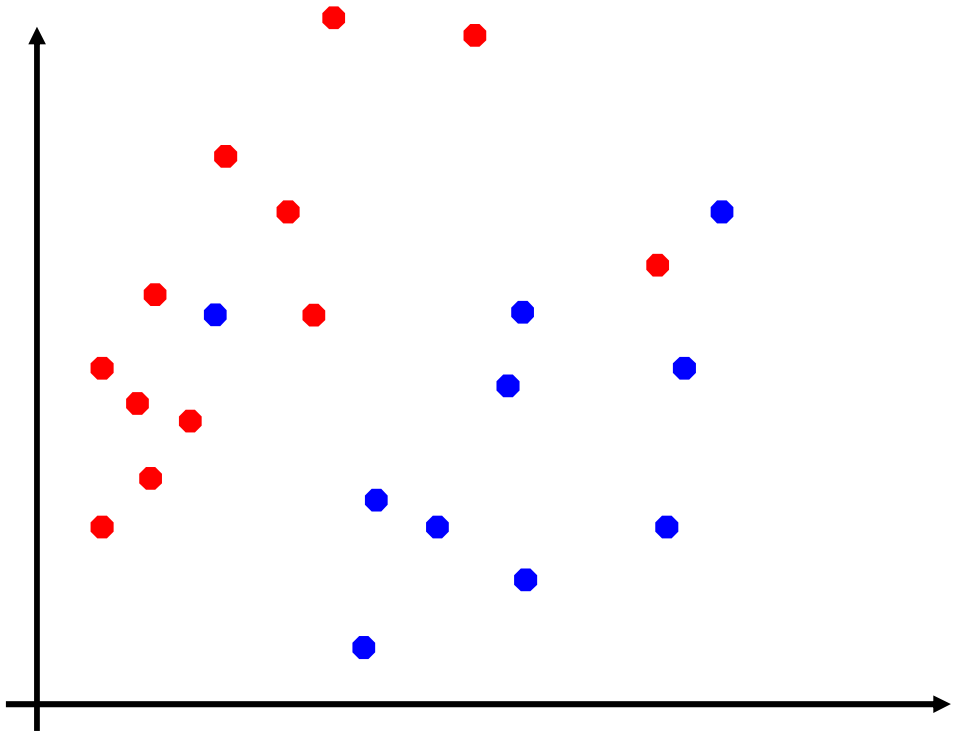
Support Vector Machines

- Nonlinearly separable problems



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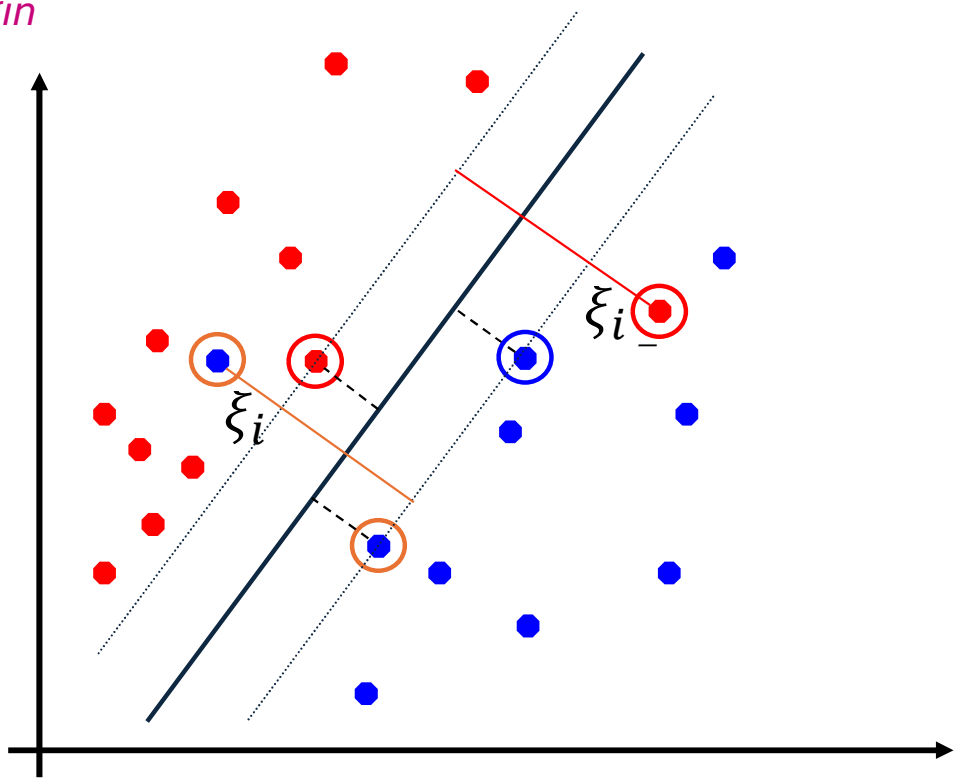
Support Vector Machines

- Nonlinearly separable problems
  - In these cases, special variables ξ_i are added to allow incorrectly classified instances.
    - Because they are difficult to classify, or have noise
    - *Soft margin*



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Support Vector Machines

- Nonlinearly separable problems
  - The cost function:
    - Find w and b such that

$$\text{minimize: } \frac{1}{2}\|w\|^2 \text{ subjecto to: } y_i(w^T x_i + b) \geq 1$$

  - is changed to:
    - Find w and b such that

$$\text{minimize: } \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{l}\xi_i \text{ subjecto to: } y_i(w^T x_i + b) \geq 1 - \xi_i\,, \quad \xi_i \geq 0$$

    - The C parameter can be seen as a way to control overfitting:
      - It is a trade-off between the relative importance of maximizing the margin and fitting the training data

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Support Vector Machines

- Nonlinearly separable problems

$$\text{minimize: } \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{l}\xi_i$$

  - C parameter:
    - High values:
      - More importance is placed on minimizing errors $C\sum_{i=1}^{l}\xi_i$ and less on maximizing margin $\frac{1}{2}\|w\|^2$
        - That is, that the errors in the classification are the minimum
        - Narrower margins
      - Greater accuracy in classification ➡ Possible overfitting
    - Low values:
      - Less importance is given to minimizing errors $C\sum_{i=1}^{l}\xi_i$ and more to maximizing margin $\frac{1}{2}\|w\|^2$
      - Look for wider margins, allow some errors in classification ➡ less overfitting, more generalization

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Support Vector Machines

- Nonlinearly separable problems
  - As before, the dual problem is constructed by associating Lagrange multipliers αi, βi :
    - Original problem:

minimize: $\frac{1}{2}\|w\|^2 + C\sum_{i=1}^{l}\xi_i$ subjecto to: $y_i(w^T x_i + b) \geq 1 - \xi_i$, $\xi_i \geq 0$

    - Dual problem:

$$L_P \equiv \frac{1}{2}\left\|\mathbf{w}\right\|^2 - C\sum_{i=1}^{l}\xi_i - \sum_{i=1}^{l}\alpha_i\left[y_i\left(\mathbf{w}\cdot\mathbf{x}_i + b\right) - 1 + \xi_i\right] - \sum_{i=1}^{l}\beta_i\xi_i$$

    - This problem must be minimized with respect to w, b and ξ, and maximized with respect to α_i and β_i

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Support Vector Machines

- Nonlinearly separable problems
  - Minimize dual problem with respect to w, by ξ :
    - Calculate the derivative of L_p with respect to w and b and set it equal to 0

$$\frac{\partial L_P}{\partial b} = 0 \Rightarrow \sum_{i=1}^{l} \alpha_i y_i = 0 \qquad \frac{\partial L_P}{\partial w} = 0 \Rightarrow w = \sum_{i=1}^{l} \alpha_i y_i x_i$$

      - Same as in the linear case

$$\frac{\partial L_P}{\partial \xi} = 0 \Rightarrow \alpha_i + \beta_i = C$$

    - Substituting in the previous equation:

$$L_D = \sum_{i=1}^{l} \alpha_i - \frac{1}{2} \sum_{i=1}^{l} \sum_{j=1}^{l} \alpha_i \alpha_j y_i y_j \mathbf{x}_i^T \mathbf{x}_j$$

    - which must be maximized with respect to α_i, with the constraint $\sum_{i=1}^{l} \alpha_i y_i = 0$

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Support Vector Machines

- Nonlinearly separable problems
  - The dual problem becomes

Find $\alpha_1 \ldots \alpha_N$ such that

maximize: $$\sum_{i=1}^{L} \alpha_i - \frac{1}{2} \sum_{i=1}^{L} \sum_{j=1}^{L} \alpha_i \alpha_j y_i y_j x_i^T x_j$$

identical to the previous one

subject to: $$\sum_{i=1}^{L} y_i \alpha_i = 0$$

$$0 < \alpha_i < C \qquad \text{i = 1, ..., L}$$

- Same techniques to find the ai
- Again, samples x_i with nonzero α_i will be support vectors

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Support Vector Machines

- Nonlinearly separable problems
  - Once the αi have been found, the solution would be similar:
    - w: $$w = \sum_{i=1}^{l} \alpha_i y_i \mathbf{x}_i$$
    - b: for any support vector ($\alpha_i > 0$):

$$y_k(w^T x_k + b) = 1 - \xi_k \Rightarrow w^T x_k + b = {1-\xi_k}/{y_k} = y_k(1-\xi_k) \Rightarrow$$

$$\Rightarrow b = y_k(1-\xi_k) - w^T x_k \qquad b = y_k(1-\xi_k) - \sum_{i=1}^{l} \alpha_i y_i x_i^T x_k$$

  - As before, it is not necessary to calculate *w* to perform the classification:

$$f(x) = \sum_{i=1}^{l} \alpha_i y_i x_i^T x + b$$

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Support Vector Machines

- Nonlinearly separable problems
  - Again, if the $\alpha_i \geq 0$, the Kuhn-Tucker theorem can be applied to convert the inequalities:

$$y_i(w^T x_i + b) \geq 1 - \xi_i$$

in several equations of equality for each sample i:

$$\alpha_i(y_i(w^T x_i + b) - 1 + \xi_i) = 0 \qquad (C - \alpha_i)\xi_i = 0$$

  - All x_i where $\alpha_i > 0$ are the support vectors
    - If $\alpha_i \neq 0$, then

$$y_i(w^T x_i + b) - 1 + \xi_i = 0 \quad \Rightarrow \quad y_i(w^T x_i + b) = 1 - \xi_i$$

    - and therefore x_i could be at the limit of the decision margin
      - If $\xi_i > 0$, x_i is not at the edge of the margin, but it is a support vector

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Support Vector Machines

- Nonlinearly separable problems
  - The points x_i with $\alpha_i > 0$ are the support vectors.
    - Those who touch and define the limits of the margin
  - The support vectors are the critical elements of the training set and these are the closest to the decision boundary.
  - In a non-separable case there are two types:
    - $0 < \alpha_i < C$:
      - Since $(C - \alpha_i)\xi_i = 0$ and $\alpha_i < C$, it follows that $\xi_i = 0$
      - In this case, the support vector x_i satisfies the equalities

$$y_i(w \cdot x_i + b) = 1 \qquad \xi_i = 0$$

      - The support vector is at the margin boundary

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Support Vector Machines

- Nonlinearly separable problems
  - The points x_i with $\alpha_i > 0$ are the support vectors.
    - Those who touch and define the limits of the margin
  - The support vectors are the critical elements of the training set and these are the closest to the decision boundary.
  - In a non-separable case there are two types:
    - $\alpha_i = C$
      - In this case $\xi_i \neq 0$
      - Therefore, the corresponding x_i does not satisfy $y_i(w^T x_i + b) = 1$
      - But it does satisfy $y_i(w^T x_i + b) = 1 - \xi_i$
      - That is, it is a classification error
      - But if the error value ξ_i is added to it, this x_i is "displaced" to the margin limit

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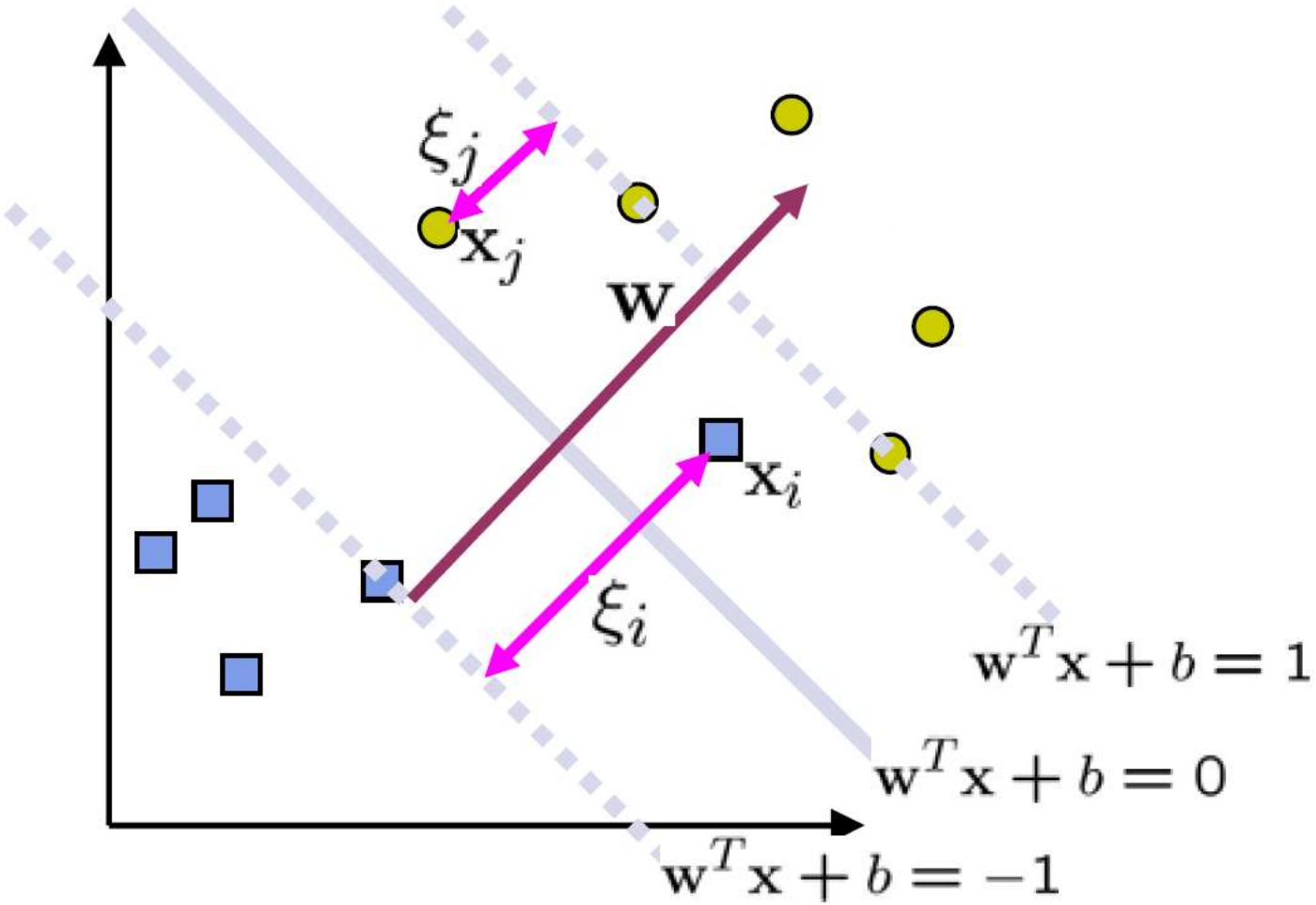
Support Vector Machines

- Nonlinearly separable problems



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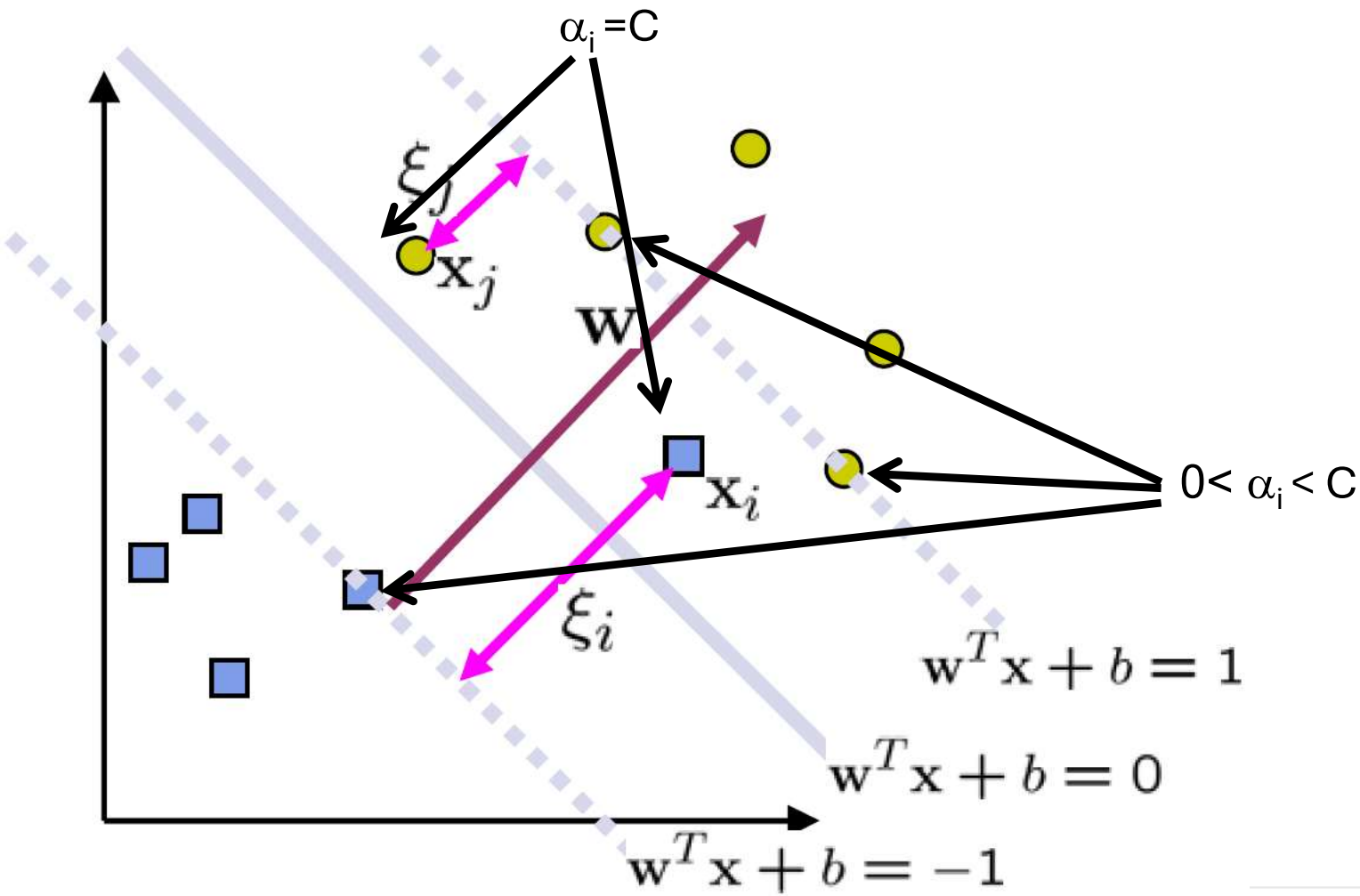
Support Vector Machines

- Nonlinearly separable problems



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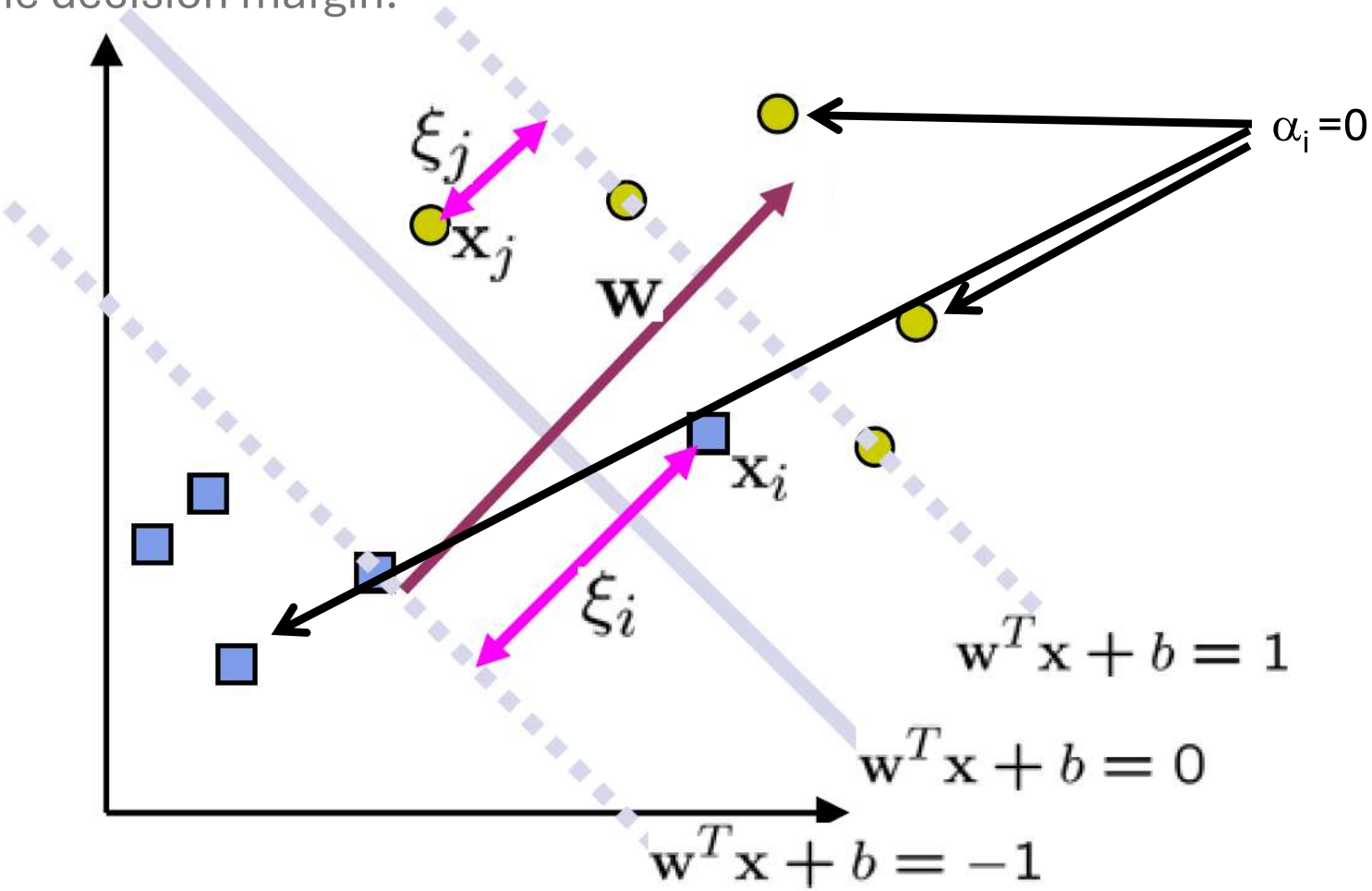
Support Vector Machines

- Nonlinearly separable problems
  - Furthermore, the cases ai = 0 correspond to points that are clearly far from the decision margin:



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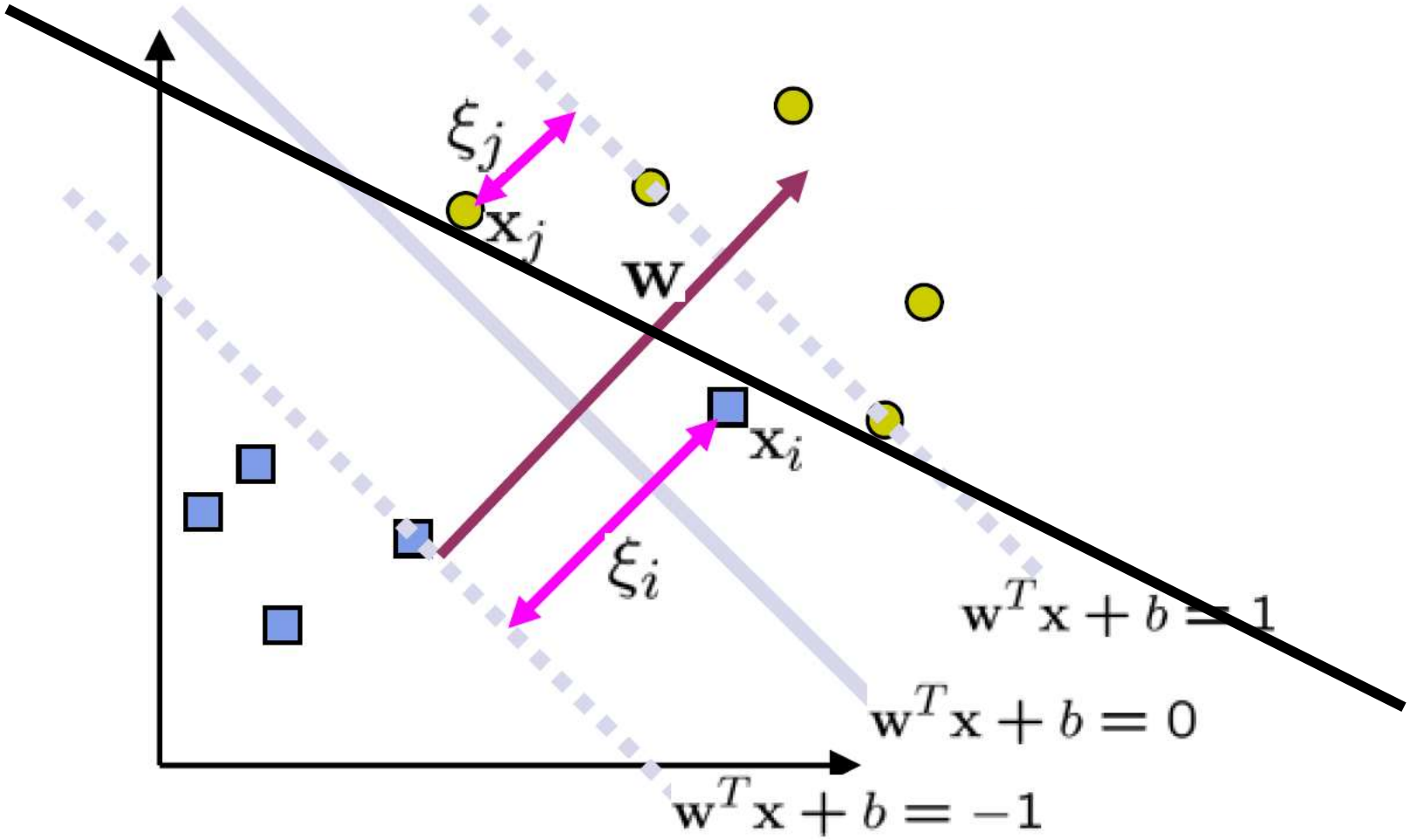
Support Vector Machines

- Linearly and Nonlinearly Separable Problems
  - This problem is linearly separable:



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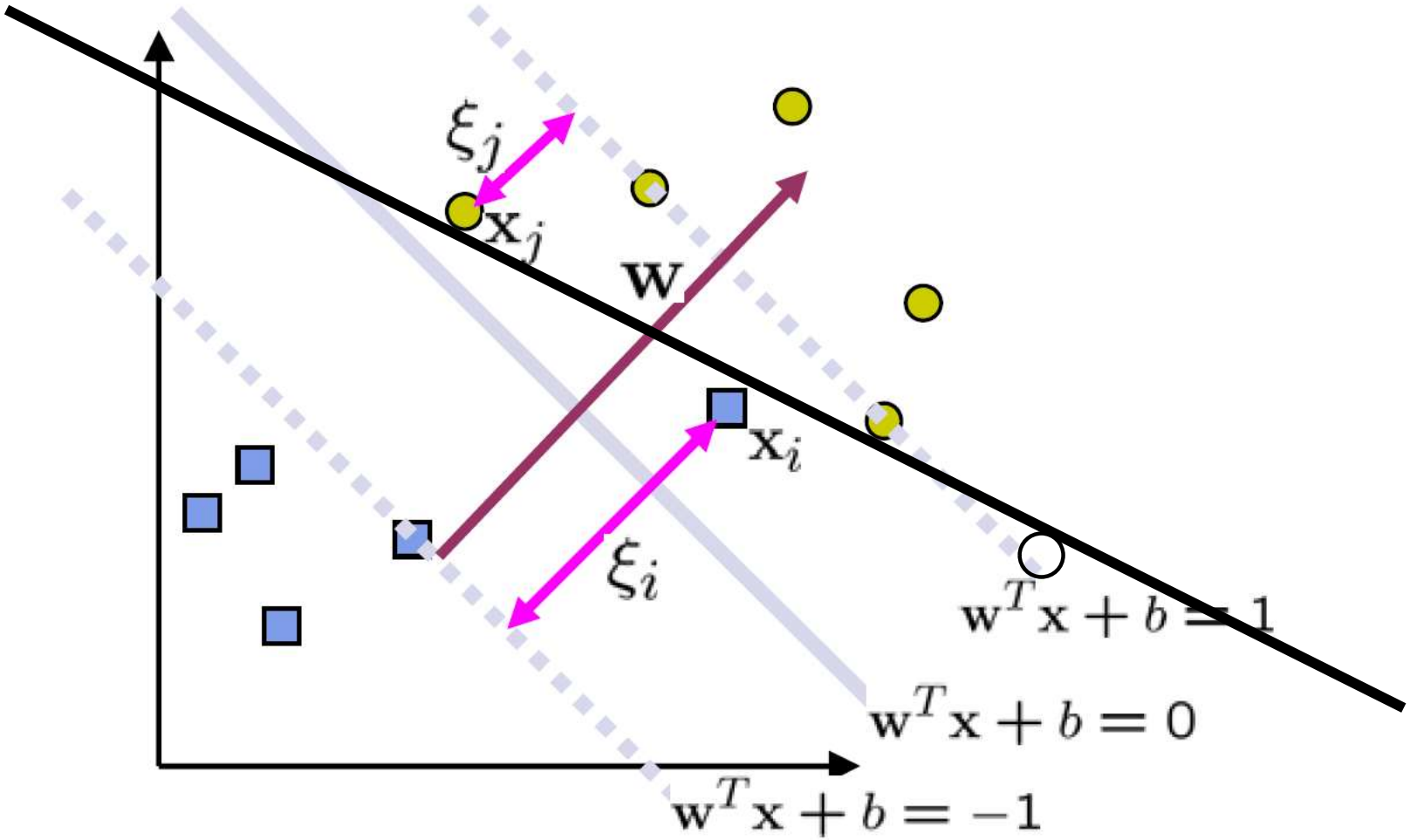
Support Vector Machines

- Linearly and Nonlinearly Separable Problems
  - But what would happen to this new instance?



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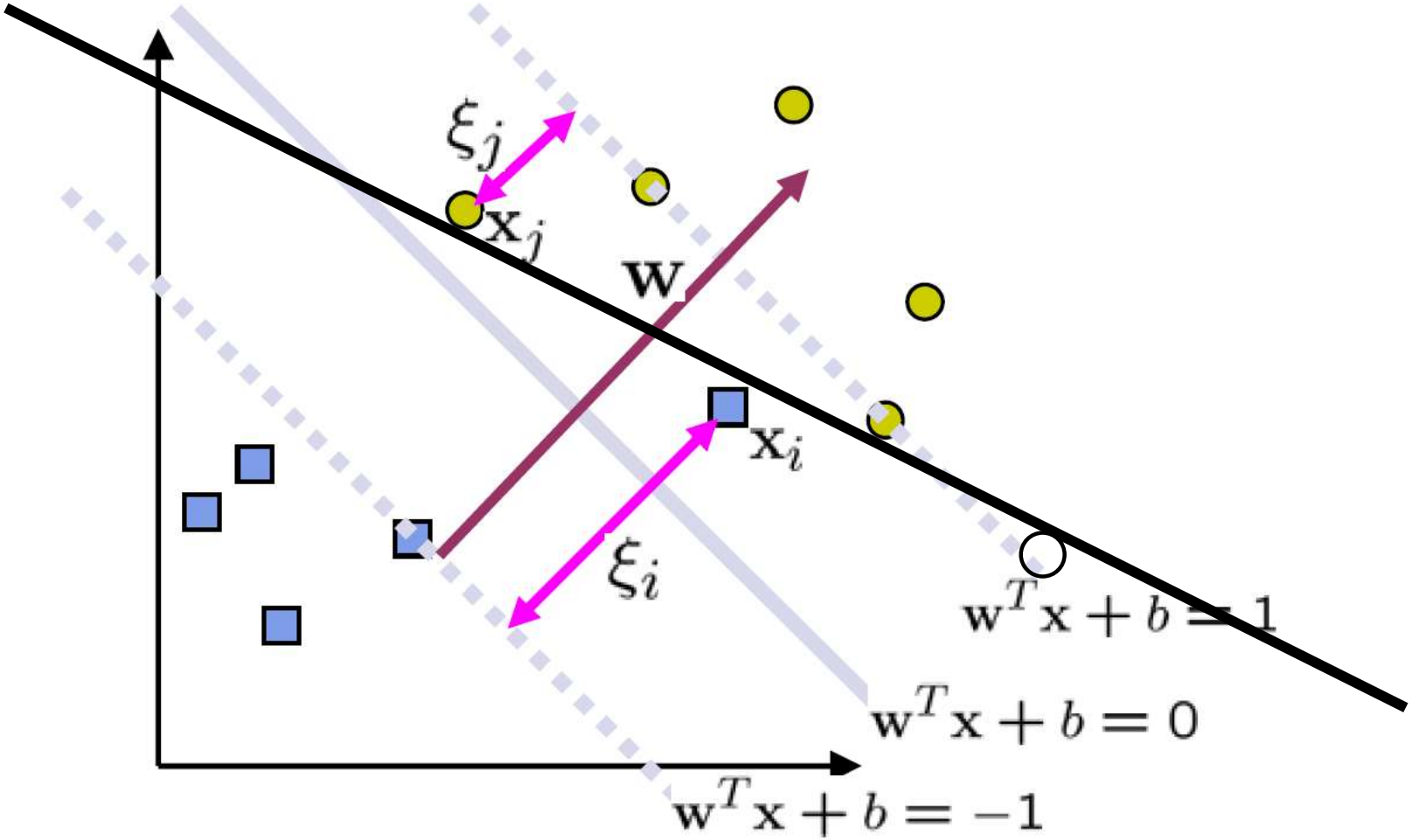
Support Vector Machines

- Linearly and Nonlinearly Separable Problems
  - Margin too narrow
    - Overfitting: Generalization Problems



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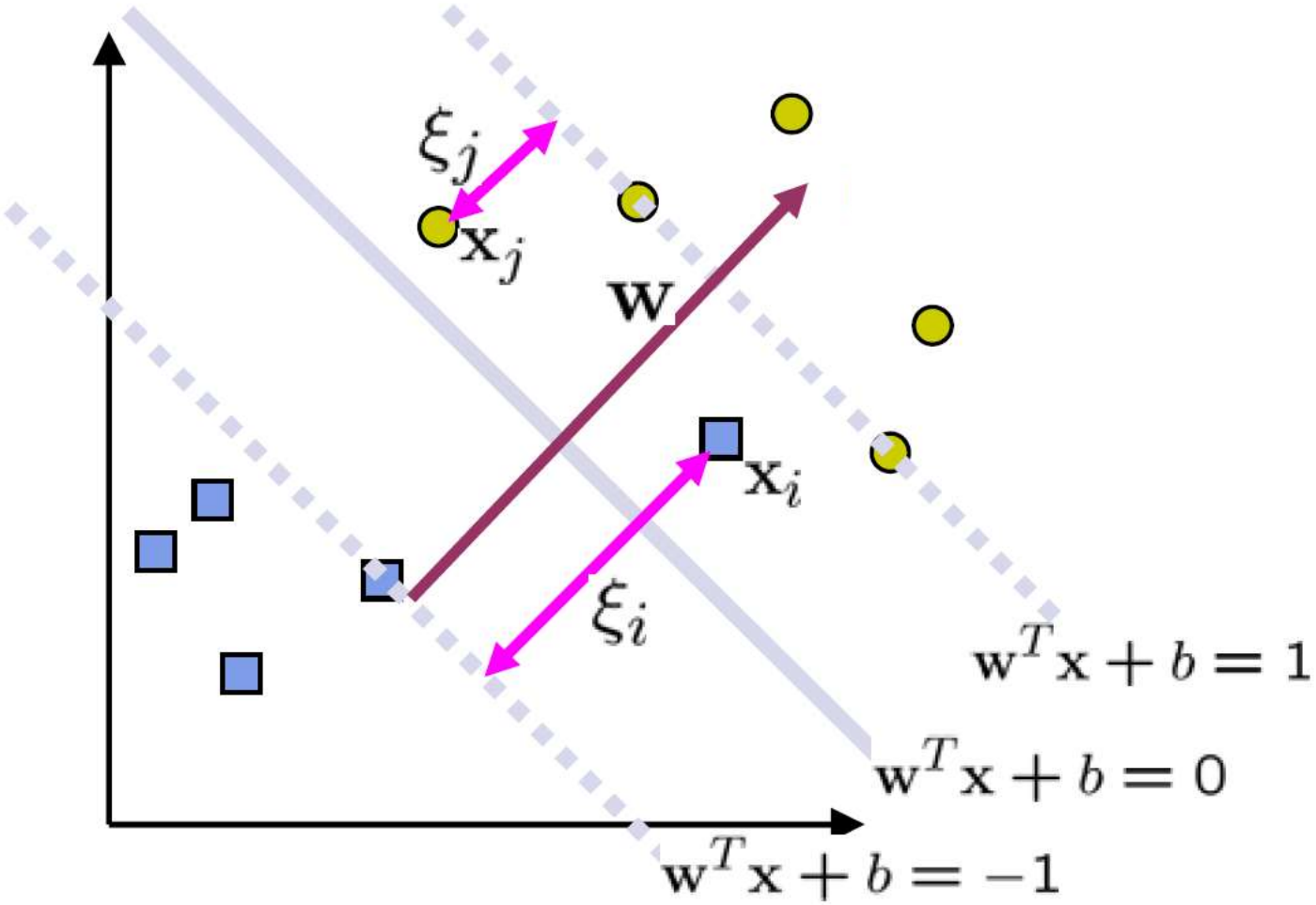
Support Vector Machines

- Linearly and Nonlinearly Separable Problems
  - Allow some error (ξ) to avoid overfitting: C low



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Support Vector Machines

- Linear SVMs:
  - The classifier is a separating hyperplane
  - The most important training samples are support vectors
    - They define the hyperplane
  - Quadratic optimization algorithms allow to identify which training samples x_i are support vectors, with nonzero Lagrange multipliers α_i
  - In both the dual formulation of the problem and the solution, only **scalar products** appear:

$$\sum_{i=1}^{L} \alpha_i - \frac{1}{2} \sum_{i=1}^{L} \sum_{j=1}^{L} \alpha_i \alpha_j y_i y_j \boxed{x_i^T x_j} \qquad f(x) = \sum_{i=1}^{l} \alpha_i y_i \boxed{x_i^T x} + b$$

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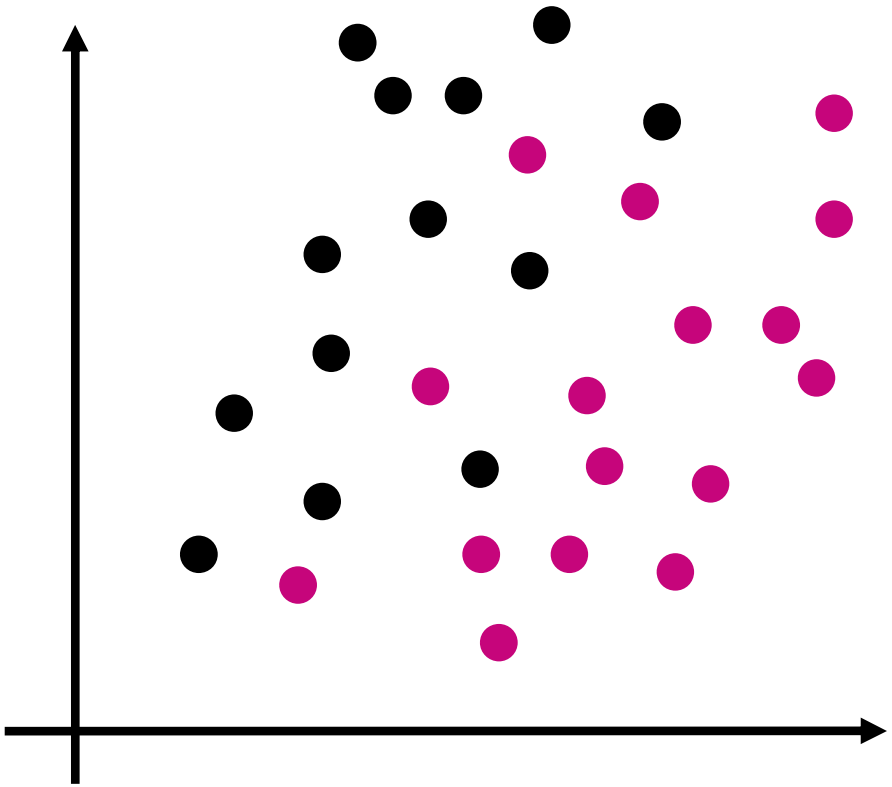
Support Vector Machines

- Nonlinear SVM
  - The above model works fine with datasets with some noise



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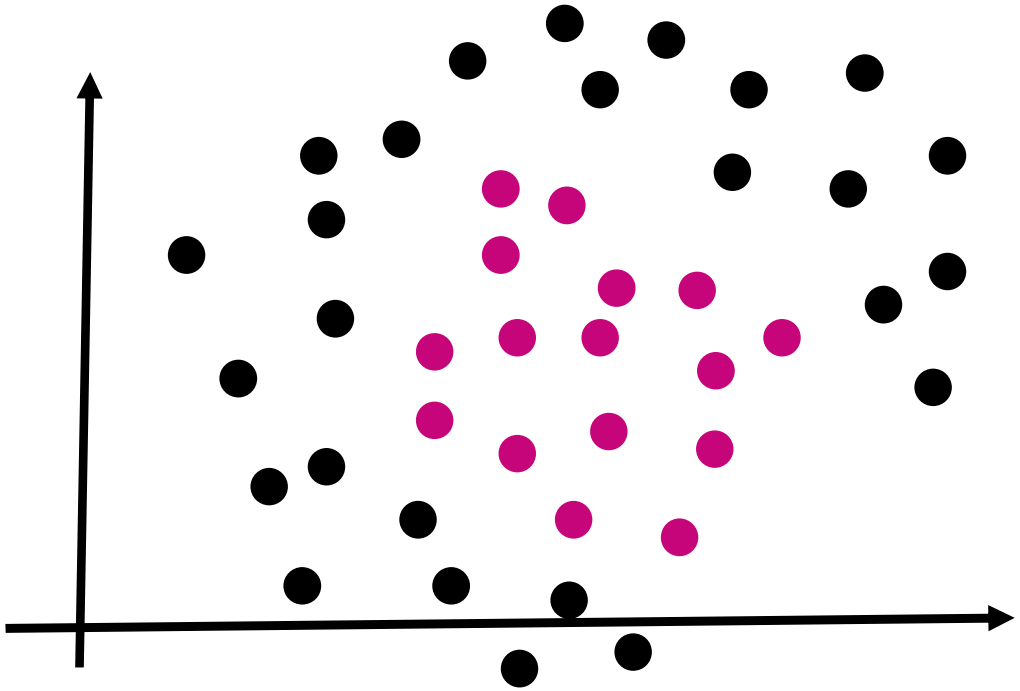
Support Vector Machines

- Nonlinear SVM
  - But, what happens if the data has a distribution that is very difficult to separate by a hyperplane?
    - For example:
      - (2 dimensions)



    - For example:
      - (1 dimension)

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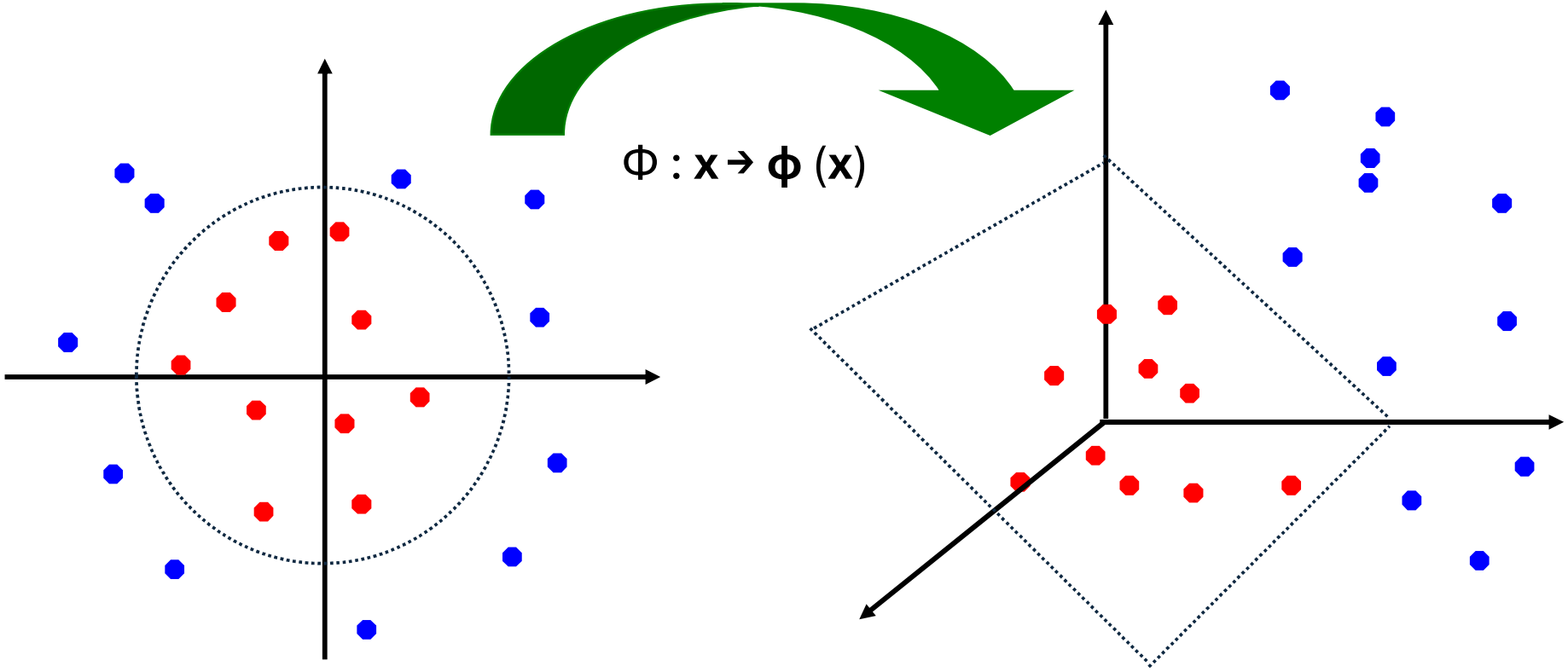
Support Vector Machines

- Nonlinear SVM
  - Solution: Map the data to a higher dimensional space where the training set is separable



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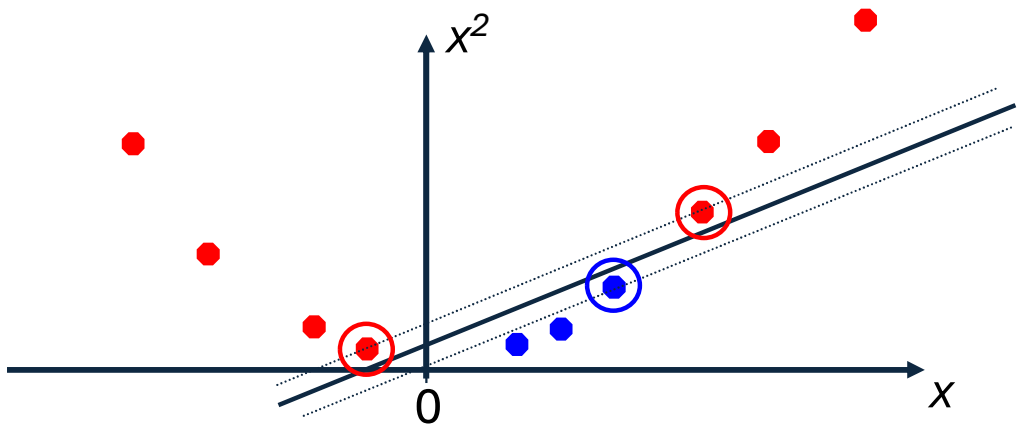
Support Vector Machines

- Nonlinear SVM
  - For example:

Non-linearly separable

$\Phi : \mathbf{x} \rightarrow \boldsymbol{\phi}(\mathbf{x})$

$\boldsymbol{\phi}(\mathbf{x}) = (x, x^2)$



linearly separable

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Support Vector Machines

- Nonlinear SVM
  - Linear SVMs are based on the dot products:

$$f(x) = \sum_{i=1}^{l} \alpha_i y_i x_i^T x + b$$

  - The idea is to project each sample through a transformation $\phi: x \rightarrow \varphi(x)$ to a new space of higher dimensionality in such a way that in this new space the samples are linearly separable.
    - Or a linear separation can be applied tolerating some error
    - The previous equations are applied (linear separation) in that new space:

$$f(x) = \sum_{i=1}^{l} \alpha_i y_i \varphi(x_i)^T \varphi(x) + b$$

      - The scalar products are performed in the new space
    - Which is this transformation? $\phi: x \rightarrow \varphi(x)$

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Support Vector Machines

- Nonlinear SVM
  - Instead of defining this transformation $\phi: x \rightarrow \varphi(x)$, we can define the dot product in this new space, but operating with the values in the original space:

$$K(\boldsymbol{x}_i, \boldsymbol{x}_j) = \boldsymbol{\varphi}(\boldsymbol{x}_i)^{\boldsymbol{T}} \boldsymbol{\varphi}(\boldsymbol{x}_j)$$

  - A *kernel* function is a function that is equivalent to a dot product in a given space
    - But with the values of x_i and x_j in the original space

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Support Vector Machines

- Nonlinear SVM
  - The kernel trick :
    - In the original space, we would calculate the dot product
      - $x_i^T x$
    - However, since the problem is complex, the vectors are projected into a new space
      - $\Phi : x \rightarrow \varphi(x)$
    - Therefore , it will be necessary calculate the dot product on that space $\varphi(x_i\ )^T \varphi(x)$
      - It would involve projecting both vectors to the new space and calculate its dot product
    - However, we can define a function that calculates the dot product in the new space $\varphi(xi)^T\, \varphi(x)$, from the original vectors
      - In this way, we don't need to calculate the projections in the new space
        - No need to calculate $\varphi(x)$ to know the new space
      - Kernel function: $\varphi(x_1)^T\, \varphi(x_2) = K(x_1, x_2)$
    - Therefore, the kernel function implicitly maps the data to a higher dimensional space
      - Without the need to compute explicitly each $\varphi(x)$
      - “The kernel trick”, common to all kernel-based methods

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Support Vector Machines

- Kernel-based methods
  - These are a series of pattern recognition and analysis methods, most known by SVMs
  - The main feature of these methods is the different approach they use
    - These methods map the data into higher dimensional spaces in the hope that in this space the data will become more easily separable, or better structured.
    - No restrictions on how to map
    - This mapping function, however, hardly needs to be computed thanks to a tool called **the kernel trick**

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Support Vector Machines

- Kernel-based methods
  - The kernel trick is a mathematical tool that can be applied to any algorithm that **only depends on the dot product of two vectors**
  - When a scalar product is to be done, it is replaced by a kernel function
  - If applied properly, these **linear algorithms become non-linear algorithms**
    - Those nonlinear algorithms are equivalent to the original linear ones operating on the space ϕ
    - However, since kernels are used, the function $\boldsymbol{\varphi}(\boldsymbol{x})$ does not need to be explicitly computed.

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Support Vector Machines

- Kernel-based methods
  - Choosing an appropriate kernel depends on the problem to be solved
  - For example:
    - A polynomial kernel allows modeling conjunctions of the values up to the order of the polynomial
    - A radial basis function allows the construction of circles (or hyperspheres)
    - A linear kernel allows the construction of lines (or hyperplanes)
  - Also, setting the parameters can be a tedious process

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Support Vector Machines

- Kernel-based methods
  - Some commonly used kernels are:
    - Linear: the simplest $\varphi(x) = x$
      - Kernel algorithms using this type are usually the same as the corresponding non- kernel algorithms.
        - For example, KPCA (kernel PCA, version of PCA with kernels) is the same as PCA when the kernel is linear
        - An SVM with a linear kernel is the same as a linear SVM

$$k(x, y) = x^T y + c$$

    - Polynomial
      - They are suitable for problems where all the training data is normalized.

$$k(x, y) = (\alpha x^T y + c)^d$$

      - Adjustable parameters are the slope α, the constant term c, and the degree of the polynomial d

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Support Vector Machines

- Kernel-based methods
  - Some commonly used kernels are:
    - Gaussian – This is an example of a radial basis function kernel

$$k(x,y) = \exp\left(-\frac{\| x - y \|^2}{2\sigma^2}\right)$$

ϕ (x) infinite-dimensional: each point is mapped to a (Gaussian) function

      - The sigma parameter plays an important role in the operation of the kernel , and its value must be carefully set:
        - If overestimated, the exponential will behave almost linearly, and the projection into higher dimensional space will start to lose its nonlinear power.
        - On the contrary, if it takes values too low, the decision boundaries will become very sensitive to noise in the training data.

    - Exponential: this is also a radial basis function kernel
      - Very similar to the previous

$$k(x,y) = \exp\left(-\frac{\| x - y \|}{2\sigma^2}\right)$$

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Support Vector Machines

- Kernel-based methods
  - Some commonly used kernels are:
    - Laplacian – also radial basis function kernel
      - Fully equivalent to the exponential, except for being less sensitive to changes in the sigma parameter

$$k(x, y) = \exp\left(-\frac{\| x - y \|}{\sigma}\right)$$

      - Observations made on the sigma parameters in the kernel Gaussian can also be applied to the exponential and the Laplacian

    - Hyperbolic tangent
      - Also known as sigmoid kernel or multilayer perceptron kernel
      - It comes from ANNs (used as transfer function)
      - An SVM that uses a sigmoid kernel is equivalent to a two-layer perceptron

$$k(x, y) = \tanh(\alpha x^T y + c)$$

      - Two parameters: the slope α , and the constant c.
        - A common value for alpha is $1/N$, where N is the number of dimensions.

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Support Vector Machines

- Kernel-based methods
  - Other kernels:
    - ANOVA
    - rational quadratic
    - Multiquadratic
    - Inverse Multiquadratic
    - Circular
    - Spherical
    - Logarithmic
    - splines
    - B -Spline
    - Cauchy
    - Chi squared
    - Bayesian
    - Wavelets
    - etc.

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Support Vector Machines

- Nonlinear SVM
  - The cost function:
    - Find w and b such that

$$\text{minimize: } \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{l}\xi_i \text{ subjecto to: } y_i(w^T x_i + b) \geq 1 - \xi_i\,, \quad \xi_i \geq 0$$

  - is changed to:
    - Find w and b such that

$$\text{minimize: } \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{l}\xi_i \text{ subjecto to: } y_i(w^T \varphi(x_i) + b) \geq 1 - \xi_i\,, \quad \xi_i \geq 0$$

  - $\varphi(x)$ performs a transformation of the data to a higher dimensional space
    - Search for the hyperplane that separates the data in that new space
    - However, it is not necessary to know this transformation
      - It is enough to know a function, kernel, that computes the dot product

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Support Vector Machines

- Nonlinear SVM
  - The dual problem would become:

Find $\alpha_1 \ldots \alpha_N$ such that they maximize:

$$\sum_{i=1}^{L} \alpha_i - \frac{1}{2} \sum_{i=1}^{L} \sum_{j=1}^{L} \alpha_i \alpha_j y_i y_j \varphi(x_i) \varphi(x_j)$$

subject to: $\sum_{i=1}^{L} y_i \alpha_i = 0 \quad 0 < \alpha_i < C$ i = 1, ..., L

Find $\alpha_1 \ldots \alpha_N$ such that they maximize:

$$\sum_{i=1}^{L} \alpha_i - \frac{1}{2} \sum_{i=1}^{L} \sum_{j=1}^{L} \alpha_i \alpha_j y_i y_j K(x_i, x_j)$$

subject to: $\sum_{i=1}^{L} y_i \alpha_i = 0 \quad 0 < \alpha_i < C$ i = 1, ..., L

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Support Vector Machines

- Nonlinear SVM
  - Solution to the problem:
    - Decision function:

$$f(x) = \sum_{i=1}^{l} \alpha_i y_i \varphi(x_i)^T \varphi(x) + b$$

$$f(x) = \sum_{i=1}^{l} \alpha_i y_i K(x_i, x) + b$$

- The optimization technique to find the α_i remains the same

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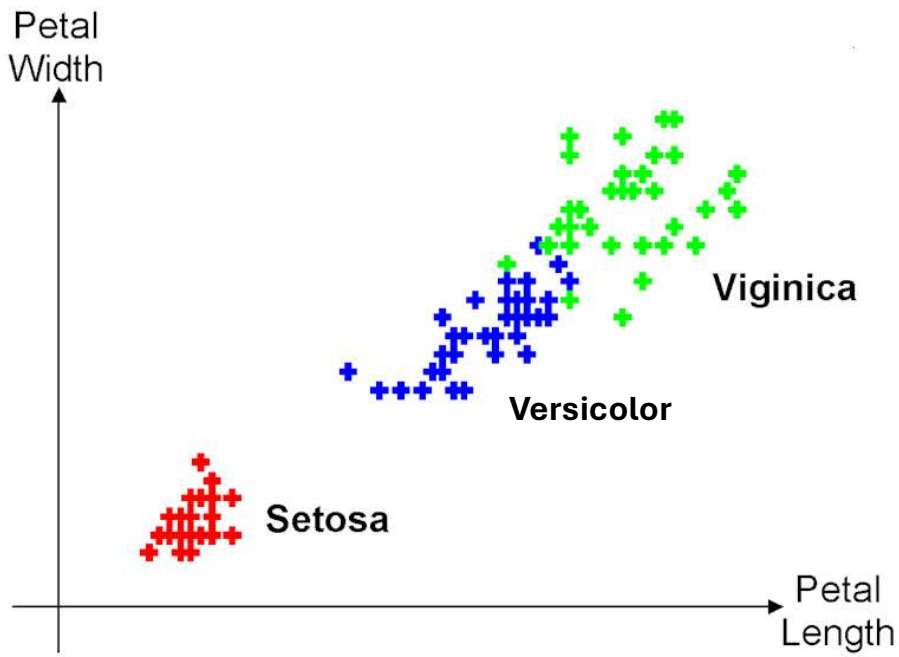
Support Vector Machines

- Example: Iris flower classification problem:
  - Well-known classification dataset
  - Contains data on 150 iris flowers, each with 4 attributes, classifiable into 3 possible classes: setosa, virginica and versicolor
    - Samples can be represented by the 2 attributes that containing the most information
    - Train with only those two
    - attributes
  - **Strategy "one vs. all"**



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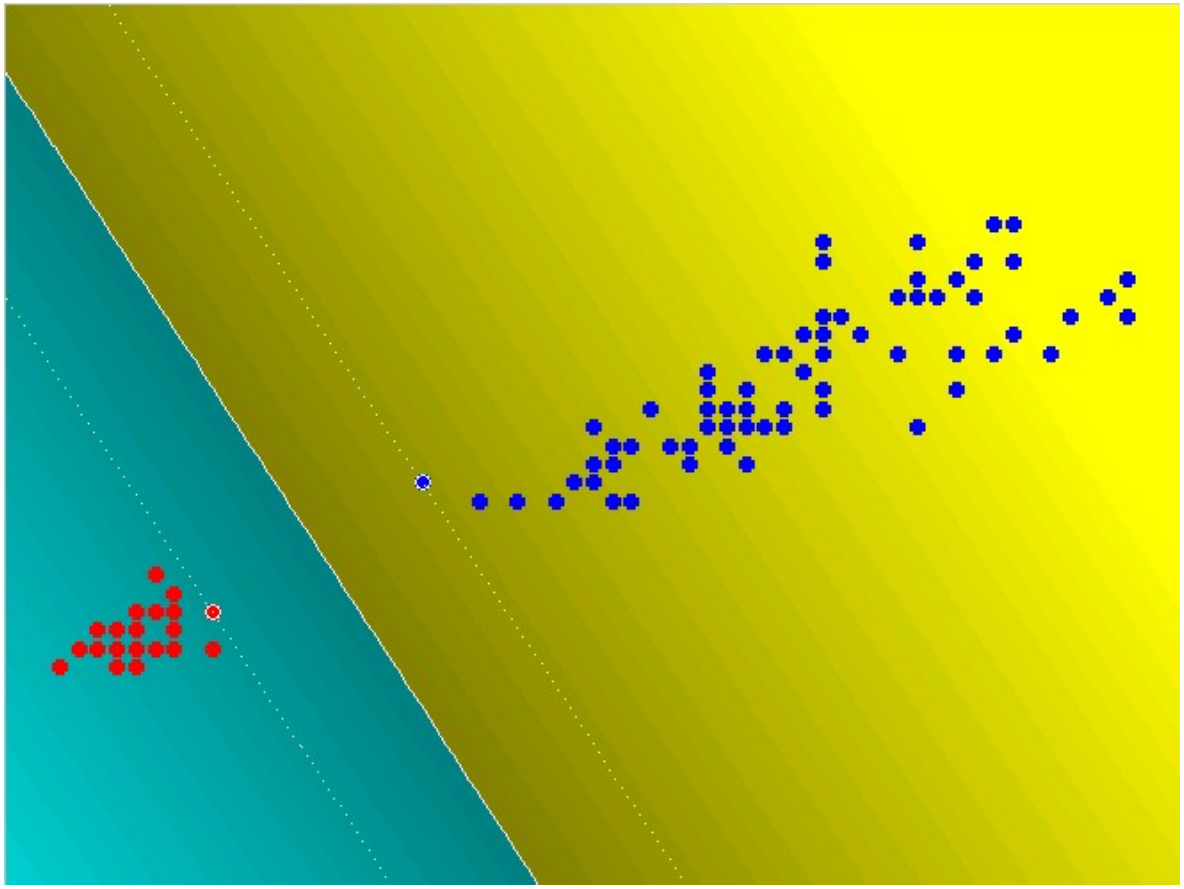
Support Vector Machines

- Example: Iris flower classification problem:
  - Separation of "setosa" using a linear SVM (C = inf)



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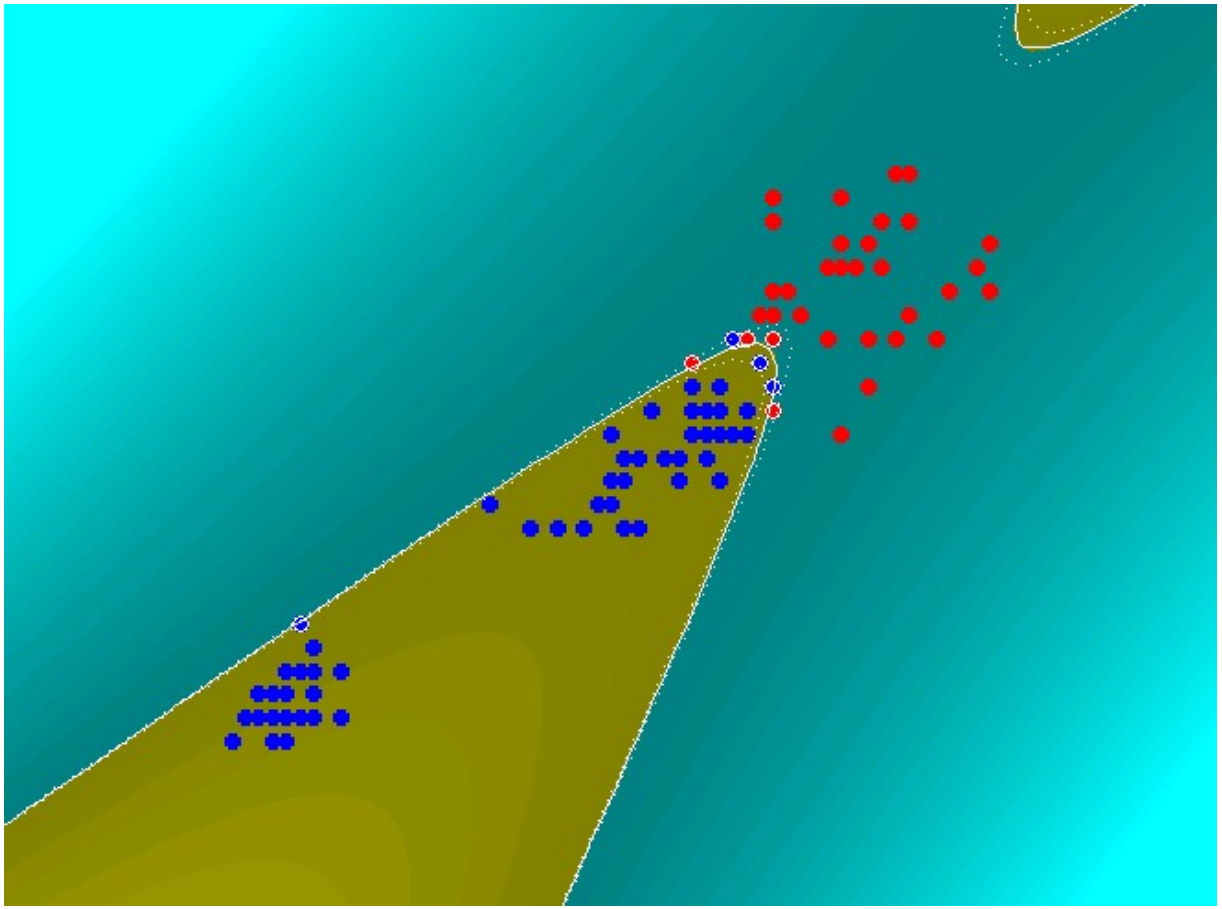
Support Vector Machines

- Example: Iris flower classification problem:
  - Separation of "Virginica" with a polynomial SVM (degree 2, C = inf)



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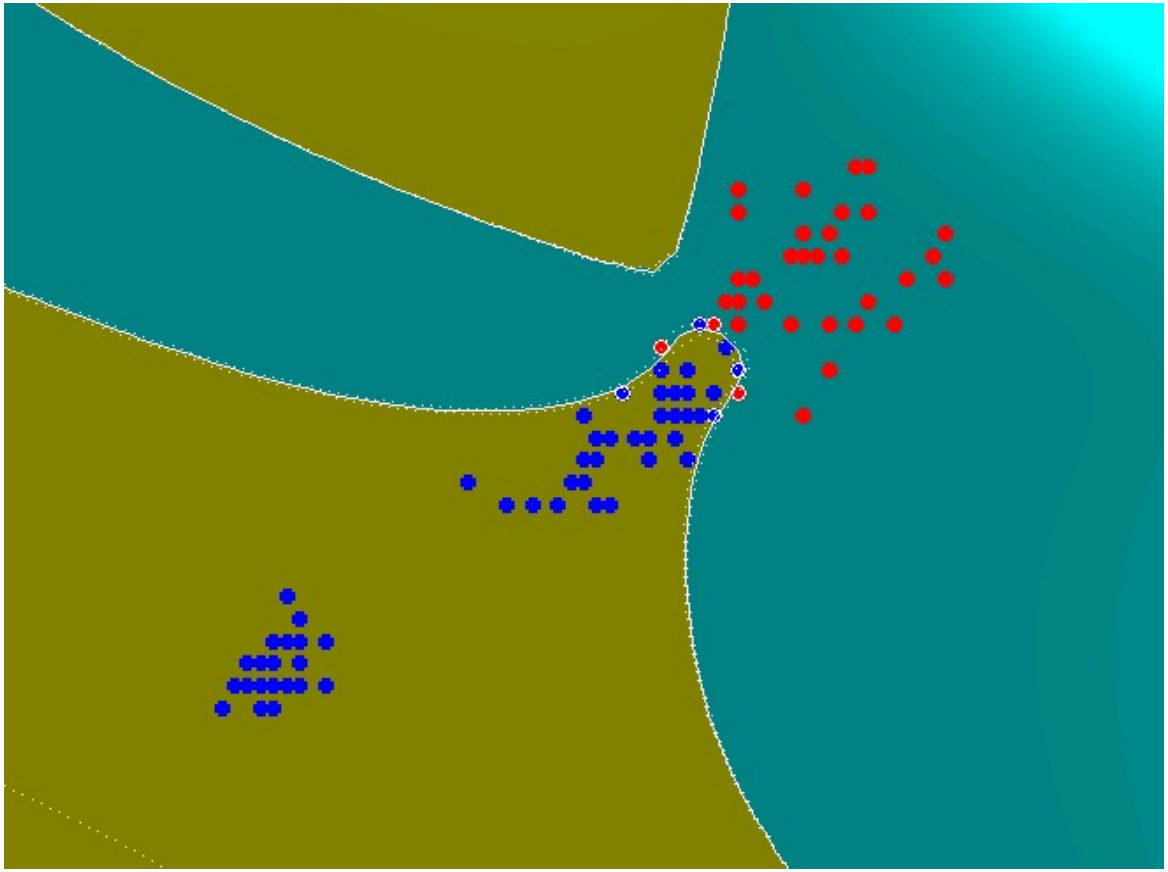
Support Vector Machines

- Example: Iris flower classification problem:
  - Separation of "Virginica" with a polynomial SVM (degree 10, C = inf)
    - Hyperplane in a 55-dimensional space
    - Overfit



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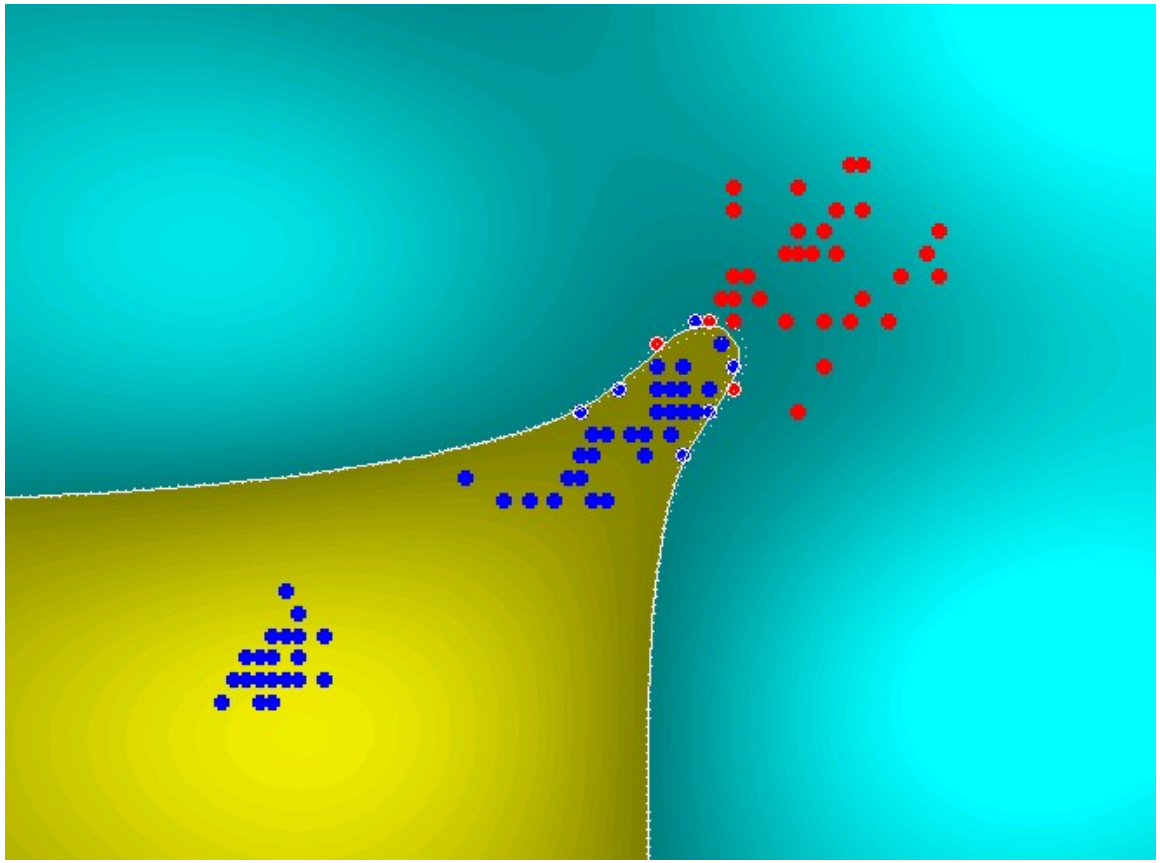
Support Vector Machines

- Example: Iris flower classification problem:
  - Separation of "Virginica" with a radial basis function SVM (σ = 1.0, C = inf)
    - Quite similar to the polynomial of degree 2



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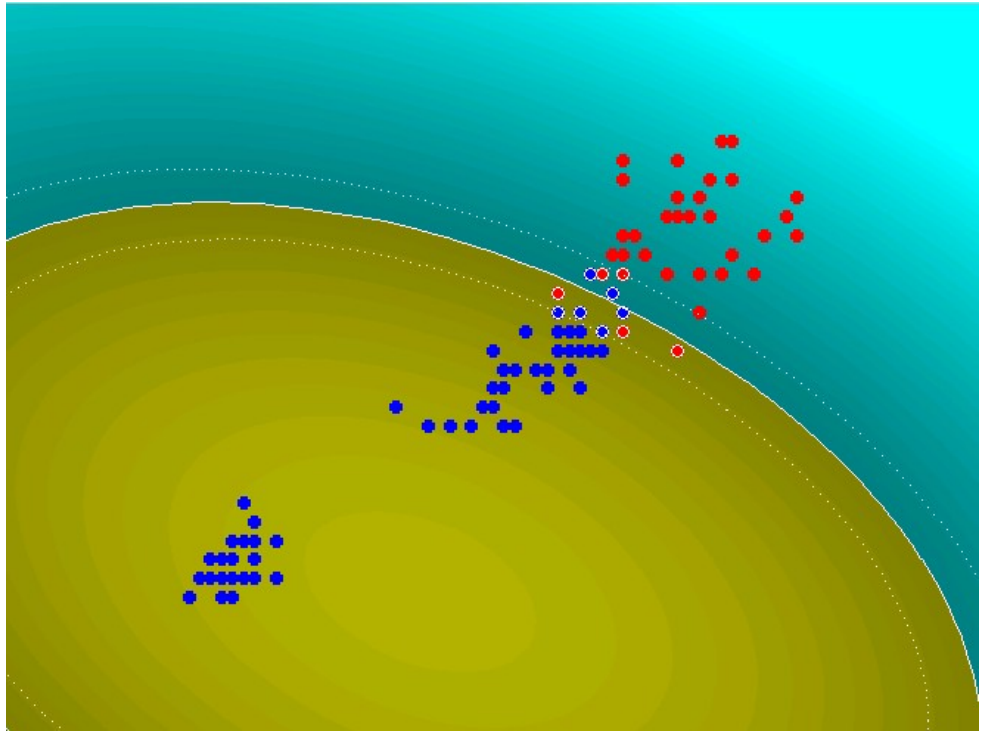
Support Vector Machines

- Example: Iris flower classification problem:
  - " Virginica" with a polynomial SVM (degree 2, C = 10)
    - C=10 → Some error tolerance in classification
    - Seems to give a good generalization
      - This emphasizes the importance of tolerating certain errors in classification.
      - This is necessary due to the non-separable nature of the data using only two attributes.



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Support Vector Machines

- Example: Iris flower classification problem:
  - To show the effect of error tolerance on classification at the classifier boundaries, different examples are shown.
    - Spline, with various degrees of tolerance
    - Separation of “ Versicolor”
    - High values of C give tighter limits.

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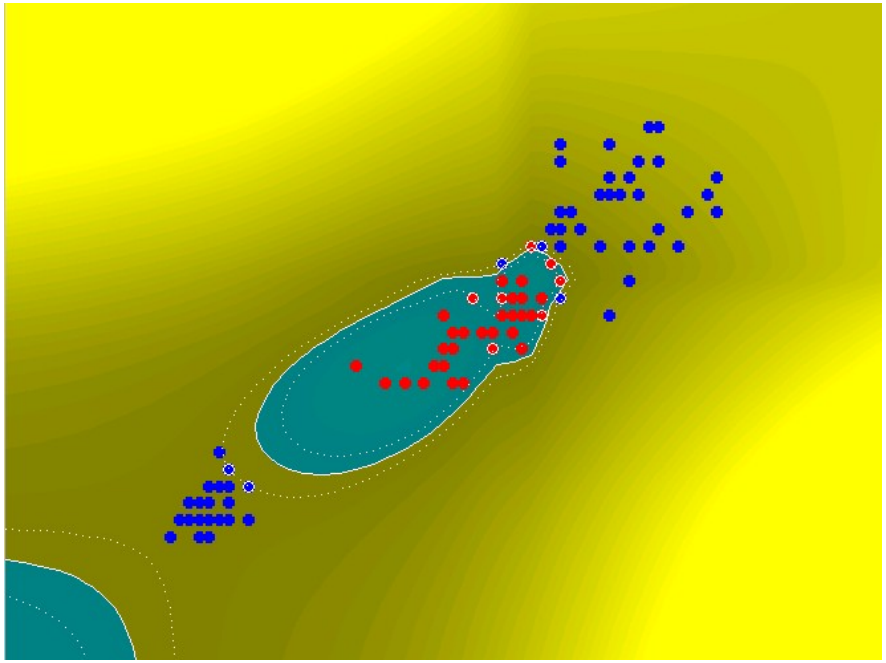
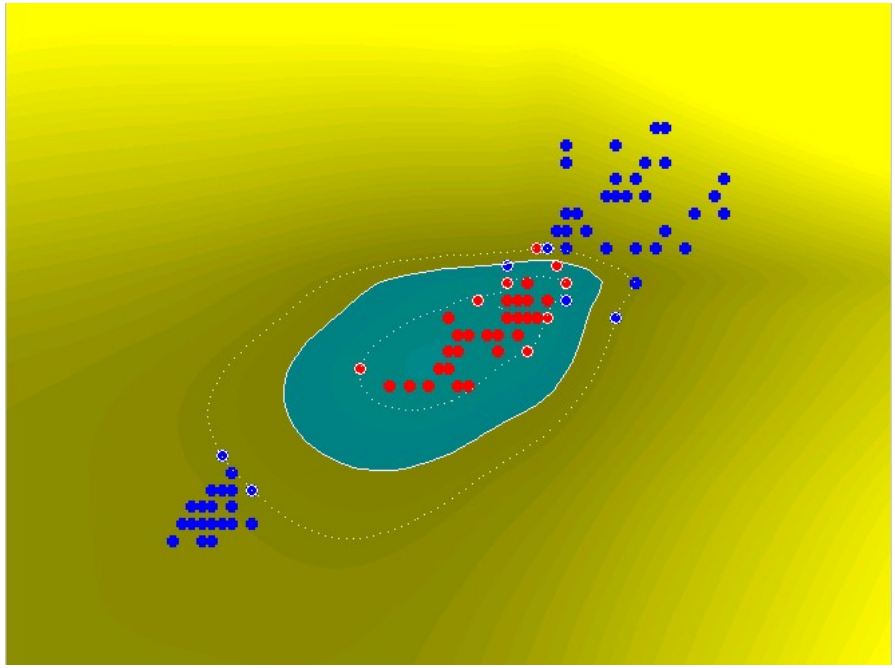
Support Vector Machines

- Example: Iris flower classification problem:

C = Inf



C = 1000



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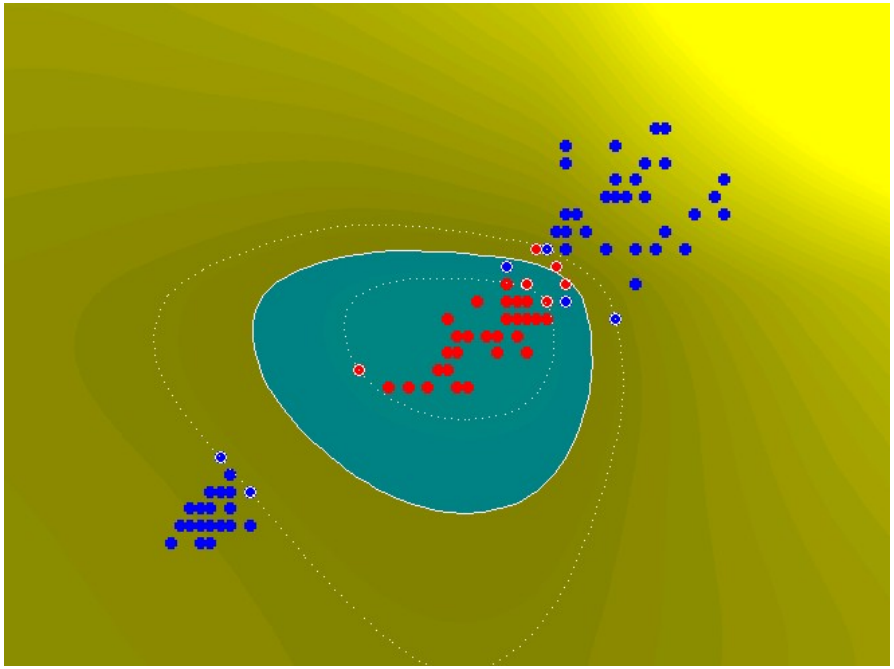
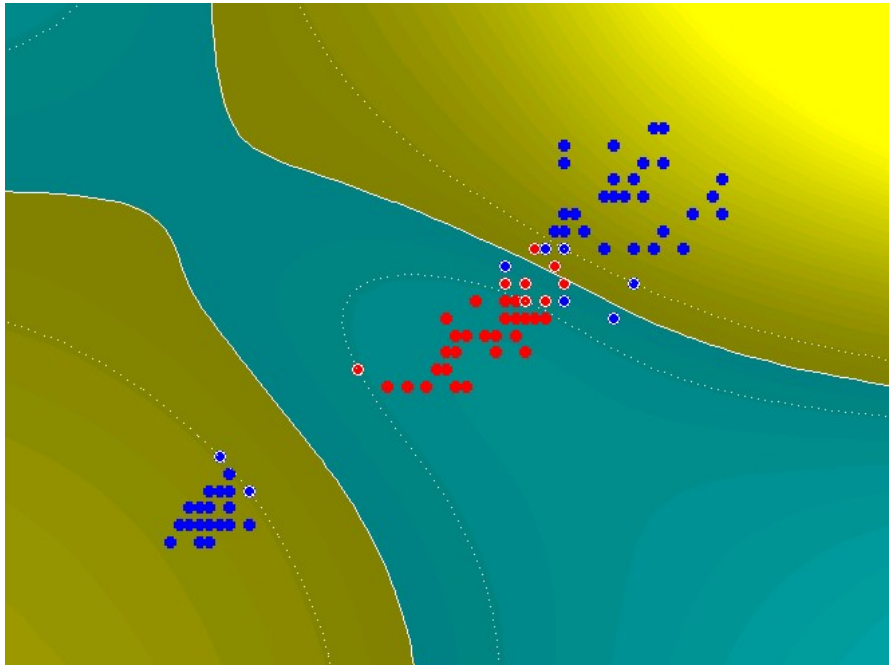
Support Vector Machines

- Example: Iris flower classification problem:

C = 100



C = 10



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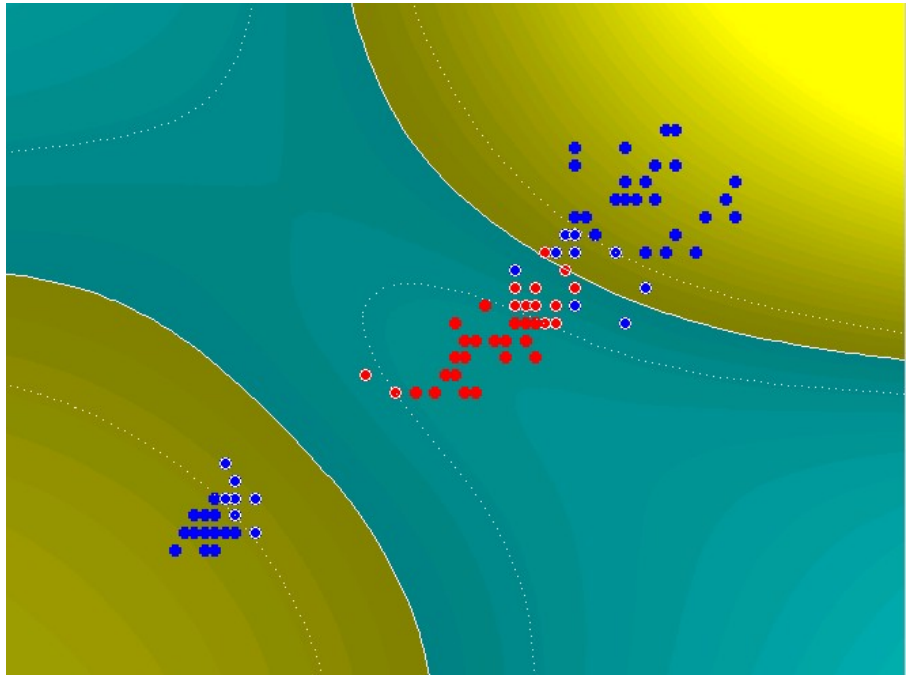
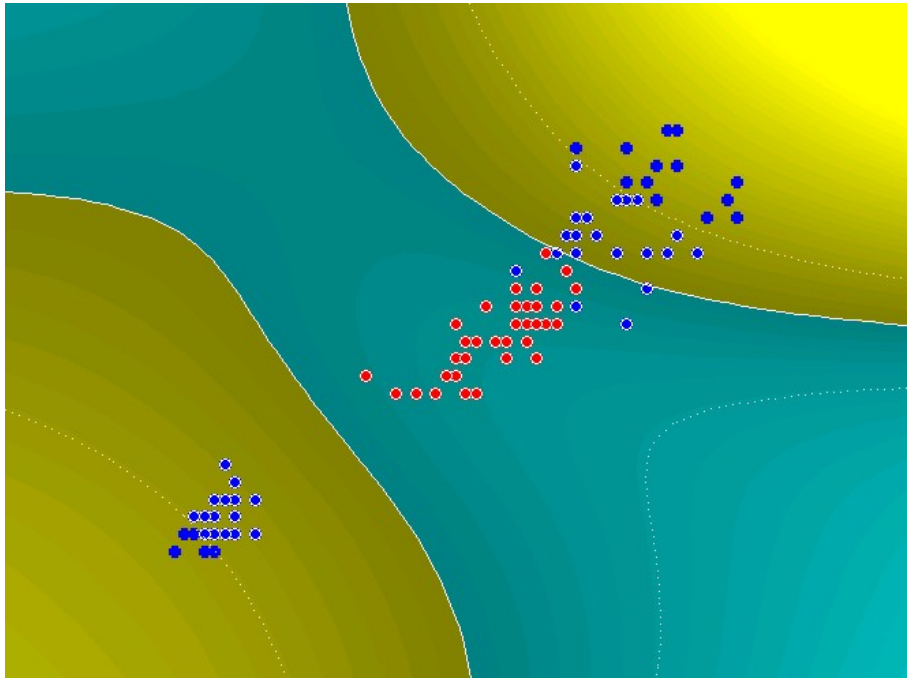
Support Vector Machines

- Example: Iris flower classification problem:

C = 1



C = 0.1



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Support Vector Machines

- SVMs for regression:
  - The approach for a classification problem is:

$$\text{minimize: } \|w\|^2 \text{ subjecto to: } y_i(w^T x_i + b) \geq 1 \quad \forall i = 1,\ldots,l$$

  - Instead, for a regression problem it is:

$$\text{minimize: } \|w\|^2 \text{ subjecto to: } \begin{array}{l} (w^T x_i + b) - y_i \leq \varepsilon \\ y_i - (w^T x_i + b) \leq \varepsilon \end{array} \quad \forall i = 1,\ldots,l$$

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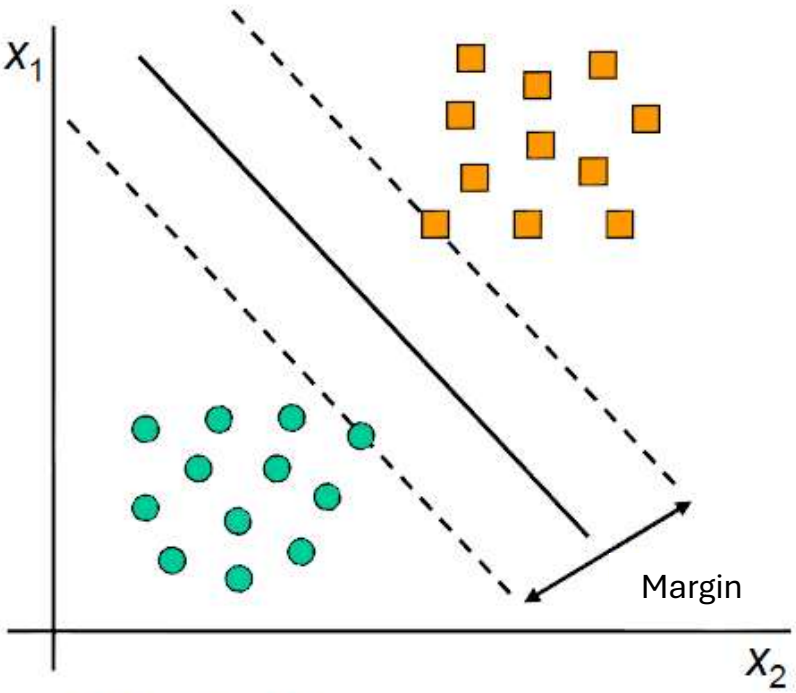
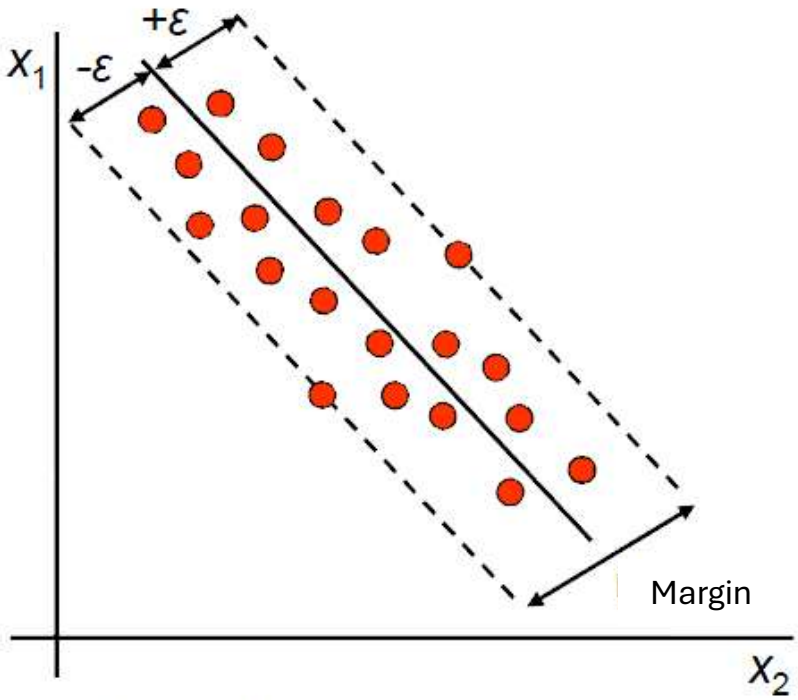
Support Vector Machines

- SVMs for regression:
  - Approach difference:



Classification:
All of the data outside the margin



Regression:
All of the data inside the margin

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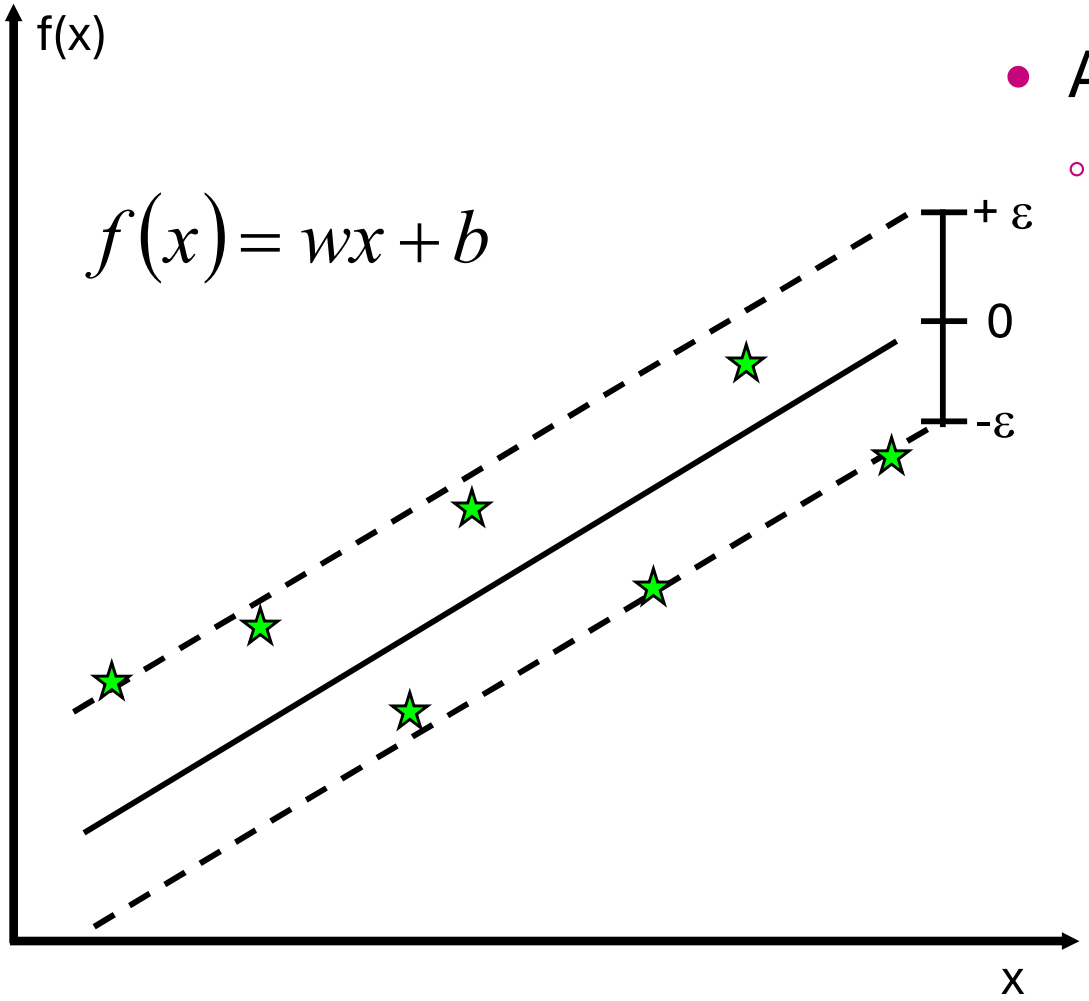
Support Vector Machines

- SVMs for regression:



- Approach:
  - Find the hyperplane f(x)= wx+b that approximates the points as best as possible

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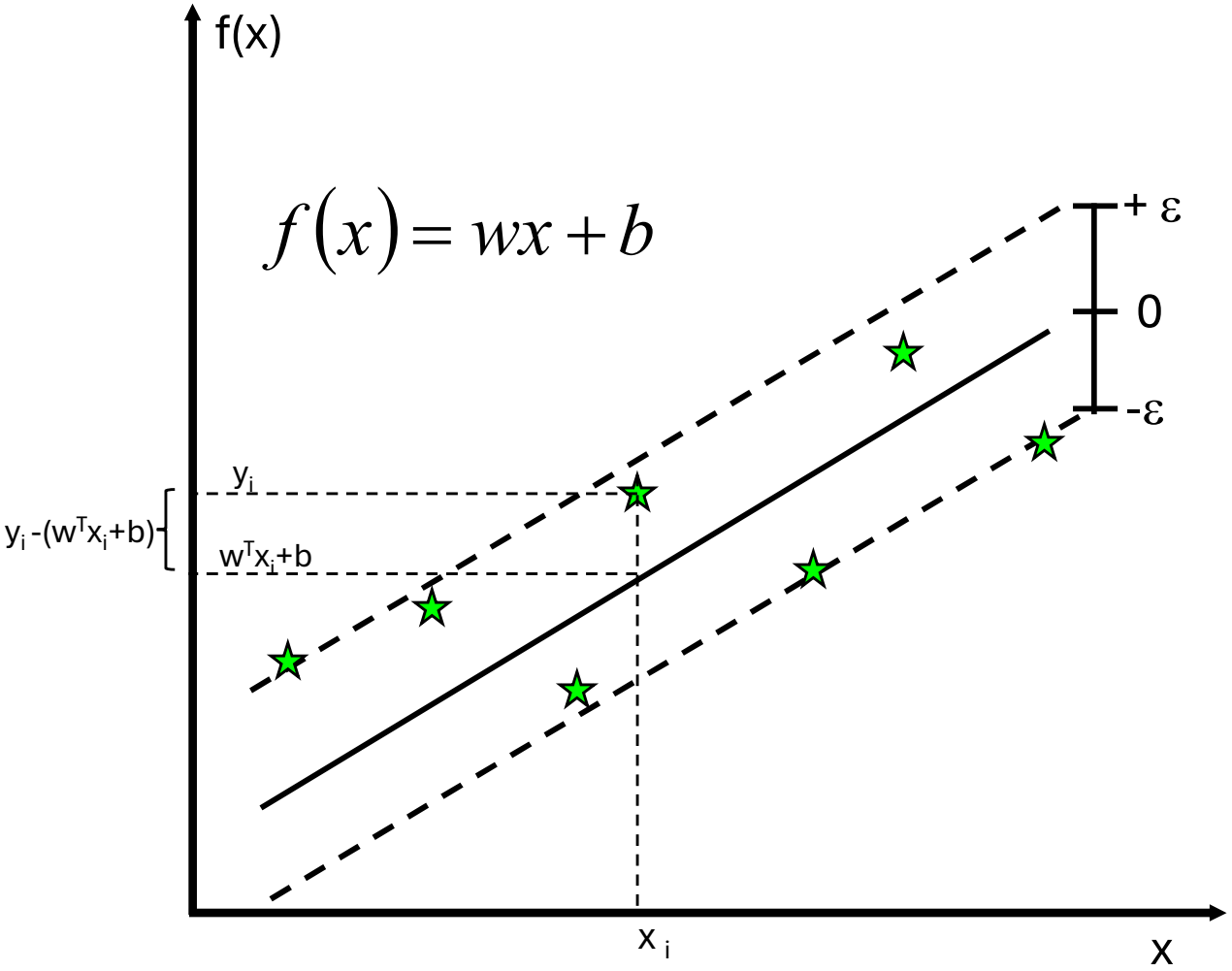
Support Vector Machines

- SVMs for regression:



- Solution :

$$Min \frac{1}{2} w^T w$$

- Constraints:

$$y_i - w^T x_i - b \leq \varepsilon$$

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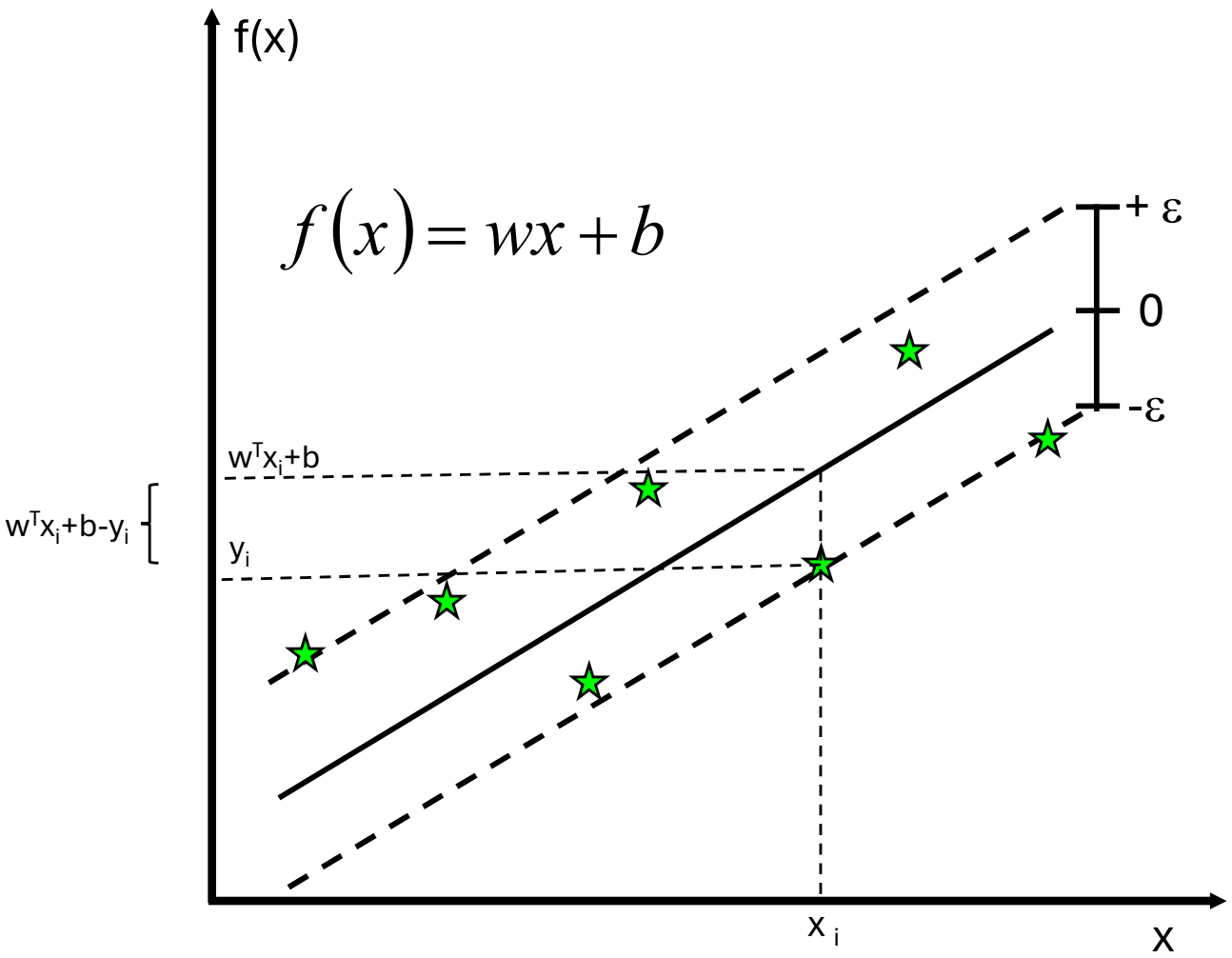
Support Vector Machines

- SVMs for regression:



- Solution :

$$Min \frac{1}{2} w^T w$$

- Restrictions :

$$w^T x_i + b - y_i \leq \varepsilon$$

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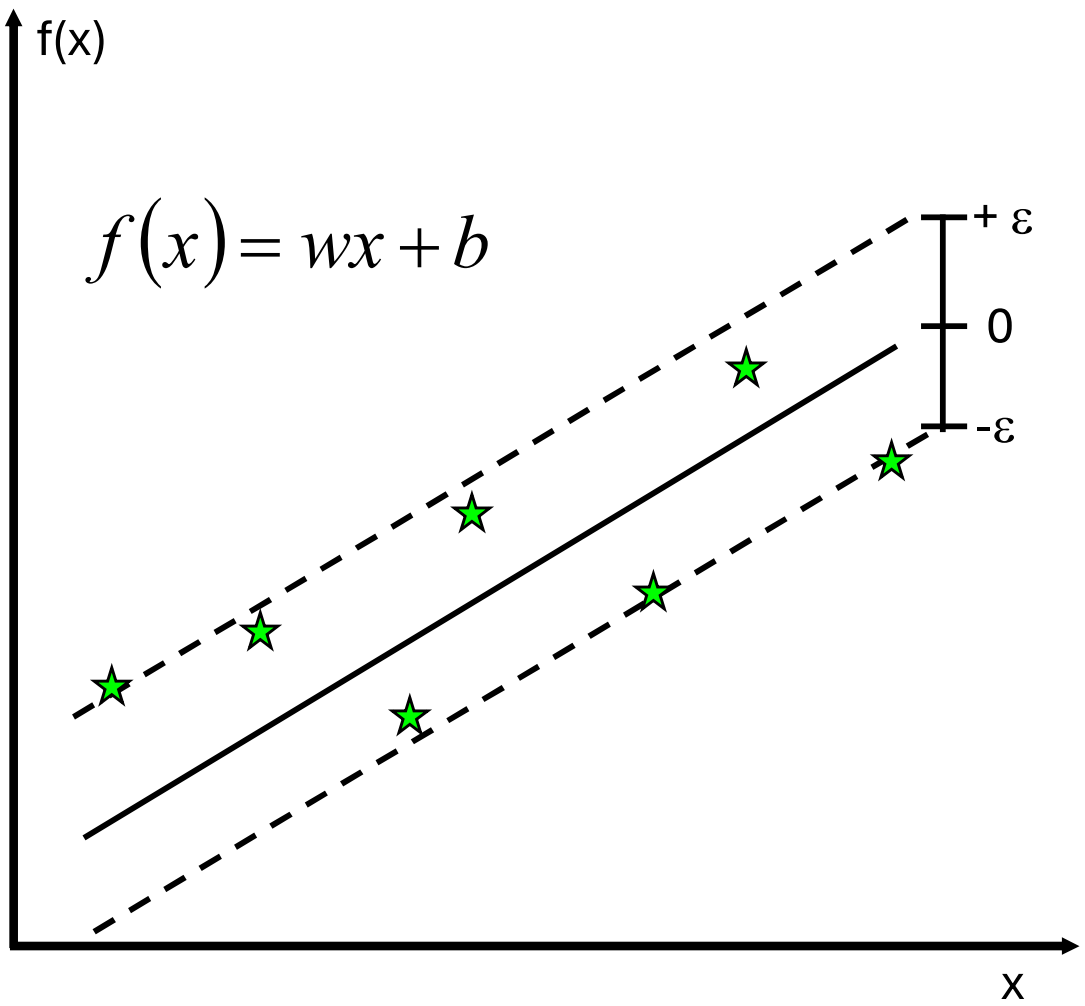
Support Vector Machines

- SVMs for regression:



- Solution :

$$Min \frac{1}{2} w^T w$$

- Restrictions :

$$y_i - w^T x_i - b \leq \varepsilon$$

$$w^T x_i + b - y_i \leq \varepsilon$$

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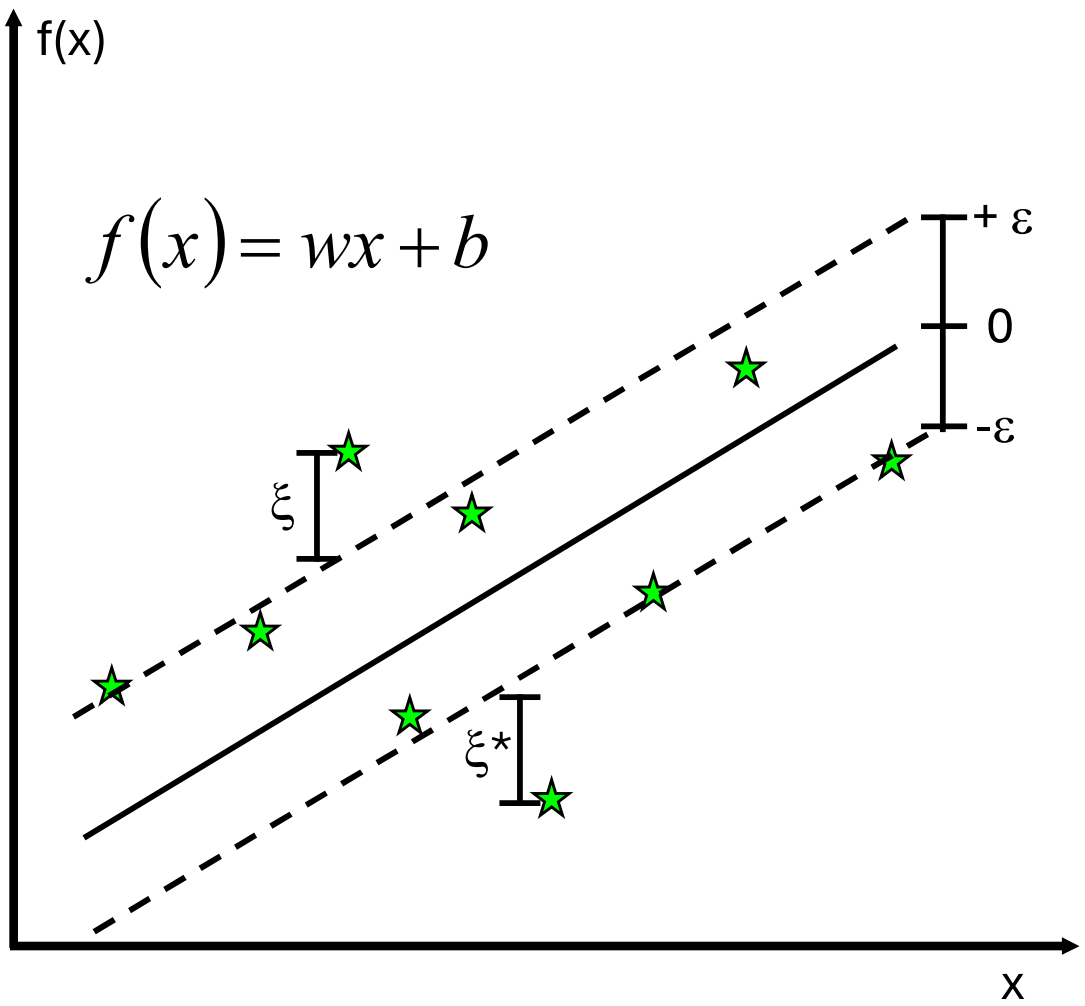
Support Vector Machines

- SVMs for regression:



- Minimize :

$$\frac{1}{2} w^T w + \boxed{C \sum_{i=1}^{N} \left(\xi_i + \xi_i^* \right)}$$

- Restrictions :

$$y_i - w^T x_i - b \leq \varepsilon + \xi_i$$

$$w^T x_i + b - y_i \leq \varepsilon + \xi_i^*$$

$$\xi_i, \xi_i^* \geq 0$$

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Support Vector Machines

- SVMs for regression:
  - Applying Lagrange :

$$L = \boxed{\frac{1}{2} w^T w + C \sum_{i=1}^{N} \left(\xi_i + \xi_i^* \right)}$$

Target

$$\boxed{\begin{aligned} &- \sum_{i=1}^{N} \alpha_i \left(\varepsilon + \xi_i - y_i + w^T x_i + b \right) \\ &- \sum_{i=1}^{N} \alpha_i^* \left(\varepsilon + \xi_i^* - y_i + w^T x_i + b \right) \\ &- \sum_{i=1}^{N} \left(\eta_i \xi_i + \eta_i^* \xi_i^* \right) \end{aligned}}$$

Constraints

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Support Vector Machines

- SVMs for regression:
  - Resulting regression equation:

$$y(x) = \sum_{i=1}^{N} \left(\alpha_i - \alpha_i^*\right) \cdot \langle x_i, x \rangle + b$$

$\langle x_i, x \rangle$: dot product between both vectors

  - Non-linear extension

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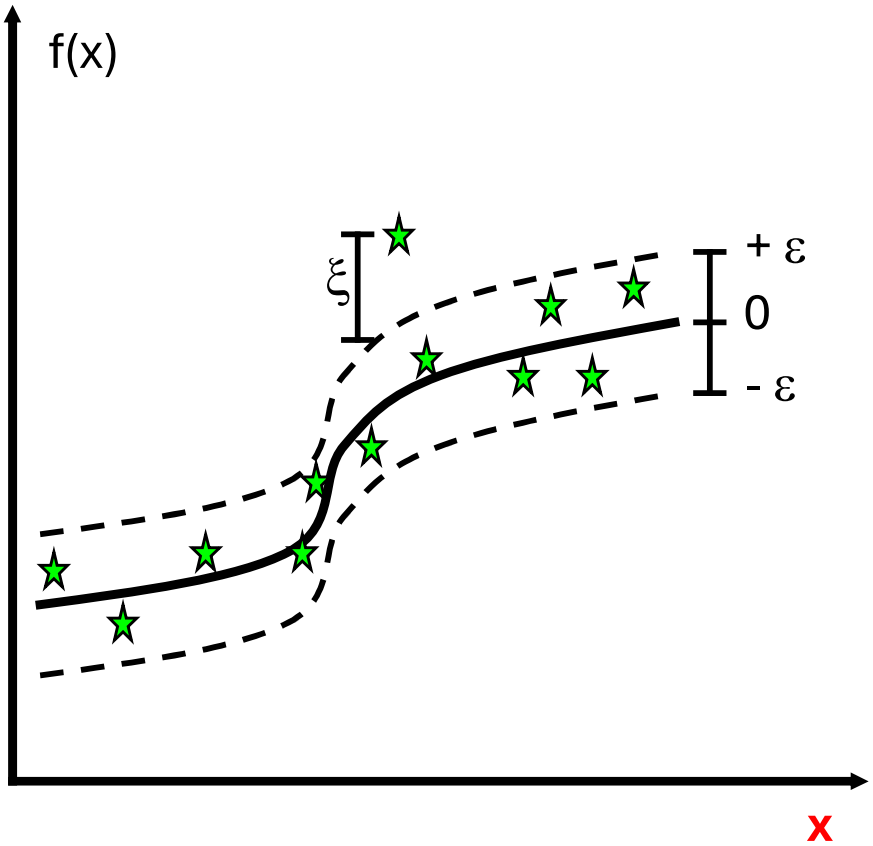
Support Vector Machines

- SVMs for regression:
  - Non-linear regression:
    - What if the points cannot be approximated by a hyperplane or the shape sought is not a hyperplane?



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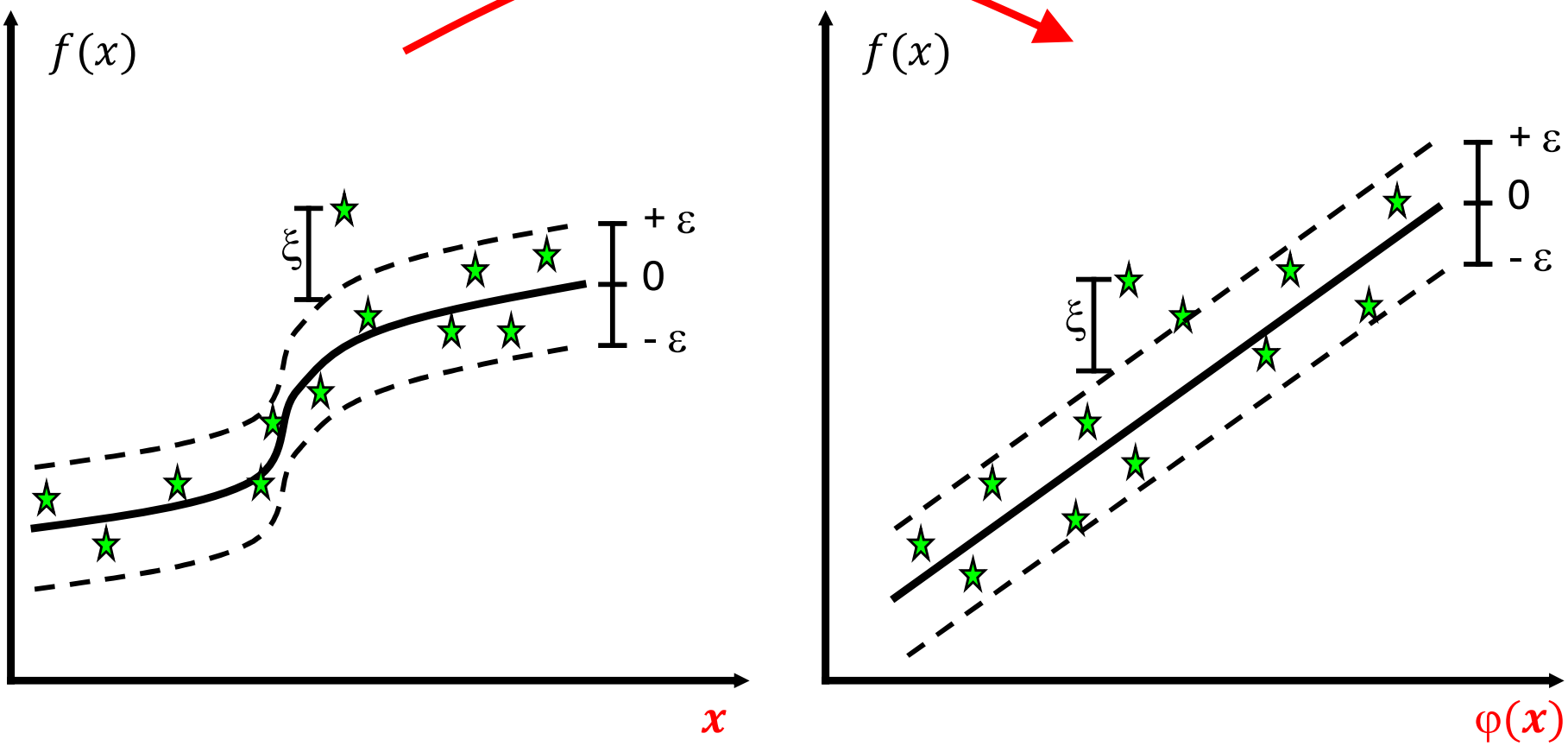
Support Vector Machines

- SVMs for regression:
  - Nonlinear regression:

Projection in another space where they can be approached with a hyperplane



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Support Vector Machines

- SVMs for regression:
  - Regression formulas:
    - Linear:

$$y(x) = \sum_{i=1}^{N} \left(\alpha_i - \alpha_i^*\right) \cdot \langle x_i, x \rangle + b$$

$\langle x_i, x_j \rangle$: dot product between both vectors

    - Nonlinear:

$$y(x) = \sum_{i=1}^{N} \left(\alpha_i - \alpha_i^*\right) \cdot \langle \varphi(x_i), \varphi(x) \rangle + b$$

    - General:

$$y(x) = \sum_{i=1}^{N} \left(\alpha_i - \alpha_i^*\right) \cdot K(x_i, x) + b$$

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Support Vector Machines

- SVMs for regression:
  - Kernel types:
    - Linear
    - Polynomial

$$K(x, x_i) = \langle x, x_i \rangle^d$$

    - Radial Basis

$$K(x, x_i) = \exp\left(-\frac{\|x - x_i\|^2}{2\sigma^2} \right)$$

    - Exponential RBF

$$K(x, x_i) = \exp\left(-\frac{\|x - x_i\|}{2\sigma^2} \right)$$

    - Etc.

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Support Vector Machines

- The SVMs were originally proposed by Boser, Guyon and Vapnik in 1992, and became very popular in the late 1990s.
- SVMs are currently among the best performing classifier systems for a great number of classification tasks, from text to genomic data
- SVMs can be applied to data complex types, which are based on feature vectors (e.g. graphs, sequences, relational data) by designing kernel functions for such data .
- SVMs have been extended to a number of tasks such as regression **[Vapnik et al. '97]**, Principal Component Analysis **[Scholkopf et al. '99]**, etc...

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Support Vector Machines

- SVMs are basically classifiers for 2 classes.
  - The formulation of the QP algorithm can be changed to allow multi-class classification .
  - More commonly, the data is "intelligently" split into two parts in different ways and an SVM is trained for each way of splitting.
  - Multi-class classification is done by combining the outputs of all classifiers
- The modeling of the SVM is such that it does not need all the available points to find a solution to the problem of maximizing the separation between classes

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Support Vector Machines

- Advantages:
  - Training is relatively easy
  - There are no local optimums, as in neural networks
  - They scale relatively well for data in high dimensional spaces
  - The trade-off between classifier complexity and error can be explicitly controlled
  - Non-traditional data such as strings and trees can be used as input to the SVM, instead of feature vectors
- Weaknesses:
  - A "good" kernel function is needed, that is, efficient methodologies are needed to tune the initialization parameters of the SVM.
  - Setting SVM hyperarameters remains a laborious process
    - Select a specific kernel and parameters is usually done through trial and error.

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